

BERT and Cross-Encoders, End to End

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BERT, End to End: Architecture, Fully Expanded Worked Examples, and Parameter Scaling

A complete reference: what BERT is and why it works, followed by every single calculation of one real encoder layer processing the sentence “I love AI” — no step abbreviated, no matrix skipped — and closing with how the architecture scales in real BERT-base/large, then extended to a cross-encoder reranker.

Part 1 — What BERT Is and Why It Works

Before BERT, models like word2vec assigned each word a single fixed vector regardless of context — “bank” got the same embedding in “river bank” and “bank account.” BERT produces **contextual** embeddings: every token’s representation is recomputed by looking at *all* other tokens around it, in both directions simultaneously — hence **B**idirectional **E**ncoder **R**epresentations from **T**ransformers.

Tokenization

BERT uses **WordPiece** subword tokenization: common words stay whole, rare words split into known pieces (“unbelievable” → [“un”, “##believable”]), so the model never hits a truly “unknown” word. Two special tokens frame every input:

- [CLS] — prepended to the start; its final representation is used as a whole-sequence summary vector (fed to a classifier head, or a reranker’s relevance score).
- [SEP] — separates segments (e.g., Query [SEP] Document).

Input representation

Each token’s input vector is the **sum** of three learned embeddings:

$$\text{Input}(\text{token}_i) = \text{TokenEmbedding}(i) + \text{SegmentEmbedding}(i) + \text{PositionEmbedding}(i)$$

- **Token embedding** — one learned vector per vocabulary entry (a 30522×768 lookup table in real BERT-base).
- **Segment embedding** — marks which of two sentences a token belongs to (Segment A / Segment B).
- **Position embedding** — real BERT uses a *learned* per-position vector (not a formula); without positional information, self-attention has no notion of word order at all — “dog bites man” and “man bites dog” would look identical to it.

Self-attention — the core mechanism

For every token, self-attention builds a new representation as a weighted combination of *every other token's* representation, with weights learned to reflect relevance. Each token's embedding is projected through three learned weight matrices:

$$Q = XW_q \quad (\text{Query — “what am I looking for”})$$

$$K = XW_k \quad (\text{Key — “what do I contain”})$$

$$V = XW_v \quad (\text{Value — “what do I offer if selected”})$$

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

d_k is the per-head key dimension (e.g. 64 in real BERT); dividing by $\sqrt{d_k}$ prevents dot products from growing large enough to push softmax into near-zero-gradient regions.

Multi-head attention

Instead of one attention computation, BERT runs h parallel heads with independent W_q, W_k, W_v , each operating in a smaller $d_k = d_{\text{model}}/h$ dimensional space. Each head can specialize — one tracking subject-verb agreement, another coreference, another local/positional patterns. Outputs are concatenated and passed through one more linear layer W_o .

One full encoder layer

$$Z = \text{LayerNorm}(X + \text{MultiHead}(X)) \quad (\text{residual + norm})$$

$$\text{Output} = \text{LayerNorm}(Z + \text{FFN}(Z)) \quad (\text{residual + norm})$$

$$\text{FFN}(z) = \text{GELU}(zW_1)W_2 \quad (\text{per-token MLP, expands 4x then back})$$

- **Residual connections** let gradients flow directly through the network, which is what makes stacking 12–24 layers trainable without vanishing gradients.
- **LayerNorm** rescales each token's vector to mean 0, variance 1, stabilizing training — it adds no new information, just keeps activations in a consistent range.

BERT-base stacks 12 of these layers; each layer's output feeds the next layer's input as its new X .

Pretraining objectives

- **Masked Language Modeling (MLM)** — mask 15% of input tokens, predict the original token using both left *and* right context (this bidirectionality is exactly what a left-to-right model like GPT cannot do without cheating). Of the masked 15%: 80% become [MASK], 10% become a random token, 10% stay unchanged — preventing the model from only learning to handle the literal [MASK] token, which never appears at real inference time.
- **Next Sentence Prediction (NSP)** — predict whether segment B genuinely follows segment A, or is a random unrelated sentence. (Later work — RoBERTa — found this contributes little; most modern BERT variants drop it.)

Fine-tuning

The same pretrained weights get a small task-specific head bolted on: - **Sentence-level classification / reranking:** final [CLS] vector $\rightarrow$ linear layer $\rightarrow$ score. - **Token-level tasks (NER, QA span extraction):** every token's final vector $\rightarrow$ linear layer. - **Sentence-pair tasks (cross-encoders):** [CLS] Query [SEP] Document [SEP] $\rightarrow$ because self-attention lets every token attend across the [SEP] boundary, the model directly models query-document token interactions — exactly why cross-encoders outperform bi-encoders (which never let the two texts attend to each other) on ranking accuracy.

Part 2 — Full Worked Example: “I love AI”

Toy dimensions used throughout so every number is hand-traceable: $d_{model} = 4$, $num_heads = 2$, $d_k = d_v = 2$ per head, $d_{ff} = 8$. All weight matrices are hand-picked illustrative values (in real BERT they are learned via backpropagation over millions of sentences) — the *mechanics* below are identical to real BERT at any scale.

Step 1 — Tokenization

[CLS] I love AI [SEP] — 5 tokens (all common enough to stay whole under WordPiece).

Steps 2-5 — Embeddings

Token, segment, and position embeddings are looked up and summed into the input matrix X . Every calculation for all 5 tokens is fully expanded in **Appendix A**.

Steps 6-11 — Attention, residual, LayerNorm, FFN

Every step of the encoder layer — Q/K/V projection for both heads, the full bidirectional attention score/softmax/weighted-sum matrices, the W_o output projection, both residual+LayerNorm steps, and the complete feed-forward network — is fully expanded in **Appendix A**, with every weight matrix shown before it's used and every result shown as a complete matrix afterward.

Part 3 — How Many Heads BERT Actually Uses, and What Scales With It

	BERT-base	BERT-large
Layers (L)	12	24
Hidden size (d_{model})	768	1024
Attention heads (A)	12	16
Dim per head ($d_k = d_{model}/A$)	64	64
FFN inner dim (d_{ff})	3072 ($4 \times d_{model}$)	4096 ($4 \times d_{model}$)
Total parameters	~110M	~340M

The head count is constrained: d_{model} must divide evenly by num_heads , since each head gets $d_k = d_{model}/num_heads$ ($768/12 = 64$, $1024/16 = 64$). Both land on $d_k = 64$ — not a coincidence, but an empirically-settled convention: when researchers scale d_{model} up, they scale num_heads up proportionally to keep each head's dimension fixed, rather than shrinking it.

More heads (at fixed d_{model}) adds zero extra parameters. The full W_q across all heads combined is really one $d_{model} \times d_{model}$ matrix; splitting it into h heads of d_{model}/h dims each is a *reshape*, not new weights. Per layer:

$$\text{Params(attention)} = 4 \times d_{model}^2 \quad (W_q + W_k + W_v + W_o, \text{ each } d_{model} \times d_{model})$$

$$\text{Params(FFN)} = 2 \times d_{\text{model}} \times d_{\text{ff}} \quad (W_1 + W_2)$$

For BERT-base: $4 \times 768^2 = 2,359,296$ (attention) + $2 \times 768 \times 3072 = 4,718,592$ (FFN) $\approx 7.08M$ per layer $\times 12$ layers $\approx 85M$, plus token/position/segment embedding tables ($\sim 23M$ for a 30522-word vocab at 768 dims) $\rightarrow \sim \mathbf{110M}$ total, matching the published figure.

What changes as you add more heads (d_{model} fixed): no new parameters — you’re carving the same total dimensionality into more, narrower “views.” Each head can specialize in a different relationship (subject-verb agreement, coreference, local position), at the cost of each head having less room to represent what it tracks. Too many heads with too little d_k each hurts — there’s a practical floor below which a head can’t represent much, which is why nobody uses e.g. 768 heads of dim 1. Too few heads means less specialization diversity even though each is individually more expressive.

What changes as d_{model} scales up (base $\rightarrow$ large): genuinely more parameters — quadratically in the attention block, linearly in the FFN — which is why BERT-large has roughly 3x the parameters of BERT-base despite only $\sim 1.33x$ the hidden size, and needs meaningfully more compute and data to train well.

Part 4 — From BERT to a Cross-Encoder: Scoring a Query-Document Pair

A cross-encoder reranker is BERT with one addition: a small classifier head on top of [CLS], applied to **a query and a document packed into one sequence** rather than a single sentence. Every layer inside — embeddings, multi-head self-attention, residual+LayerNorm, feed-forward network — is identical to what Parts 1–2 already covered. This section walks the exact same architecture through a query-document pair: **query = “AI risky”, document = “AI safe”**.

Step 1 — Tokenize query + document as one sequence

Sequence: [CLS] AI risky [SEP] AI safe [SEP] (7 tokens)
 Segment A = [CLS] AI risky [SEP] (query side)
 Segment B = AI safe [SEP] (document side)

Note “AI” and [SEP] each appear twice — same vocabulary lookup vector both times, but they end up with different final representations because their segment and position differ, and because attention pulls in different surrounding context each time.

Steps 2-11 — Embeddings, attention, residual+LayerNorm, FFN

Every step is identical in mechanism to Part 2 — fully expanded for all 7 tokens in **Appendix B**. The segment embedding is what tells the model “these first 4 tokens are the query, these last 3 are the document” — a bi-encoder never needs this because it never sees both texts in one sequence.

Step 12 — Classifier head on [CLS] (the piece plain BERT doesn’t have)

Only the [CLS] vector is used. One more linear layer, learned during fine-tuning on labeled query-document relevance pairs (e.g. MS MARCO triples), turns it into a score:

$$\begin{aligned} \text{logit} &= \text{CLS} \cdot W_{\text{clf}} + b_{\text{clf}} \\ \text{score} &= \text{sigmoid}(\text{logit}) = \frac{1}{1 + e^{-\text{logit}}} \end{aligned}$$

Fully worked in **Appendix B**, ending in a single relevance score for the “AI risky” / “AI safe” pair. With hand-picked (untrained) weights, the specific score is not meaningful — only the mechanism connecting an attended-over [CLS] vector to a final relevance score is real. In an actually fine-tuned reranker, W_{clf} (and every layer before it) has been trained so this exact pipeline produces high scores for genuinely relevant query-document pairs and low scores for irrelevant ones.

Appendix A — Fully Expanded Per-Token Arithmetic, in LaTeX Matrix Form: “I love AI”

X (input embedding = token + segment + position) (shape 5×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
I	1.0915	0.6903	−0.0400	1.0500
love	0.8593	−0.0661	0.2700	1.1498
AI	0.4911	−1.1400	0.1800	1.2496
[SEP]	−0.7068	−0.6036	0.1900	0.9492

Head 1: Q, K, V projections

W_{q_1} (shape 4×2):

1.0000	0.0000
0.0000	2.0000
1.0000	1.0000
0.0000	0.0000

X (every token as a row) (shape 5×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
I	1.0915	0.6903	−0.0400	1.0500
love	0.8593	−0.0661	0.2700	1.1498
AI	0.4911	−1.1400	0.1800	1.2496
[SEP]	−0.7068	−0.6036	0.1900	0.9492

Expanded arithmetic, $Q = X \cdot W_{q_1}$, every token:

[CLS] (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$Q[0] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

$$Q[1] = (0.1500 \times 0.0000) + (1.0500 \times 2.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 2.1500$$

I (X row = $[1.0915, 0.6903, −0.0400, 1.0500]$)

$$Q[0] = (1.0915 \times 1.0000) + (0.6903 \times 0.0000) + (−0.0400 \times 1.0000) + (1.0500 \times 0.0000) = 1.0515$$

$$Q[1] = (1.0915 \times 0.0000) + (0.6903 \times 2.0000) + (−0.0400 \times 1.0000) + (1.0500 \times 0.0000) = 1.3406$$

love (X row = $[0.8593, −0.0661, 0.2700, 1.1498]$)

$$Q[0] = (0.8593 \times 1.0000) + (−0.0661 \times 0.0000) + (0.2700 \times 1.0000) + (1.1498 \times 0.0000) = 1.1293$$

$$Q[1] = (0.8593 \times 0.0000) + (−0.0661 \times 2.0000) + (0.2700 \times 1.0000) + (1.1498 \times 0.0000) = 0.1377$$

AI (X row = $[0.4911, −1.1400, 0.1800, 1.2496]$)

$$Q[0] = (0.4911 \times 1.0000) + (−1.1400 \times 0.0000) + (0.1800 \times 1.0000) + (1.2496 \times 0.0000) = 0.6711$$

$$Q[1] = (0.4911 \times 0.0000) + (−1.1400 \times 2.0000) + (0.1800 \times 1.0000) + (1.2496 \times 0.0000) = −2.1000$$

[SEP] (X row = $[−0.7068, −0.6036, 0.1900, 0.9492]$)

$$Q[0] = (-0.7068 \times 1.0000) + (-0.6036 \times 0.0000) + (0.1900 \times 1.0000) + (0.9492 \times 0.0000) = -0.5168$$

$$Q[1] = (-0.7068 \times 0.0000) + (-0.6036 \times 2.0000) + (0.1900 \times 1.0000) + (0.9492 \times 0.0000) = -1.0173$$

Resulting Q matrix (shape 5×2):

$$\begin{array}{c|cc} \text{[CLS]} & 0.2000 & 2.1500 \\ \text{I} & 1.0515 & 1.3406 \\ \text{love} & 1.1293 & 0.1377 \\ \text{AI} & 0.6711 & -2.1000 \\ \text{[SEP]} & -0.5168 & -1.0173 \end{array}$$

Wk_1 (shape 4×2):

$$\begin{bmatrix} 2.0000 & 0.0000 \\ 0.0000 & 1.0000 \\ 1.0000 & 0.0000 \\ 0.0000 & 1.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 5×4):

$$\begin{array}{c|cccc} \text{[CLS]} & 0.1500 & 1.0500 & 0.0500 & 1.1500 \\ \text{I} & 1.0915 & 0.6903 & -0.0400 & 1.0500 \\ \text{love} & 0.8593 & -0.0661 & 0.2700 & 1.1498 \\ \text{AI} & 0.4911 & -1.1400 & 0.1800 & 1.2496 \\ \text{[SEP]} & -0.7068 & -0.6036 & 0.1900 & 0.9492 \end{array}$$

Expanded arithmetic, $K = X \cdot Wk_1$, every token:

$[CLS]$ (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$K[0] = (0.1500 \times 2.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.3500$$

$$K[1] = (0.1500 \times 0.0000) + (1.0500 \times 1.0000) + (0.0500 \times 0.0000) + (1.1500 \times 1.0000) = 2.2000$$

I (X row = $[1.0915, 0.6903, -0.0400, 1.0500]$)

$$K[0] = (1.0915 \times 2.0000) + (0.6903 \times 0.0000) + (-0.0400 \times 1.0000) + (1.0500 \times 0.0000) = 2.1429$$

$$K[1] = (1.0915 \times 0.0000) + (0.6903 \times 1.0000) + (-0.0400 \times 0.0000) + (1.0500 \times 1.0000) = 1.7403$$

$love$ (X row = $[0.8593, -0.0661, 0.2700, 1.1498]$)

$$K[0] = (0.8593 \times 2.0000) + (-0.0661 \times 0.0000) + (0.2700 \times 1.0000) + (1.1498 \times 0.0000) = 1.9886$$

$$K[1] = (0.8593 \times 0.0000) + (-0.0661 \times 1.0000) + (0.2700 \times 0.0000) + (1.1498 \times 1.0000) = 1.0837$$

AI (X row = $[0.4911, -1.1400, 0.1800, 1.2496]$)

$$K[0] = (0.4911 \times 2.0000) + (-1.1400 \times 0.0000) + (0.1800 \times 1.0000) + (1.2496 \times 0.0000) = 1.1622$$

$$K[1] = (0.4911 \times 0.0000) + (-1.1400 \times 1.0000) + (0.1800 \times 0.0000) + (1.2496 \times 1.0000) = 0.1096$$

[SEP] (X row = $[-0.7068, -0.6036, 0.1900, 0.9492]$)

$$K[0] = (-0.7068 \times 2.0000) + (-0.6036 \times 0.0000) + (0.1900 \times 1.0000) + (0.9492 \times 0.0000) = -1.2236$$

$$K[1] = (-0.7068 \times 0.0000) + (-0.6036 \times 1.0000) + (0.1900 \times 0.0000) + (0.9492 \times 1.0000) = 0.3456$$

Resulting K matrix (shape 5×2):

$$\begin{array}{c|cc} \text{[CLS]} & 0.3500 & 2.2000 \\ \text{I} & 2.1429 & 1.7403 \\ \text{love} & 1.9886 & 1.0837 \\ \text{AI} & 1.1622 & 0.1096 \\ \text{[SEP]} & -1.2236 & 0.3456 \end{array}$$

Wv_1 (shape 4×2):

$$\begin{bmatrix} 0.1000 & 0.1000 \\ 0.1000 & 0.0000 \\ 0.0000 & 0.2000 \\ 0.2000 & 0.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 5×4):

$$\begin{array}{c|cccc} \text{[CLS]} & 0.1500 & 1.0500 & 0.0500 & 1.1500 \\ \text{I} & 1.0915 & 0.6903 & -0.0400 & 1.0500 \\ \text{love} & 0.8593 & -0.0661 & 0.2700 & 1.1498 \\ \text{AI} & 0.4911 & -1.1400 & 0.1800 & 1.2496 \\ \text{[SEP]} & -0.7068 & -0.6036 & 0.1900 & 0.9492 \end{array}$$

Expanded arithmetic, $V = X \cdot Wv_1$, every token:

[CLS] (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$V[0] = (0.1500 \times 0.1000) + (1.0500 \times 0.1000) + (0.0500 \times 0.0000) + (1.1500 \times 0.2000) = 0.3500$$

$$V[1] = (0.1500 \times 0.1000) + (1.0500 \times 0.0000) + (0.0500 \times 0.2000) + (1.1500 \times 0.0000) = 0.0250$$

I (X row = $[1.0915, 0.6903, -0.0400, 1.0500]$)

$$V[0] = (1.0915 \times 0.1000) + (0.6903 \times 0.1000) + (-0.0400 \times 0.0000) + (1.0500 \times 0.2000) = 0.3882$$

$$V[1] = (1.0915 \times 0.1000) + (0.6903 \times 0.0000) + (-0.0400 \times 0.2000) + (1.0500 \times 0.0000) = 0.1011$$

love (X row = $[0.8593, -0.0661, 0.2700, 1.1498]$)

$$V[0] = (0.8593 \times 0.1000) + (-0.0661 \times 0.1000) + (0.2700 \times 0.0000) + (1.1498 \times 0.2000) = 0.3093$$

$$V[1] = (0.8593 \times 0.1000) + (-0.0661 \times 0.0000) + (0.2700 \times 0.2000) + (1.1498 \times 0.0000) = 0.1399$$

AI (X row = $[0.4911, -1.1400, 0.1800, 1.2496]$)

$$V[0] = (0.4911 \times 0.1000) + (-1.1400 \times 0.1000) + (0.1800 \times 0.0000) + (1.2496 \times 0.2000) = 0.1850$$

$$V[1] = (0.4911 \times 0.1000) + (-1.1400 \times 0.0000) + (0.1800 \times 0.2000) + (1.2496 \times 0.0000) = 0.0851$$

$$[SEP] \text{ (} X \text{ row} = [-0.7068, -0.6036, 0.1900, 0.9492])$$

$$V[0] = (-0.7068 \times 0.1000) + (-0.6036 \times 0.1000) + (0.1900 \times 0.0000) + (0.9492 \times 0.2000) = 0.0588$$

$$V[1] = (-0.7068 \times 0.1000) + (-0.6036 \times 0.0000) + (0.1900 \times 0.2000) + (0.9492 \times 0.0000) = -0.0327$$

Resulting V matrix (shape 5×2):

$$\begin{array}{c|cc} [CLS] & 0.3500 & 0.0250 \\ I & 0.3882 & 0.1011 \\ love & 0.3093 & 0.1399 \\ AI & 0.1850 & 0.0851 \\ [SEP] & 0.0588 & -0.0327 \end{array}$$

Head 2: Q , K , V projections

W_{q_2} (shape 4×2):

$$\begin{bmatrix} 0.0000 & 1.0000 \\ 2.0000 & 0.0000 \\ 0.0000 & 1.0000 \\ 1.0000 & 0.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 5×4):

$$\begin{array}{c|cccc} [CLS] & 0.1500 & 1.0500 & 0.0500 & 1.1500 \\ I & 1.0915 & 0.6903 & -0.0400 & 1.0500 \\ love & 0.8593 & -0.0661 & 0.2700 & 1.1498 \\ AI & 0.4911 & -1.1400 & 0.1800 & 1.2496 \\ [SEP] & -0.7068 & -0.6036 & 0.1900 & 0.9492 \end{array}$$

Expanded arithmetic, $Q = X \cdot W_{q_2}$, every token:

$$[CLS] \text{ (} X \text{ row} = [0.1500, 1.0500, 0.0500, 1.1500])$$

$$Q[0] = (0.1500 \times 0.0000) + (1.0500 \times 2.0000) + (0.0500 \times 0.0000) + (1.1500 \times 1.0000) = 3.2500$$

$$Q[1] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

$$I \text{ (} X \text{ row} = [1.0915, 0.6903, -0.0400, 1.0500])$$

$$Q[0] = (1.0915 \times 0.0000) + (0.6903 \times 2.0000) + (-0.0400 \times 0.0000) + (1.0500 \times 1.0000) = 2.4306$$

$$Q[1] = (1.0915 \times 1.0000) + (0.6903 \times 0.0000) + (-0.0400 \times 1.0000) + (1.0500 \times 0.0000) = 1.0515$$

$$love \text{ (} X \text{ row} = [0.8593, -0.0661, 0.2700, 1.1498])$$

$$Q[0] = (0.8593 \times 0.0000) + (-0.0661 \times 2.0000) + (0.2700 \times 0.0000) + (1.1498 \times 1.0000) = 1.0175$$

$$Q[1] = (0.8593 \times 1.0000) + (-0.0661 \times 0.0000) + (0.2700 \times 1.0000) + (1.1498 \times 0.0000) = 1.1293$$

$$AI \text{ (} X \text{ row} = [0.4911, -1.1400, 0.1800, 1.2496])$$

$$Q[0] = (0.4911 \times 0.0000) + (-1.1400 \times 2.0000) + (0.1800 \times 0.0000) + (1.2496 \times 1.0000) = -1.0304$$

$$Q[1] = (0.4911 \times 1.0000) + (-1.1400 \times 0.0000) + (0.1800 \times 1.0000) + (1.2496 \times 0.0000) = 0.6711$$

[SEP] (X row = $[-0.7068, -0.6036, 0.1900, 0.9492]$)

$$Q[0] = (-0.7068 \times 0.0000) + (-0.6036 \times 2.0000) + (0.1900 \times 0.0000) + (0.9492 \times 1.0000) = -0.2581$$

$$Q[1] = (-0.7068 \times 1.0000) + (-0.6036 \times 0.0000) + (0.1900 \times 1.0000) + (0.9492 \times 0.0000) = -0.5168$$

Resulting Q matrix (shape 5×2):

$$\begin{array}{c|cc} \text{[CLS]} & 3.2500 & 0.2000 \\ \text{I} & 2.4306 & 1.0515 \\ \text{love} & 1.0175 & 1.1293 \\ \text{AI} & -1.0304 & 0.6711 \\ \text{[SEP]} & -0.2581 & -0.5168 \end{array}$$

Wk_2 (shape 4×2):

$$\begin{bmatrix} 1.0000 & 1.0000 \\ 0.0000 & 2.0000 \\ 1.0000 & 0.0000 \\ 0.0000 & 0.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 5×4):

$$\begin{array}{c|cccc} \text{[CLS]} & 0.1500 & 1.0500 & 0.0500 & 1.1500 \\ \text{I} & 1.0915 & 0.6903 & -0.0400 & 1.0500 \\ \text{love} & 0.8593 & -0.0661 & 0.2700 & 1.1498 \\ \text{AI} & 0.4911 & -1.1400 & 0.1800 & 1.2496 \\ \text{[SEP]} & -0.7068 & -0.6036 & 0.1900 & 0.9492 \end{array}$$

Expanded arithmetic, $K = X \cdot Wk_2$, every token:

[CLS] (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$K[0] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

$$K[1] = (0.1500 \times 1.0000) + (1.0500 \times 2.0000) + (0.0500 \times 0.0000) + (1.1500 \times 0.0000) = 2.2500$$

I (X row = $[1.0915, 0.6903, -0.0400, 1.0500]$)

$$K[0] = (1.0915 \times 1.0000) + (0.6903 \times 0.0000) + (-0.0400 \times 1.0000) + (1.0500 \times 0.0000) = 1.0515$$

$$K[1] = (1.0915 \times 1.0000) + (0.6903 \times 2.0000) + (-0.0400 \times 0.0000) + (1.0500 \times 0.0000) = 2.4721$$

love (X row = $[0.8593, -0.0661, 0.2700, 1.1498]$)

$$K[0] = (0.8593 \times 1.0000) + (-0.0661 \times 0.0000) + (0.2700 \times 1.0000) + (1.1498 \times 0.0000) = 1.1293$$

$$K[1] = (0.8593 \times 1.0000) + (-0.0661 \times 2.0000) + (0.2700 \times 0.0000) + (1.1498 \times 0.0000) = 0.7270$$

AI (X row = [0.4911, -1.1400, 0.1800, 1.2496])

$$K[0] = (0.4911 \times 1.0000) + (-1.1400 \times 0.0000) + (0.1800 \times 1.0000) + (1.2496 \times 0.0000) = 0.6711$$

$$K[1] = (0.4911 \times 1.0000) + (-1.1400 \times 2.0000) + (0.1800 \times 0.0000) + (1.2496 \times 0.0000) = -1.7889$$

$[SEP]$ (X row = [-0.7068, -0.6036, 0.1900, 0.9492])

$$K[0] = (-0.7068 \times 1.0000) + (-0.6036 \times 0.0000) + (0.1900 \times 1.0000) + (0.9492 \times 0.0000) = -0.5168$$

$$K[1] = (-0.7068 \times 1.0000) + (-0.6036 \times 2.0000) + (0.1900 \times 0.0000) + (0.9492 \times 0.0000) = -1.9141$$

Resulting K matrix (shape 5×2):

[CLS]	0.2000	2.2500
I	1.0515	2.4721
love	1.1293	0.7270
AI	0.6711	-1.7889
[SEP]	-0.5168	-1.9141

Wv_2 (shape 4×2):

0.0000	0.2000
0.1000	0.1000
0.0000	0.1000
0.1000	0.0000

X (**every token as a row**) (shape 5×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
I	1.0915	0.6903	-0.0400	1.0500
love	0.8593	-0.0661	0.2700	1.1498
AI	0.4911	-1.1400	0.1800	1.2496
[SEP]	-0.7068	-0.6036	0.1900	0.9492

Expanded arithmetic, $V = X \cdot Wv_2$, every token:

$[CLS]$ (X row = [0.1500, 1.0500, 0.0500, 1.1500])

$$V[0] = (0.1500 \times 0.0000) + (1.0500 \times 0.1000) + (0.0500 \times 0.0000) + (1.1500 \times 0.1000) = 0.2200$$

$$V[1] = (0.1500 \times 0.2000) + (1.0500 \times 0.1000) + (0.0500 \times 0.1000) + (1.1500 \times 0.0000) = 0.1400$$

I (X row = [1.0915, 0.6903, -0.0400, 1.0500])

$$V[0] = (1.0915 \times 0.0000) + (0.6903 \times 0.1000) + (-0.0400 \times 0.0000) + (1.0500 \times 0.1000) = 0.1740$$

$$V[1] = (1.0915 \times 0.2000) + (0.6903 \times 0.1000) + (-0.0400 \times 0.1000) + (1.0500 \times 0.0000) = 0.2833$$

$love$ (X row = [0.8593, -0.0661, 0.2700, 1.1498])

$$V[0] = (0.8593 \times 0.0000) + (-0.0661 \times 0.1000) + (0.2700 \times 0.0000) + (1.1498 \times 0.1000) = 0.1084$$

$$V[1] = (0.8593 \times 0.2000) + (-0.0661 \times 0.1000) + (0.2700 \times 0.1000) + (1.1498 \times 0.0000) = 0.1922$$

AI (X row = [0.4911, -1.1400, 0.1800, 1.2496])

$$V[0] = (0.4911 \times 0.0000) + (-1.1400 \times 0.1000) + (0.1800 \times 0.0000) + (1.2496 \times 0.1000) = 0.0110$$

$$V[1] = (0.4911 \times 0.2000) + (-1.1400 \times 0.1000) + (0.1800 \times 0.1000) + (1.2496 \times 0.0000) = 0.0022$$

[SEP] (X row = [-0.7068, -0.6036, 0.1900, 0.9492])

$$V[0] = (-0.7068 \times 0.0000) + (-0.6036 \times 0.1000) + (0.1900 \times 0.0000) + (0.9492 \times 0.1000) = 0.0346$$

$$V[1] = (-0.7068 \times 0.2000) + (-0.6036 \times 0.1000) + (0.1900 \times 0.1000) + (0.9492 \times 0.0000) = -0.1827$$

Resulting V matrix (shape 5×2):

[CLS]	0.2200	0.1400
I	0.1740	0.2833
love	0.1084	0.1922
AI	0.0110	0.0022
[SEP]	0.0346	-0.1827

Head 1: full bidirectional attention

Head 1: Q matrix (shape 5×2):

[CLS]	0.2000	2.1500
I	1.0515	1.3406
love	1.1293	0.1377
AI	0.6711	-2.1000
[SEP]	-0.5168	-1.0173

Head 1: K matrix (shape 5×2):

[CLS]	0.3500	2.2000
I	2.1429	1.7403
love	1.9886	1.0837
AI	1.1622	0.1096
[SEP]	-1.2236	0.3456

Head 1: V matrix (shape 5×2):

[CLS]	0.3500	0.0250
I	0.3882	0.1011
love	0.3093	0.1399
AI	0.1850	0.0851
[SEP]	0.0588	-0.0327

Attention scores (bidirectional) $\text{score}(i, j) = (Q_i \cdot K_j) / \sqrt{2} = /1.4142$:

Query: [CLS]

$$\text{score}([\text{CLS}], [\text{CLS}]) = \frac{(0.2000 \times 0.3500) + (2.1500 \times 2.2000)}{1.4142} = 3.3941$$

$$\begin{aligned}
\text{score}([\text{CLS}], \text{I}) &= \frac{(0.2000 \times 2.1429) + (2.1500 \times 1.7403)}{1.4142} = 2.9487 \\
\text{score}([\text{CLS}], \text{love}) &= \frac{(0.2000 \times 1.9886) + (2.1500 \times 1.0837)}{1.4142} = 1.9287 \\
\text{score}([\text{CLS}], \text{AI}) &= \frac{(0.2000 \times 1.1622) + (2.1500 \times 0.1096)}{1.4142} = 0.3309 \\
\text{score}([\text{CLS}], [\text{SEP}]) &= \frac{(0.2000 \times -1.2236) + (2.1500 \times 0.3456)}{1.4142} = 0.3523
\end{aligned}$$

Query: I

$$\begin{aligned}
\text{score}(\text{I}, [\text{CLS}]) &= \frac{(1.0515 \times 0.3500) + (1.3406 \times 2.2000)}{1.4142} = 2.3457 \\
\text{score}(\text{I}, \text{I}) &= \frac{(1.0515 \times 2.1429) + (1.3406 \times 1.7403)}{1.4142} = 3.2430 \\
\text{score}(\text{I}, \text{love}) &= \frac{(1.0515 \times 1.9886) + (1.3406 \times 1.0837)}{1.4142} = 2.5058 \\
\text{score}(\text{I}, \text{AI}) &= \frac{(1.0515 \times 1.1622) + (1.3406 \times 0.1096)}{1.4142} = 0.9680 \\
\text{score}(\text{I}, [\text{SEP}]) &= \frac{(1.0515 \times -1.2236) + (1.3406 \times 0.3456)}{1.4142} = -0.5822
\end{aligned}$$

Query: love

$$\begin{aligned}
\text{score}(\text{love}, [\text{CLS}]) &= \frac{(1.1293 \times 0.3500) + (0.1377 \times 2.2000)}{1.4142} = 0.4937 \\
\text{score}(\text{love}, \text{I}) &= \frac{(1.1293 \times 2.1429) + (0.1377 \times 1.7403)}{1.4142} = 1.8807 \\
\text{score}(\text{love}, \text{love}) &= \frac{(1.1293 \times 1.9886) + (0.1377 \times 1.0837)}{1.4142} = 1.6935 \\
\text{score}(\text{love}, \text{AI}) &= \frac{(1.1293 \times 1.1622) + (0.1377 \times 0.1096)}{1.4142} = 0.9388 \\
\text{score}(\text{love}, [\text{SEP}]) &= \frac{(1.1293 \times -1.2236) + (0.1377 \times 0.3456)}{1.4142} = -0.9434
\end{aligned}$$

Query: AI

$$\begin{aligned}
\text{score}(\text{AI}, [\text{CLS}]) &= \frac{(0.6711 \times 0.3500) + (-2.1000 \times 2.2000)}{1.4142} = -3.1007 \\
\text{score}(\text{AI}, \text{I}) &= \frac{(0.6711 \times 2.1429) + (-2.1000 \times 1.7403)}{1.4142} = -1.5672 \\
\text{score}(\text{AI}, \text{love}) &= \frac{(0.6711 \times 1.9886) + (-2.1000 \times 1.0837)}{1.4142} = -0.6654 \\
\text{score}(\text{AI}, \text{AI}) &= \frac{(0.6711 \times 1.1622) + (-2.1000 \times 0.1096)}{1.4142} = 0.3889 \\
\text{score}(\text{AI}, [\text{SEP}]) &= \frac{(0.6711 \times -1.2236) + (-2.1000 \times 0.3456)}{1.4142} = -1.0938
\end{aligned}$$

Query: [SEP]

$$\begin{aligned}
\text{score}([SEP], [CLS]) &= \frac{(-0.5168 \times 0.3500) + (-1.0173 \times 2.2000)}{1.4142} = -1.7104 \\
\text{score}([SEP], I) &= \frac{(-0.5168 \times 2.1429) + (-1.0173 \times 1.7403)}{1.4142} = -2.0350 \\
\text{score}([SEP], \text{love}) &= \frac{(-0.5168 \times 1.9886) + (-1.0173 \times 1.0837)}{1.4142} = -1.5062 \\
\text{score}([SEP], AI) &= \frac{(-0.5168 \times 1.1622) + (-1.0173 \times 0.1096)}{1.4142} = -0.5035 \\
\text{score}([SEP], [SEP]) &= \frac{(-0.5168 \times -1.2236) + (-1.0173 \times 0.3456)}{1.4142} = 0.1986
\end{aligned}$$

Head 1: raw score matrix (shape 5×5):

	<i>[CLS]</i>	<i>I</i>	<i>love</i>	<i>AI</i>	<i>[SEP]</i>
<i>[CLS]</i>	3.3941	2.9487	1.9287	0.3309	0.3523
<i>I</i>	2.3457	3.2430	2.5058	0.9680	-0.5822
<i>love</i>	0.4937	1.8807	1.6935	0.9388	-0.9434
<i>AI</i>	-3.1007	-1.5672	-0.6654	0.3889	-1.0938
<i>[SEP]</i>	-1.7104	-2.0350	-1.5062	-0.5035	0.1986

Softmax over every row:

[CLS]:

$$\begin{aligned}
\text{exponentials} &= [e^{3.3941} = 1.0000, e^{2.9487} = 0.6406, e^{1.9287} = 0.2310, e^{0.3309} = 0.0467, \\
&\quad e^{0.3523} = 0.0477] \\
\text{sum} &= 1.9660 \\
\text{weights} &= [\frac{1.0000}{1.9660} = 0.5086, \frac{0.6406}{1.9660} = 0.3258, \frac{0.2310}{1.9660} = 0.1175, \frac{0.0467}{1.9660} = 0.0238, \\
&\quad \frac{0.0477}{1.9660} = 0.0243]
\end{aligned}$$

I:

$$\begin{aligned}
\text{exponentials} &= [e^{2.3457} = 0.4077, e^{3.2430} = 1.0000, e^{2.5058} = 0.4785, e^{0.9680} = 0.1028, \\
&\quad e^{-0.5822} = 0.0218] \\
\text{sum} &= 2.0108 \\
\text{weights} &= [\frac{0.4077}{2.0108} = 0.2028, \frac{1.0000}{2.0108} = 0.4973, \frac{0.4785}{2.0108} = 0.2379, \frac{0.1028}{2.0108} = 0.0511, \\
&\quad \frac{0.0218}{2.0108} = 0.0108]
\end{aligned}$$

love:

$$\begin{aligned}
\text{exponentials} &= [e^{0.4937} = 0.2498, e^{1.8807} = 1.0000, e^{1.6935} = 0.8293, e^{0.9388} = 0.3899, \\
&\quad e^{-0.9434} = 0.0594] \\
\text{sum} &= 2.5284 \\
\text{weights} &= [\frac{0.2498}{2.5284} = 0.0988, \frac{1.0000}{2.5284} = 0.3955, \frac{0.8293}{2.5284} = 0.3280, \frac{0.3899}{2.5284} = 0.1542, \\
&\quad \frac{0.0594}{2.5284} = 0.0235]
\end{aligned}$$

AI:

$$\begin{aligned}
\text{exponentials} &= [e^{-3.1007} = 0.0305, e^{-1.5672} = 0.1414, e^{-0.6654} = 0.3484, e^{0.3889} = 1.0000, \\
&\quad e^{-1.0938} = 0.2270]
\end{aligned}$$

$$\begin{aligned} \text{sum} &= 1.7474 \\ \text{weights} &= \left[\frac{0.0305}{1.7474} = 0.0175, \frac{0.1414}{1.7474} = 0.0809, \frac{0.3484}{1.7474} = 0.1994, \frac{1.0000}{1.7474} = 0.5723, \right. \\ &\quad \left. \frac{0.2270}{1.7474} = 0.1299 \right] \end{aligned}$$

[SEP]:

$$\begin{aligned} \text{exponentials} &= [e^{-1.7104} = 0.1482, e^{-2.0350} = 0.1071, e^{-1.5062} = 0.1818, e^{-0.5035} = 0.4955, \\ &\quad e^{0.1986} = 1.0000] \end{aligned}$$

$$\begin{aligned} \text{sum} &= 1.9327 \\ \text{weights} &= \left[\frac{0.1482}{1.9327} = 0.0767, \frac{0.1071}{1.9327} = 0.0554, \frac{0.1818}{1.9327} = 0.0941, \frac{0.4955}{1.9327} = 0.2564, \right. \\ &\quad \left. \frac{1.0000}{1.9327} = 0.5174 \right] \end{aligned}$$

Head 1: full softmax weight matrix (shape 5×5):

	[CLS]	I	love	AI	[SEP]
[CLS]	0.5086	0.3258	0.1175	0.0238	0.0243
I	0.2028	0.4973	0.2379	0.0511	0.0108
love	0.0988	0.3955	0.3280	0.1542	0.0235
AI	0.0175	0.0809	0.1994	0.5723	0.1299
[SEP]	0.0767	0.0554	0.0941	0.2564	0.5174

Attention output = weighted sum of V:

$$\begin{aligned} [\text{CLS}]_{[0]} &= (0.5086 \times 0.3500) + (0.3258 \times 0.3882) + (0.1175 \times 0.3093) + (0.0238 \times 0.1850) \\ &\quad + (0.0243 \times 0.0588) = 0.3467 \end{aligned}$$

$$\begin{aligned} [\text{CLS}]_{[1]} &= (0.5086 \times 0.0250) + (0.3258 \times 0.1011) + (0.1175 \times 0.1399) + (0.0238 \times 0.0851) \\ &\quad + (0.0243 \times -0.0327) = 0.0633 \end{aligned}$$

$$\begin{aligned} \text{I}_{[0]} &= (0.2028 \times 0.3500) + (0.4973 \times 0.3882) + (0.2379 \times 0.3093) + (0.0511 \times 0.1850) \\ &\quad + (0.0108 \times 0.0588) = 0.3477 \end{aligned}$$

$$\begin{aligned} \text{I}_{[1]} &= (0.2028 \times 0.0250) + (0.4973 \times 0.1011) + (0.2379 \times 0.1399) + (0.0511 \times 0.0851) \\ &\quad + (0.0108 \times -0.0327) = 0.0927 \end{aligned}$$

$$\begin{aligned} \text{love}_{[0]} &= (0.0988 \times 0.3500) + (0.3955 \times 0.3882) + (0.3280 \times 0.3093) + (0.1542 \times 0.1850) \\ &\quad + (0.0235 \times 0.0588) = 0.3195 \end{aligned}$$

$$\begin{aligned} \text{love}_{[1]} &= (0.0988 \times 0.0250) + (0.3955 \times 0.1011) + (0.3280 \times 0.1399) + (0.1542 \times 0.0851) \\ &\quad + (0.0235 \times -0.0327) = 0.1007 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[0]} &= (0.0175 \times 0.3500) + (0.0809 \times 0.3882) + (0.1994 \times 0.3093) + (0.5723 \times 0.1850) \\ &\quad + (0.1299 \times 0.0588) = 0.2127 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[1]} &= (0.0175 \times 0.0250) + (0.0809 \times 0.1011) + (0.1994 \times 0.1399) + (0.5723 \times 0.0851) \\ &\quad + (0.1299 \times -0.0327) = 0.0810 \end{aligned}$$

$$[\text{SEP}]_{[0]} = (0.0767 \times 0.3500) + (0.0554 \times 0.3882) + (0.0941 \times 0.3093) + (0.2564 \times 0.1850) \\ + (0.5174 \times 0.0588) = 0.1553$$

$$[\text{SEP}]_{[1]} = (0.0767 \times 0.0250) + (0.0554 \times 0.1011) + (0.0941 \times 0.1399) + (0.2564 \times 0.0851) \\ + (0.5174 \times -0.0327) = 0.0256$$

Head 1: resulting attention-output matrix (shape 5×2):

[CLS]	0.3467	0.0633
I	0.3477	0.0927
love	0.3195	0.1007
AI	0.2127	0.0810
[SEP]	0.1553	0.0256

Head 2: full bidirectional attention

Head 2: Q matrix (shape 5×2):

[CLS]	3.2500	0.2000
I	2.4306	1.0515
love	1.0175	1.1293
AI	-1.0304	0.6711
[SEP]	-0.2581	-0.5168

Head 2: K matrix (shape 5×2):

[CLS]	0.2000	2.2500
I	1.0515	2.4721
love	1.1293	0.7270
AI	0.6711	-1.7889
[SEP]	-0.5168	-1.9141

Head 2: V matrix (shape 5×2):

[CLS]	0.2200	0.1400
I	0.1740	0.2833
love	0.1084	0.1922
AI	0.0110	0.0022
[SEP]	0.0346	-0.1827

Attention scores (bidirectional) $\text{score}(i, j) = (Q_i \cdot K_j) / \sqrt{2} = /1.4142$:

Query: [CLS]

$$\text{score}([\text{CLS}], [\text{CLS}]) = \frac{(3.2500 \times 0.2000) + (0.2000 \times 2.2500)}{1.4142} = 0.7778$$

$$\text{score}([\text{CLS}], \text{I}) = \frac{(3.2500 \times 1.0515) + (0.2000 \times 2.4721)}{1.4142} = 2.7660$$

$$\text{score}([\text{CLS}], \text{love}) = \frac{(3.2500 \times 1.1293) + (0.2000 \times 0.7270)}{1.4142} = 2.6980$$

$$\text{score}([\text{CLS}], \text{AI}) = \frac{(3.2500 \times 0.6711) + (0.2000 \times -1.7889)}{1.4142} = 1.2893$$

$$\text{score}([\text{CLS}], [\text{SEP}]) = \frac{(3.2500 \times -0.5168) + (0.2000 \times -1.9141)}{1.4142} = -1.4584$$

Query: I

$$\text{score}(\text{I}, [\text{CLS}]) = \frac{(2.4306 \times 0.2000) + (1.0515 \times 2.2500)}{1.4142} = 2.0166$$

$$\text{score}(\text{I}, \text{I}) = \frac{(2.4306 \times 1.0515) + (1.0515 \times 2.4721)}{1.4142} = 3.6451$$

$$\text{score}(\text{I}, \text{love}) = \frac{(2.4306 \times 1.1293) + (1.0515 \times 0.7270)}{1.4142} = 2.4814$$

$$\text{score}(\text{I}, \text{AI}) = \frac{(2.4306 \times 0.6711) + (1.0515 \times -1.7889)}{1.4142} = -0.1766$$

$$\text{score}(\text{I}, [\text{SEP}]) = \frac{(2.4306 \times -0.5168) + (1.0515 \times -1.9141)}{1.4142} = -2.3114$$

Query: love

$$\text{score}(\text{love}, [\text{CLS}]) = \frac{(1.0175 \times 0.2000) + (1.1293 \times 2.2500)}{1.4142} = 1.9406$$

$$\text{score}(\text{love}, \text{I}) = \frac{(1.0175 \times 1.0515) + (1.1293 \times 2.4721)}{1.4142} = 2.7306$$

$$\text{score}(\text{love}, \text{love}) = \frac{(1.0175 \times 1.1293) + (1.1293 \times 0.7270)}{1.4142} = 1.3930$$

$$\text{score}(\text{love}, \text{AI}) = \frac{(1.0175 \times 0.6711) + (1.1293 \times -1.7889)}{1.4142} = -0.9456$$

$$\text{score}(\text{love}, [\text{SEP}]) = \frac{(1.0175 \times -0.5168) + (1.1293 \times -1.9141)}{1.4142} = -1.9003$$

Query: AI

$$\text{score}(\text{AI}, [\text{CLS}]) = \frac{(-1.0304 \times 0.2000) + (0.6711 \times 2.2500)}{1.4142} = 0.9220$$

$$\text{score}(\text{AI}, \text{I}) = \frac{(-1.0304 \times 1.0515) + (0.6711 \times 2.4721)}{1.4142} = 0.4070$$

$$\text{score}(\text{AI}, \text{love}) = \frac{(-1.0304 \times 1.1293) + (0.6711 \times 0.7270)}{1.4142} = -0.4778$$

$$\text{score}(\text{AI}, \text{AI}) = \frac{(-1.0304 \times 0.6711) + (0.6711 \times -1.7889)}{1.4142} = -1.3379$$

$$\text{score}(\text{AI}, [\text{SEP}]) = \frac{(-1.0304 \times -0.5168) + (0.6711 \times -1.9141)}{1.4142} = -0.5318$$

Query: [SEP]

$$\text{score}([\text{SEP}], [\text{CLS}]) = \frac{(-0.2581 \times 0.2000) + (-0.5168 \times 2.2500)}{1.4142} = -0.8587$$

$$\text{score}([\text{SEP}], \text{I}) = \frac{(-0.2581 \times 1.0515) + (-0.5168 \times 2.4721)}{1.4142} = -1.0953$$

$$\text{score}([\text{SEP}], \text{love}) = \frac{(-0.2581 \times 1.1293) + (-0.5168 \times 0.7270)}{1.4142} = -0.4718$$

$$\text{score}([\text{SEP}], \text{AI}) = \frac{(-0.2581 \times 0.6711) + (-0.5168 \times -1.7889)}{1.4142} = 0.5313$$

$$\text{score}([\text{SEP}], [\text{SEP}]) = \frac{(-0.2581 \times -0.5168) + (-0.5168 \times -1.9141)}{1.4142} = 0.7938$$

Head 2: raw score matrix (shape 5×5):

	<i>[CLS]</i>	<i>I</i>	<i>love</i>	<i>AI</i>	<i>[SEP]</i>
<i>[CLS]</i>	0.7778	2.7660	2.6980	1.2893	-1.4584
<i>I</i>	2.0166	3.6451	2.4814	-0.1766	-2.3114
<i>love</i>	1.9406	2.7306	1.3930	-0.9456	-1.9003
<i>AI</i>	0.9220	0.4070	-0.4778	-1.3379	-0.5318
<i>[SEP]</i>	-0.8587	-1.0953	-0.4718	0.5313	0.7938

Softmax over every row:

[CLS]:

$$\text{exponentials} = [e^{0.7778} = 0.1369, e^{2.7660} = 1.0000, e^{2.6980} = 0.9343, e^{1.2893} = 0.2284, e^{-1.4584} = 0.0146]$$

$$\text{sum} = 2.3143$$

$$\text{weights} = [\frac{0.1369}{2.3143} = 0.0592, \frac{1.0000}{2.3143} = 0.4321, \frac{0.9343}{2.3143} = 0.4037, \frac{0.2284}{2.3143} = 0.0987, \frac{0.0146}{2.3143} = 0.0063]$$

I:

$$\text{exponentials} = [e^{2.0166} = 0.1962, e^{3.6451} = 1.0000, e^{2.4814} = 0.3123, e^{-0.1766} = 0.0219, e^{-2.3114} = 0.0026]$$

$$\text{sum} = 1.5330$$

$$\text{weights} = [\frac{0.1962}{1.5330} = 0.1280, \frac{1.0000}{1.5330} = 0.6523, \frac{0.3123}{1.5330} = 0.2037, \frac{0.0219}{1.5330} = 0.0143, \frac{0.0026}{1.5330} = 0.0017]$$

love:

$$\text{exponentials} = [e^{1.9406} = 0.4539, e^{2.7306} = 1.0000, e^{1.3930} = 0.2625, e^{-0.9456} = 0.0253, e^{-1.9003} = 0.0097]$$

$$\text{sum} = 1.7514$$

$$\text{weights} = [\frac{0.4539}{1.7514} = 0.2591, \frac{1.0000}{1.7514} = 0.5710, \frac{0.2625}{1.7514} = 0.1499, \frac{0.0253}{1.7514} = 0.0145, \frac{0.0097}{1.7514} = 0.0056]$$

AI:

$$\text{exponentials} = [e^{0.9220} = 1.0000, e^{0.4070} = 0.5975, e^{-0.4778} = 0.2466, e^{-1.3379} = 0.1044, e^{-0.5318} = 0.2337]$$

$$\text{sum} = 2.1822$$

$$\text{weights} = [\frac{1.0000}{2.1822} = 0.4583, \frac{0.5975}{2.1822} = 0.2738, \frac{0.2466}{2.1822} = 0.1130, \frac{0.1044}{2.1822} = 0.0478, \frac{0.2337}{2.1822} = 0.1071]$$

[SEP]:

$$\text{exponentials} = [e^{-0.8587} = 0.1916, e^{-1.0953} = 0.1512, e^{-0.4718} = 0.2821, e^{0.5313} = 0.7691, \\ e^{0.7938} = 1.0000]$$

$$\text{sum} = 2.3939 \\ \text{weights} = [\frac{0.1916}{2.3939} = 0.0800, \frac{0.1512}{2.3939} = 0.0632, \frac{0.2821}{2.3939} = 0.1178, \frac{0.7691}{2.3939} = 0.3213, \\ \frac{1.0000}{2.3939} = 0.4177]$$

Head 2: full softmax weight matrix (shape 5×5):

	[CLS]	I	love	AI	[SEP]
[CLS]	0.0592	0.4321	0.4037	0.0987	0.0063
I	0.1280	0.6523	0.2037	0.0143	0.0017
love	0.2591	0.5710	0.1499	0.0145	0.0056
AI	0.4583	0.2738	0.1130	0.0478	0.1071
[SEP]	0.0800	0.0632	0.1178	0.3213	0.4177

Attention output = weighted sum of V:

$$[\text{CLS}]_{[0]} = (0.0592 \times 0.2200) + (0.4321 \times 0.1740) + (0.4037 \times 0.1084) + (0.0987 \times 0.0110) \\ + (0.0063 \times 0.0346) = 0.1333$$

$$[\text{CLS}]_{[1]} = (0.0592 \times 0.1400) + (0.4321 \times 0.2833) + (0.4037 \times 0.1922) + (0.0987 \times 0.0022) \\ + (0.0063 \times -0.1827) = 0.2074$$

$$I_{[0]} = (0.1280 \times 0.2200) + (0.6523 \times 0.1740) + (0.2037 \times 0.1084) + (0.0143 \times 0.0110) \\ + (0.0017 \times 0.0346) = 0.1640$$

$$I_{[1]} = (0.1280 \times 0.1400) + (0.6523 \times 0.2833) + (0.2037 \times 0.1922) + (0.0143 \times 0.0022) \\ + (0.0017 \times -0.1827) = 0.2416$$

$$\text{love}_{[0]} = (0.2591 \times 0.2200) + (0.5710 \times 0.1740) + (0.1499 \times 0.1084) + (0.0145 \times 0.0110) \\ + (0.0056 \times 0.0346) = 0.1730$$

$$\text{love}_{[1]} = (0.2591 \times 0.1400) + (0.5710 \times 0.2833) + (0.1499 \times 0.1922) + (0.0145 \times 0.0022) \\ + (0.0056 \times -0.1827) = 0.2259$$

$$AI_{[0]} = (0.4583 \times 0.2200) + (0.2738 \times 0.1740) + (0.1130 \times 0.1084) + (0.0478 \times 0.0110) \\ + (0.1071 \times 0.0346) = 0.1649$$

$$AI_{[1]} = (0.4583 \times 0.1400) + (0.2738 \times 0.2833) + (0.1130 \times 0.1922) + (0.0478 \times 0.0022) \\ + (0.1071 \times -0.1827) = 0.1440$$

$$[\text{SEP}]_{[0]} = (0.0800 \times 0.2200) + (0.0632 \times 0.1740) + (0.1178 \times 0.1084) + (0.3213 \times 0.0110) \\ + (0.4177 \times 0.0346) = 0.0593$$

$$[\text{SEP}]_{[1]} = (0.0800 \times 0.1400) + (0.0632 \times 0.2833) + (0.1178 \times 0.1922) + (0.3213 \times 0.0022) \\ + (0.4177 \times -0.1827) = -0.0239$$

Head 2: resulting attention-output matrix (shape 5×2):

[CLS]	0.1333	0.2074
I	0.1640	0.2416
love	0.1730	0.2259
AI	0.1649	0.1440
[SEP]	0.0593	-0.0239

Concatenate heads, project through W_o

Concat(head1, head2) (shape 5×4):

[CLS]	0.3467	0.0633	0.1333	0.2074
I	0.3477	0.0927	0.1640	0.2416
love	0.3195	0.1007	0.1730	0.2259
AI	0.2127	0.0810	0.1649	0.1440
[SEP]	0.1553	0.0256	0.0593	-0.0239

W_o (shape 4×4):

0.3000	0.1000	0.0000	0.2000
0.1000	0.2000	0.1000	0.0000
0.2000	0.0000	0.3000	0.1000
0.0000	0.3000	0.1000	0.2000

concat (**every token as a row**) (shape 5×4):

[CLS]	0.3467	0.0633	0.1333	0.2074
I	0.3477	0.0927	0.1640	0.2416
love	0.3195	0.1007	0.1730	0.2259
AI	0.2127	0.0810	0.1649	0.1440
[SEP]	0.1553	0.0256	0.0593	-0.0239

Expanded arithmetic, $MultiHead = concat \cdot W_o$, every token:

[CLS] (*concat row* = $[0.3467, 0.0633, 0.1333, 0.2074]$)

$$MultiHead[0] = (0.3467 \times 0.3000) + (0.0633 \times 0.1000) + (0.1333 \times 0.2000) + (0.2074 \times 0.0000) = 0.1370$$

$$MultiHead[1] = (0.3467 \times 0.1000) + (0.0633 \times 0.2000) + (0.1333 \times 0.0000) + (0.2074 \times 0.3000) = 0.1095$$

$$MultiHead[2] = (0.3467 \times 0.0000) + (0.0633 \times 0.1000) + (0.1333 \times 0.3000) + (0.2074 \times 0.1000) = 0.0671$$

$$MultiHead[3] = (0.3467 \times 0.2000) + (0.0633 \times 0.0000) + (0.1333 \times 0.1000) + (0.2074 \times 0.2000) = 0.1241$$

I (*concat row* = $[0.3477, 0.0927, 0.1640, 0.2416]$)

$$MultiHead[0] = (0.3477 \times 0.3000) + (0.0927 \times 0.1000) + (0.1640 \times 0.2000) + (0.2416 \times 0.0000) = 0.1464$$

$$MultiHead[1] = (0.3477 \times 0.1000) + (0.0927 \times 0.2000) + (0.1640 \times 0.0000) + (0.2416 \times 0.3000) = 0.1258$$

$$MultiHead[2] = (0.3477 \times 0.0000) + (0.0927 \times 0.1000) + (0.1640 \times 0.3000) + (0.2416 \times 0.1000) = 0.0826$$

$$MultiHead[3] = (0.3477 \times 0.2000) + (0.0927 \times 0.0000) + (0.1640 \times 0.1000) + (0.2416 \times 0.2000) = 0.1343$$

love (*concat* row = [0.3195, 0.1007, 0.1730, 0.2259])

$$MultiHead[0] = (0.3195 \times 0.3000) + (0.1007 \times 0.1000) + (0.1730 \times 0.2000) + (0.2259 \times 0.0000) = 0.1405$$

$$MultiHead[1] = (0.3195 \times 0.1000) + (0.1007 \times 0.2000) + (0.1730 \times 0.0000) + (0.2259 \times 0.3000) = 0.1199$$

$$MultiHead[2] = (0.3195 \times 0.0000) + (0.1007 \times 0.1000) + (0.1730 \times 0.3000) + (0.2259 \times 0.1000) = 0.0845$$

$$MultiHead[3] = (0.3195 \times 0.2000) + (0.1007 \times 0.0000) + (0.1730 \times 0.1000) + (0.2259 \times 0.2000) = 0.1264$$

AI (*concat* row = [0.2127, 0.0810, 0.1649, 0.1440])

$$MultiHead[0] = (0.2127 \times 0.3000) + (0.0810 \times 0.1000) + (0.1649 \times 0.2000) + (0.1440 \times 0.0000) = 0.1049$$

$$MultiHead[1] = (0.2127 \times 0.1000) + (0.0810 \times 0.2000) + (0.1649 \times 0.0000) + (0.1440 \times 0.3000) = 0.0807$$

$$MultiHead[2] = (0.2127 \times 0.0000) + (0.0810 \times 0.1000) + (0.1649 \times 0.3000) + (0.1440 \times 0.1000) = 0.0720$$

$$MultiHead[3] = (0.2127 \times 0.2000) + (0.0810 \times 0.0000) + (0.1649 \times 0.1000) + (0.1440 \times 0.2000) = 0.0878$$

[SEP] (*concat* row = [0.1553, 0.0256, 0.0593, -0.0239])

$$MultiHead[0] = (0.1553 \times 0.3000) + (0.0256 \times 0.1000) + (0.0593 \times 0.2000) + (-0.0239 \times 0.0000) = 0.0610$$

$$MultiHead[1] = (0.1553 \times 0.1000) + (0.0256 \times 0.2000) + (0.0593 \times 0.0000) + (-0.0239 \times 0.3000) = 0.0135$$

$$MultiHead[2] = (0.1553 \times 0.0000) + (0.0256 \times 0.1000) + (0.0593 \times 0.3000) + (-0.0239 \times 0.1000) = 0.0180$$

$$MultiHead[3] = (0.1553 \times 0.2000) + (0.0256 \times 0.0000) + (0.0593 \times 0.1000) + (-0.0239 \times 0.2000) = 0.0322$$

Resulting *MultiHead* matrix (shape 5×4):

[CLS]	0.1370	0.1095	0.0671	0.1241
I	0.1464	0.1258	0.0826	0.1343
love	0.1405	0.1199	0.0845	0.1264
AI	0.1049	0.0807	0.0720	0.0878
[SEP]	0.0610	0.0135	0.0180	0.0322

Residual add: $X + MultiHead(X)$

X (original embedding) (shape 5×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
I	1.0915	0.6903	-0.0400	1.0500
love	0.8593	-0.0661	0.2700	1.1498
AI	0.4911	-1.1400	0.1800	1.2496
[SEP]	-0.7068	-0.6036	0.1900	0.9492

$MultiHead(X)$ (shape 5×4):

[CLS]	0.1370	0.1095	0.0671	0.1241
I	0.1464	0.1258	0.0826	0.1343
love	0.1405	0.1199	0.0845	0.1264
AI	0.1049	0.0807	0.0720	0.0878
[SEP]	0.0610	0.0135	0.0180	0.0322

Resulting residual_1 matrix (shape 5×4):

[CLS]	0.2870	1.1595	0.1171	1.2741
I	1.2378	0.8161	0.0426	1.1842
love	0.9998	0.0537	0.3545	1.2762
AI	0.5960	-1.0593	0.2520	1.3374
[SEP]	-0.6458	-0.5902	0.2080	0.9814

LayerNorm(residual_1) -> Z

Input matrix (every token as a row) (shape 5×4):

[CLS]	0.2870	1.1595	0.1171	1.2741
I	1.2378	0.8161	0.0426	1.1842
love	0.9998	0.0537	0.3545	1.2762
AI	0.5960	-1.0593	0.2520	1.3374
[SEP]	-0.6458	-0.5902	0.2080	0.9814

$\gamma = 1$, $\beta = 0$ (identity affine, to isolate the normalization math)

[CLS]: $v = [0.2870, 1.1595, 0.1171, 1.2741]$

$$\text{mean} = \frac{0.2870 + 1.1595 + 0.1171 + 1.2741}{4} = 0.7094$$

$$\text{deviations} = [-0.4224, 0.4501, -0.5924, 0.5647] \Rightarrow \text{variance} = \frac{1.0509}{4} = 0.2627$$

$$\text{std} = \sqrt{0.2627 + 10^{-5}} = 0.5126$$

$$\text{normalized} = \left[\frac{-0.4224}{0.5126} = -0.8242, \frac{0.4501}{0.5126} = 0.8782, \frac{-0.5924}{0.5126} = -1.1557, \frac{0.5647}{0.5126} = 1.1017 \right]$$

I: $v = [1.2378, 0.8161, 0.0426, 1.1842]$

$$\text{mean} = \frac{1.2378 + 0.8161 + 0.0426 + 1.1842}{4} = 0.8202$$

$$\text{deviations} = [0.4176, -0.0041, -0.7776, 0.3640] \Rightarrow \text{variance} = \frac{0.9116}{4} = 0.2279$$

$$\text{std} = \sqrt{0.2279 + 10^{-5}} = 0.4774$$

$$\text{normalized} = \left[\frac{0.4176}{0.4774} = 0.8749, \frac{-0.0041}{0.4774} = -0.0086, \frac{-0.7776}{0.4774} = -1.6288, \frac{0.3640}{0.4774} = 0.7625 \right]$$

love: $v = [0.9998, 0.0537, 0.3545, 1.2762]$

$$\text{mean} = \frac{0.9998 + 0.0537 + 0.3545 + 1.2762}{4} = 0.6711$$

$$\text{deviations} = [0.3287, -0.6173, -0.3165, 0.6051] \Rightarrow \text{variance} = \frac{0.9555}{4} = 0.2389$$

$$\text{std} = \sqrt{0.2389 + 10^{-5}} = 0.4888$$

$$\text{normalized} = \left[\frac{0.3287}{0.4888} = 0.6726, \frac{-0.6173}{0.4888} = -1.2631, \frac{-0.3165}{0.4888} = -0.6476, \frac{0.6051}{0.4888} = 1.2380 \right]$$

AI: $v = [0.5960, -1.0593, 0.2520, 1.3374]$

$$\begin{aligned}\text{mean} &= \frac{0.5960 + -1.0593 + 0.2520 + 1.3374}{4} = 0.2815 \\ \text{deviations} &= [0.3145, -1.3408, -0.0295, 1.0559] \Rightarrow \text{variance} = \frac{3.0125}{4} = 0.7531 \\ \text{std} &= \sqrt{0.7531 + 10^{-5}} = 0.8678 \\ \text{normalized} &= \left[\frac{0.3145}{0.8678} = 0.3624, \frac{-1.3408}{0.8678} = -1.5450, \frac{-0.0295}{0.8678} = -0.0340, \frac{1.0559}{0.8678} = 1.2167 \right] \\ [SEP]: v &= [-0.6458, -0.5902, 0.2080, 0.9814]\end{aligned}$$

$$\begin{aligned}\text{mean} &= \frac{-0.6458 + -0.5902 + 0.2080 + 0.9814}{4} = -0.0116 \\ \text{deviations} &= [-0.6341, -0.5785, 0.2196, 0.9931] \Rightarrow \text{variance} = \frac{1.7712}{4} = 0.4428 \\ \text{std} &= \sqrt{0.4428 + 10^{-5}} = 0.6654 \\ \text{normalized} &= \left[\frac{-0.6341}{0.6654} = -0.9530, \frac{-0.5785}{0.6654} = -0.8694, \frac{0.2196}{0.6654} = 0.3300, \frac{0.9931}{0.6654} = 1.4923 \right]\end{aligned}$$

Resulting normalized matrix (shape 5×4):

$$\begin{array}{c|cccc} [\text{CLS}] & -0.8242 & 0.8782 & -1.1557 & 1.1017 \\ \text{I} & 0.8749 & -0.0086 & -1.6288 & 0.7625 \\ \text{love} & 0.6726 & -1.2631 & -0.6476 & 1.2380 \\ \text{AI} & 0.3624 & -1.5450 & -0.0340 & 1.2167 \\ [\text{SEP}] & -0.9530 & -0.8694 & 0.3300 & 1.4923 \end{array}$$

FFN first linear layer: hidden = Z · W₁

W₁ (shape 4×8):

$$\begin{bmatrix} -0.0750 & 0.2700 & 0.1390 & 0.0590 & -0.2060 & -0.2060 & -0.2650 & 0.2200 \\ 0.0610 & 0.1250 & -0.2880 & 0.2820 & 0.1990 & -0.1730 & -0.1910 & -0.1900 \\ -0.1170 & 0.0150 & -0.0410 & -0.1250 & 0.0670 & -0.2160 & -0.1250 & -0.0800 \\ -0.0260 & 0.1710 & -0.1800 & 0.0090 & 0.0550 & -0.2720 & 0.0650 & -0.1980 \end{bmatrix}$$

Z (every token as a row) (shape 5×4):

$$\begin{array}{c|cccc} [\text{CLS}] & -0.8242 & 0.8782 & -1.1557 & 1.1017 \\ \text{I} & 0.8749 & -0.0086 & -1.6288 & 0.7625 \\ \text{love} & 0.6726 & -1.2631 & -0.6476 & 1.2380 \\ \text{AI} & 0.3624 & -1.5450 & -0.0340 & 1.2167 \\ [\text{SEP}] & -0.9530 & -0.8694 & 0.3300 & 1.4923 \end{array}$$

Expanded arithmetic, hidden = Z · W₁, every token:

[CLS] (Z row = [-0.8242, 0.8782, -1.1557, 1.1017])

$$\text{hidden}[0] = (-0.8242 \times -0.0750) + (0.8782 \times 0.0610) + (-1.1557 \times -0.1170) + (1.1017 \times -0.0260) = 0.2220$$

$$\text{hidden}[1] = (-0.8242 \times 0.2700) + (0.8782 \times 0.1250) + (-1.1557 \times 0.0150) + (1.1017 \times 0.1710) = 0.0583$$

$$\begin{aligned}
hidden[2] &= (-0.8242 \times 0.1390) + (0.8782 \times -0.2880) + (-1.1557 \times -0.0410) + (1.1017 \times -0.1800) = -0.5184 \\
hidden[3] &= (-0.8242 \times 0.0590) + (0.8782 \times 0.2820) + (-1.1557 \times -0.1250) + (1.1017 \times 0.0090) = 0.3534 \\
hidden[4] &= (-0.8242 \times -0.2060) + (0.8782 \times 0.1990) + (-1.1557 \times 0.0670) + (1.1017 \times 0.0550) = 0.3277 \\
hidden[5] &= (-0.8242 \times -0.2060) + (0.8782 \times -0.1730) + (-1.1557 \times -0.2160) + (1.1017 \times -0.2720) = -0.0322 \\
hidden[6] &= (-0.8242 \times -0.2650) + (0.8782 \times -0.1910) + (-1.1557 \times -0.1250) + (1.1017 \times 0.0650) = 0.2668 \\
hidden[7] &= (-0.8242 \times 0.2200) + (0.8782 \times -0.1900) + (-1.1557 \times -0.0800) + (1.1017 \times -0.1980) = -0.4739 \\
I \text{ (} Z \text{ row} &= [0.8749, -0.0086, -1.6288, 0.7625])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.8749 \times -0.0750) + (-0.0086 \times 0.0610) + (-1.6288 \times -0.1170) + (0.7625 \times -0.0260) = 0.1046 \\
hidden[1] &= (0.8749 \times 0.2700) + (-0.0086 \times 0.1250) + (-1.6288 \times 0.0150) + (0.7625 \times 0.1710) = 0.3411 \\
hidden[2] &= (0.8749 \times 0.1390) + (-0.0086 \times -0.2880) + (-1.6288 \times -0.0410) + (0.7625 \times -0.1800) = 0.0536 \\
hidden[3] &= (0.8749 \times 0.0590) + (-0.0086 \times 0.2820) + (-1.6288 \times -0.1250) + (0.7625 \times 0.0090) = 0.2597 \\
hidden[4] &= (0.8749 \times -0.2060) + (-0.0086 \times 0.1990) + (-1.6288 \times 0.0670) + (0.7625 \times 0.0550) = -0.2491 \\
hidden[5] &= (0.8749 \times -0.2060) + (-0.0086 \times -0.1730) + (-1.6288 \times -0.2160) + (0.7625 \times -0.2720) = -0.0343 \\
hidden[6] &= (0.8749 \times -0.2650) + (-0.0086 \times -0.1910) + (-1.6288 \times -0.1250) + (0.7625 \times 0.0650) = 0.0230 \\
hidden[7] &= (0.8749 \times 0.2200) + (-0.0086 \times -0.1900) + (-1.6288 \times -0.0800) + (0.7625 \times -0.1980) = 0.1734 \\
love \text{ (} Z \text{ row} &= [0.6726, -1.2631, -0.6476, 1.2380])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.6726 \times -0.0750) + (-1.2631 \times 0.0610) + (-0.6476 \times -0.1170) + (1.2380 \times -0.0260) = -0.0839 \\
hidden[1] &= (0.6726 \times 0.2700) + (-1.2631 \times 0.1250) + (-0.6476 \times 0.0150) + (1.2380 \times 0.1710) = 0.2257 \\
hidden[2] &= (0.6726 \times 0.1390) + (-1.2631 \times -0.2880) + (-0.6476 \times -0.0410) + (1.2380 \times -0.1800) = 0.2610 \\
hidden[3] &= (0.6726 \times 0.0590) + (-1.2631 \times 0.2820) + (-0.6476 \times -0.1250) + (1.2380 \times 0.0090) = -0.2244 \\
hidden[4] &= (0.6726 \times -0.2060) + (-1.2631 \times 0.1990) + (-0.6476 \times 0.0670) + (1.2380 \times 0.0550) = -0.3652 \\
hidden[5] &= (0.6726 \times -0.2060) + (-1.2631 \times -0.1730) + (-0.6476 \times -0.2160) + (1.2380 \times -0.2720) = -0.1169 \\
hidden[6] &= (0.6726 \times -0.2650) + (-1.2631 \times -0.1910) + (-0.6476 \times -0.1250) + (1.2380 \times 0.0650) = 0.2244 \\
hidden[7] &= (0.6726 \times 0.2200) + (-1.2631 \times -0.1900) + (-0.6476 \times -0.0800) + (1.2380 \times -0.1980) = 0.1946 \\
AI \text{ (} Z \text{ row} &= [0.3624, -1.5450, -0.0340, 1.2167])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.3624 \times -0.0750) + (-1.5450 \times 0.0610) + (-0.0340 \times -0.1170) + (1.2167 \times -0.0260) = -0.1491 \\
hidden[1] &= (0.3624 \times 0.2700) + (-1.5450 \times 0.1250) + (-0.0340 \times 0.0150) + (1.2167 \times 0.1710) = 0.1123 \\
hidden[2] &= (0.3624 \times 0.1390) + (-1.5450 \times -0.2880) + (-0.0340 \times -0.0410) + (1.2167 \times -0.1800) = 0.2777 \\
hidden[3] &= (0.3624 \times 0.0590) + (-1.5450 \times 0.2820) + (-0.0340 \times -0.1250) + (1.2167 \times 0.0090) = -0.3991 \\
hidden[4] &= (0.3624 \times -0.2060) + (-1.5450 \times 0.1990) + (-0.0340 \times 0.0670) + (1.2167 \times 0.0550) = -0.3175 \\
hidden[5] &= (0.3624 \times -0.2060) + (-1.5450 \times -0.1730) + (-0.0340 \times -0.2160) + (1.2167 \times -0.2720) = -0.1309 \\
hidden[6] &= (0.3624 \times -0.2650) + (-1.5450 \times -0.1910) + (-0.0340 \times -0.1250) + (1.2167 \times 0.0650) = 0.2824
\end{aligned}$$

$$hidden[7] = (0.3624 \times 0.2200) + (-1.5450 \times -0.1900) + (-0.0340 \times -0.0800) + (1.2167 \times -0.1980) = 0.1351$$

$$[SEP] \text{ (} Z \text{ row} = [-0.9530, -0.8694, 0.3300, 1.4923])$$

$$hidden[0] = (-0.9530 \times -0.0750) + (-0.8694 \times 0.0610) + (0.3300 \times -0.1170) + (1.4923 \times -0.0260) = -0.0590$$

$$hidden[1] = (-0.9530 \times 0.2700) + (-0.8694 \times 0.1250) + (0.3300 \times 0.0150) + (1.4923 \times 0.1710) = -0.1058$$

$$hidden[2] = (-0.9530 \times 0.1390) + (-0.8694 \times -0.2880) + (0.3300 \times -0.0410) + (1.4923 \times -0.1800) = -0.1642$$

$$hidden[3] = (-0.9530 \times 0.0590) + (-0.8694 \times 0.2820) + (0.3300 \times -0.1250) + (1.4923 \times 0.0090) = -0.3292$$

$$hidden[4] = (-0.9530 \times -0.2060) + (-0.8694 \times 0.1990) + (0.3300 \times 0.0670) + (1.4923 \times 0.0550) = 0.1275$$

$$hidden[5] = (-0.9530 \times -0.2060) + (-0.8694 \times -0.1730) + (0.3300 \times -0.2160) + (1.4923 \times -0.2720) = -0.1305$$

$$hidden[6] = (-0.9530 \times -0.2650) + (-0.8694 \times -0.1910) + (0.3300 \times -0.1250) + (1.4923 \times 0.0650) = 0.4743$$

$$hidden[7] = (-0.9530 \times 0.2200) + (-0.8694 \times -0.1900) + (0.3300 \times -0.0800) + (1.4923 \times -0.1980) = -0.3664$$

Resulting hidden matrix (shape 5×8):

[CLS]	0.2220	0.0583	-0.5184	0.3534	0.3277	-0.0322	0.2668	-0.4739
I	0.1046	0.3411	0.0536	0.2597	-0.2491	-0.0343	0.0230	0.1734
love	-0.0839	0.2257	0.2610	-0.2244	-0.3652	-0.1169	0.2244	0.1946
AI	-0.1491	0.1123	0.2777	-0.3991	-0.3175	-0.1309	0.2824	0.1351
[SEP]	-0.0590	-0.1058	-0.1642	-0.3292	0.1275	-0.1305	0.4743	-0.3664

GELU activation

Hidden matrix BEFORE GELU (shape 5×8):

[CLS]	0.2220	0.0583	-0.5184	0.3534	0.3277	-0.0322	0.2668	-0.4739
I	0.1046	0.3411	0.0536	0.2597	-0.2491	-0.0343	0.0230	0.1734
love	-0.0839	0.2257	0.2610	-0.2244	-0.3652	-0.1169	0.2244	0.1946
AI	-0.1491	0.1123	0.2777	-0.3991	-0.3175	-0.1309	0.2824	0.1351
[SEP]	-0.0590	-0.1058	-0.1642	-0.3292	0.1275	-0.1305	0.4743	-0.3664

$$\text{GELU}(x) = 0.5x \left(1 + \tanh \left(\sqrt{2/\pi} (x + 0.044715x^3) \right) \right)$$

[CLS]:

$$[0.2220 \rightarrow 0.1305, 0.0583 \rightarrow 0.0305, -0.5184 \rightarrow -0.1566, 0.3534 \rightarrow 0.2255,$$

$$0.3277 \rightarrow 0.2059, -0.0322 \rightarrow -0.0157, 0.2668 \rightarrow 0.1734]$$

I:

$$[0.1046 \rightarrow 0.0567, 0.3411 \rightarrow 0.2161, 0.0536 \rightarrow 0.0279, 0.2597 \rightarrow 0.1564,$$

$$-0.2491 \rightarrow -0.1001, -0.0343 \rightarrow -0.0167, 0.0230 \rightarrow 0.1734]$$

love:

$$[-0.0839 \rightarrow -0.0392, 0.2257 \rightarrow 0.1330, 0.2610 \rightarrow 0.1573, -0.2244 \rightarrow -0.0923,$$

$$-0.3652 \rightarrow -0.1306, -0.1169 \rightarrow -0.0530, 0.2244 \rightarrow 0.1946]$$

AI:

$$[-0.1491 \rightarrow -0.0657, 0.1123 \rightarrow 0.0611, 0.2777 \rightarrow 0.1693, -0.3991 \rightarrow -0.1377, \\ -0.3175 \rightarrow -0.1192, -0.1309 \rightarrow -0.0587, 0.2821 \rightarrow 0.1275]$$

[SEP]:

$$[-0.0590 \rightarrow -0.0281, -0.1058 \rightarrow -0.0485, -0.1642 \rightarrow -0.0714, -0.3292 \rightarrow -0.1221, \\ 0.1275 \rightarrow 0.0702, -0.1305 \rightarrow -0.0585, 0.4051 \rightarrow 0.1308]$$

Resulting activated matrix (shape 5×8):

[CLS]	0.1305	0.0305	-0.1566	0.2255	0.2059	-0.0157	0.1614	-0.1506
I	0.0567	0.2161	0.0279	0.1564	-0.1001	-0.0167	0.0117	0.0987
love	-0.0392	0.1330	0.1573	-0.0923	-0.1306	-0.0530	0.1321	0.1123
AI	-0.0657	0.0611	0.1693	-0.1377	-0.1192	-0.0587	0.1726	0.0748
[SEP]	-0.0281	-0.0485	-0.0714	-0.1221	0.0702	-0.0585	0.3237	-0.1308

FFN second linear layer: $\text{FFN}(\mathbf{Z}) = \text{activated} \cdot \mathbf{W}_2$

$\mathbf{W}_2$ (shape 8×4):

$$\begin{bmatrix} -0.2610 & 0.2690 & 0.2790 & 0.1850 \\ -0.1170 & -0.2410 & 0.1110 & -0.0360 \\ -0.2270 & -0.0030 & -0.2790 & 0.2460 \\ -0.1450 & 0.0980 & -0.1130 & 0.0120 \\ 0.0280 & -0.1890 & 0.2820 & 0.1650 \\ 0.2640 & 0.2370 & 0.0590 & 0.2530 \\ -0.2470 & -0.1820 & -0.2730 & -0.1050 \\ -0.0670 & -0.1370 & 0.1970 & -0.0860 \end{bmatrix}$$

activated (**every token as a row**) (shape 5×8):

[CLS]	0.1305	0.0305	-0.1566	0.2255	0.2059	-0.0157	0.1614	-0.1506
I	0.0567	0.2161	0.0279	0.1564	-0.1001	-0.0167	0.0117	0.0987
love	-0.0392	0.1330	0.1573	-0.0923	-0.1306	-0.0530	0.1321	0.1123
AI	-0.0657	0.0611	0.1693	-0.1377	-0.1192	-0.0587	0.1726	0.0748
[SEP]	-0.0281	-0.0485	-0.0714	-0.1221	0.0702	-0.0585	0.3237	-0.1308

Expanded arithmetic, $\text{FFN} = \text{activated} \cdot \mathbf{W}_2$, every token:

$$[\text{CLS}] \text{ (activated row} = [0.1305, 0.0305, -0.1566, 0.2255, 0.2059, -0.0157, 0.1614, -0.1506])$$

$$\begin{aligned} \text{FFN}[0] &= (0.1305 \times -0.2610) + (0.0305 \times -0.1170) + (-0.1566 \times -0.2270) + (0.2255 \times -0.1450) \\ &\quad + (0.2059 \times 0.0280) + (-0.0157 \times 0.2640) + (0.1614 \times -0.2470) + (-0.1506 \times -0.0670) = -0.0629 \\ \text{FFN}[1] &= (0.1305 \times 0.2690) + (0.0305 \times -0.2410) + (-0.1566 \times -0.0030) + (0.2255 \times 0.0980) \\ &\quad + (0.2059 \times -0.1890) + (-0.0157 \times 0.2370) + (0.1614 \times -0.1820) + (-0.1506 \times -0.1370) = -0.0011 \\ \text{FFN}[2] &= (0.1305 \times 0.2790) + (0.0305 \times 0.1110) + (-0.1566 \times -0.2790) + (0.2255 \times -0.1130) \\ &\quad + (0.2059 \times 0.2820) + (-0.0157 \times 0.0590) + (0.1614 \times -0.2730) + (-0.1506 \times 0.1970) = 0.0414 \\ \text{FFN}[3] &= (0.1305 \times 0.1850) + (0.0305 \times -0.0360) + (-0.1566 \times 0.2460) + (0.2255 \times 0.0120) \\ &\quad + (0.2059 \times 0.1650) + (-0.0157 \times 0.2530) + (0.1614 \times -0.1050) + (-0.1506 \times -0.0860) = 0.0132 \end{aligned}$$

I (*activated row* = [0.0567, 0.2161, 0.0279, 0.1564, -0.1001, -0.0167, 0.0117, 0.0987])

$$FFN[0] = (0.0567 \times -0.2610) + (0.2161 \times -0.1170) + (0.0279 \times -0.2270) + (0.1564 \times -0.1450) \\ + (-0.1001 \times 0.0280) + (-0.0167 \times 0.2640) + (0.0117 \times -0.2470) + (0.0987 \times -0.0670) = -0.0858$$

$$FFN[1] = (0.0567 \times 0.2690) + (0.2161 \times -0.2410) + (0.0279 \times -0.0030) + (0.1564 \times 0.0980) \\ + (-0.1001 \times -0.1890) + (-0.0167 \times 0.2370) + (0.0117 \times -0.1820) + (0.0987 \times -0.1370) = -0.0223$$

$$FFN[2] = (0.0567 \times 0.2790) + (0.2161 \times 0.1110) + (0.0279 \times -0.2790) + (0.1564 \times -0.1130) \\ + (-0.1001 \times 0.2820) + (-0.0167 \times 0.0590) + (0.0117 \times -0.2730) + (0.0987 \times 0.1970) = 0.0014$$

$$FFN[3] = (0.0567 \times 0.1850) + (0.2161 \times -0.0360) + (0.0279 \times 0.2460) + (0.1564 \times 0.0120) \\ + (-0.1001 \times 0.1650) + (-0.0167 \times 0.2530) + (0.0117 \times -0.1050) + (0.0987 \times -0.0860) = -0.0190$$

$love$ (*activated row* = [-0.0392, 0.1330, 0.1573, -0.0923, -0.1306, -0.0530, 0.1321, 0.1123])

$$FFN[0] = (-0.0392 \times -0.2610) + (0.1330 \times -0.1170) + (0.1573 \times -0.2270) + (-0.0923 \times -0.1450) \\ + (-0.1306 \times 0.0280) + (-0.0530 \times 0.2640) + (0.1321 \times -0.2470) + (0.1123 \times -0.0670) = -0.0855$$

$$FFN[1] = (-0.0392 \times 0.2690) + (0.1330 \times -0.2410) + (0.1573 \times -0.0030) + (-0.0923 \times 0.0980) \\ + (-0.1306 \times -0.1890) + (-0.0530 \times 0.2370) + (0.1321 \times -0.1820) + (0.1123 \times -0.1370) = -0.0794$$

$$FFN[2] = (-0.0392 \times 0.2790) + (0.1330 \times 0.1110) + (0.1573 \times -0.2790) + (-0.0923 \times -0.1130) \\ + (-0.1306 \times 0.2820) + (-0.0530 \times 0.0590) + (0.1321 \times -0.2730) + (0.1123 \times 0.1970) = -0.0835$$

$$FFN[3] = (-0.0392 \times 0.1850) + (0.1330 \times -0.0360) + (0.1573 \times 0.2460) + (-0.0923 \times 0.0120) \\ + (-0.1306 \times 0.1650) + (-0.0530 \times 0.2530) + (0.1321 \times -0.1050) + (0.1123 \times -0.0860) = -0.0329$$

AI (*activated row* = [-0.0657, 0.0611, 0.1693, -0.1377, -0.1192, -0.0587, 0.1726, 0.0748])

$$FFN[0] = (-0.0657 \times -0.2610) + (0.0611 \times -0.1170) + (0.1693 \times -0.2270) + (-0.1377 \times -0.1450) \\ + (-0.1192 \times 0.0280) + (-0.0587 \times 0.2640) + (0.1726 \times -0.2470) + (0.0748 \times -0.0670) = -0.0749$$

$$FFN[1] = (-0.0657 \times 0.2690) + (0.0611 \times -0.2410) + (0.1693 \times -0.0030) + (-0.1377 \times 0.0980) \\ + (-0.1192 \times -0.1890) + (-0.0587 \times 0.2370) + (0.1726 \times -0.1820) + (0.0748 \times -0.1370) = -0.0794$$

$$FFN[2] = (-0.0657 \times 0.2790) + (0.0611 \times 0.1110) + (0.1693 \times -0.2790) + (-0.1377 \times -0.1130) \\ + (-0.1192 \times 0.2820) + (-0.0587 \times 0.0590) + (0.1726 \times -0.2730) + (0.0748 \times 0.1970) = -0.1127$$

$$FFN[3] = (-0.0657 \times 0.1850) + (0.0611 \times -0.0360) + (0.1693 \times 0.2460) + (-0.1377 \times 0.0120) \\ + (-0.1192 \times 0.1650) + (-0.0587 \times 0.2530) + (0.1726 \times -0.1050) + (0.0748 \times -0.0860) = -0.0334$$

$[SEP]$ (*activated row* = [-0.0281, -0.0485, -0.0714, -0.1221, 0.0702, -0.0585, 0.3237, -0.1308])

$$FFN[0] = (-0.0281 \times -0.2610) + (-0.0485 \times -0.1170) + (-0.0714 \times -0.2270) + (-0.1221 \times -0.1450) \\ + (0.0702 \times 0.0280) + (-0.0585 \times 0.2640) + (0.3237 \times -0.2470) + (-0.1308 \times -0.0670) = -0.0377$$

$$FFN[1] = (-0.0281 \times 0.2690) + (-0.0485 \times -0.2410) + (-0.0714 \times -0.0030) + (-0.1221 \times 0.0980) \\ + (0.0702 \times -0.1890) + (-0.0585 \times 0.2370) + (0.3237 \times -0.1820) + (-0.1308 \times -0.1370) = -0.0757$$

$$FFN[2] = (-0.0281 \times 0.2790) + (-0.0485 \times 0.1110) + (-0.0714 \times -0.2790) + (-0.1221 \times -0.1130) \\ + (0.0702 \times 0.2820) + (-0.0585 \times 0.0590) + (0.3237 \times -0.2730) + (-0.1308 \times 0.1970) = -0.0773$$

$$FFN[3] = (-0.0281 \times 0.1850) + (-0.0485 \times -0.0360) + (-0.0714 \times 0.2460) + (-0.1221 \times 0.0120) \\ + (0.0702 \times 0.1650) + (-0.0585 \times 0.2530) + (0.3237 \times -0.1050) + (-0.1308 \times -0.0860) = -0.0484$$

Resulting FFN matrix (shape 5×4):

[CLS]	-0.0629	-0.0011	0.0414	0.0132
I	-0.0858	-0.0223	0.0014	-0.0190
love	-0.0855	-0.0794	-0.0835	-0.0329
AI	-0.0749	-0.0794	-0.1127	-0.0334
[SEP]	-0.0377	-0.0757	-0.0773	-0.0484

Residual add: $Z + FFN(Z)$

Z (shape 5×4):

[CLS]	-0.8242	0.8782	-1.1557	1.1017
I	0.8749	-0.0086	-1.6288	0.7625
love	0.6726	-1.2631	-0.6476	1.2380
AI	0.3624	-1.5450	-0.0340	1.2167
[SEP]	-0.9530	-0.8694	0.3300	1.4923

$FFN(Z)$ (shape 5×4):

[CLS]	-0.0629	-0.0011	0.0414	0.0132
I	-0.0858	-0.0223	0.0014	-0.0190
love	-0.0855	-0.0794	-0.0835	-0.0329
AI	-0.0749	-0.0794	-0.1127	-0.0334
[SEP]	-0.0377	-0.0757	-0.0773	-0.0484

Resulting residual_2 matrix (shape 5×4):

[CLS]	-0.8871	0.8771	-1.1143	1.1149
I	0.7891	-0.0309	-1.6274	0.7435
love	0.5871	-1.3425	-0.7311	1.2051
AI	0.2875	-1.6245	-0.1467	1.1832
[SEP]	-0.9907	-0.9451	0.2527	1.4439

Final LayerNorm(residual_2) -> FINAL OUTPUT

Input matrix (every token as a row) (shape 5×4):

[CLS]	-0.8871	0.8771	-1.1143	1.1149
I	0.7891	-0.0309	-1.6274	0.7435
love	0.5871	-1.3425	-0.7311	1.2051
AI	0.2875	-1.6245	-0.1467	1.1832
[SEP]	-0.9907	-0.9451	0.2527	1.4439

$\gamma = 1$, $\beta = 0$ (identity affine, to isolate the normalization math)

$[CLS]: v = [-0.8871, 0.8771, -1.1143, 1.1149]$

$$\text{mean} = \frac{-0.8871 + 0.8771 + -1.1143 + 1.1149}{4} = -0.0023$$

$$\text{deviations} = [-0.8848, 0.8794, -1.1120, 1.1173] \Rightarrow \text{variance} = \frac{4.0409}{4} = 1.0102$$

$$\text{std} = \sqrt{1.0102 + 10^{-5}} = 1.0051$$

$$\text{normalized} = [\frac{-0.8848}{1.0051} = -0.8803, \frac{0.8794}{1.0051} = 0.8750, \frac{-1.1120}{1.0051} = -1.1063, \frac{1.1173}{1.0051} = 1.1116]$$

$$I: v = [0.7891, -0.0309, -1.6274, 0.7435]$$

$$\text{mean} = \frac{0.7891 + -0.0309 + -1.6274 + 0.7435}{4} = -0.0314$$

$$\text{deviations} = [0.8205, 0.0006, -1.5960, 0.7750] \Rightarrow \text{variance} = \frac{3.8210}{4} = 0.9552$$

$$\text{std} = \sqrt{0.9552 + 10^{-5}} = 0.9774$$

$$\text{normalized} = [\frac{0.8205}{0.9774} = 0.8395, \frac{0.0006}{0.9774} = 0.0006, \frac{-1.5960}{0.9774} = -1.6330, \frac{0.7750}{0.9774} = 0.7929]$$

$$love: v = [0.5871, -1.3425, -0.7311, 1.2051]$$

$$\text{mean} = \frac{0.5871 + -1.3425 + -0.7311 + 1.2051}{4} = -0.0703$$

$$\text{deviations} = [0.6575, -1.2722, -0.6607, 1.2755] \Rightarrow \text{variance} = \frac{4.1140}{4} = 1.0285$$

$$\text{std} = \sqrt{1.0285 + 10^{-5}} = 1.0142$$

$$\text{normalized} = [\frac{0.6575}{1.0142} = 0.6483, \frac{-1.2722}{1.0142} = -1.2544, \frac{-0.6607}{1.0142} = -0.6515, \frac{1.2755}{1.0142} = 1.2577]$$

$$AI: v = [0.2875, -1.6245, -0.1467, 1.1832]$$

$$\text{mean} = \frac{0.2875 + -1.6245 + -0.1467 + 1.1832}{4} = -0.0751$$

$$\text{deviations} = [0.3626, -1.5494, -0.0716, 1.2584] \Rightarrow \text{variance} = \frac{4.1206}{4} = 1.0302$$

$$\text{std} = \sqrt{1.0302 + 10^{-5}} = 1.0150$$

$$\text{normalized} = [\frac{0.3626}{1.0150} = 0.3572, \frac{-1.5494}{1.0150} = -1.5265, \frac{-0.0716}{1.0150} = -0.0705, \frac{1.2584}{1.0150} = 1.2398]$$

$$[SEP]: v = [-0.9907, -0.9451, 0.2527, 1.4439]$$

$$\text{mean} = \frac{-0.9907 + -0.9451 + 0.2527 + 1.4439}{4} = -0.0598$$

$$\text{deviations} = [-0.9309, -0.8853, 0.3125, 1.5037] \Rightarrow \text{variance} = \frac{4.0092}{4} = 1.0023$$

$$\text{std} = \sqrt{1.0023 + 10^{-5}} = 1.0011$$

$$\text{normalized} = [\frac{-0.9309}{1.0011} = -0.9298, \frac{-0.8853}{1.0011} = -0.8843, \frac{0.3125}{1.0011} = 0.3122, \frac{1.5037}{1.0011} = 1.5020]$$

Resulting normalized matrix (shape 5×4):

[CLS]	−0.8803	0.8750	−1.1063	1.1116
I	0.8395	0.0006	−1.6330	0.7929
love	0.6483	−1.2544	−0.6515	1.2577
AI	0.3572	−1.5265	−0.0705	1.2398
[SEP]	−0.9298	−0.8843	0.3122	1.5020

Appendix B — Fully Expanded Per-Token Arithmetic, in LaTeX Matrix Form: Cross-Encoder on “AI risky” / “AI safe”

X (input embedding = token + segment + position) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	−0.1161	−0.0800	1.0998
[SEP]	0.1911	−0.9400	0.1800	0.9496
AI	−0.5068	−0.9036	0.0900	1.1492
safe	−0.8589	0.4337	0.2500	0.8988
[SEP]	−0.3294	0.9102	0.1100	0.8482

Head 1: Q, K, V projections

W_{q_1} (shape 4×2):

1.0000	0.0000
0.0000	2.0000
1.0000	1.0000
0.0000	0.0000

X (every token as a row) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	−0.1161	−0.0800	1.0998
[SEP]	0.1911	−0.9400	0.1800	0.9496
AI	−0.5068	−0.9036	0.0900	1.1492
safe	−0.8589	0.4337	0.2500	0.8988
[SEP]	−0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $Q = X \cdot W_{q_1}$, every token:

$[CLS]$ (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$Q[0] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

$$Q[1] = (0.1500 \times 0.0000) + (1.0500 \times 2.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 2.1500$$

AI (X row = $[1.1915, 0.3903, 0.1600, 1.2500]$)

$$Q[0] = (1.1915 \times 1.0000) + (0.3903 \times 0.0000) + (0.1600 \times 1.0000) + (1.2500 \times 0.0000) = 1.3515$$

$$Q[1] = (1.1915 \times 0.0000) + (0.3903 \times 2.0000) + (0.1600 \times 1.0000) + (1.2500 \times 0.0000) = 0.9406$$

risky (X row = [0.7593, -0.1161, -0.0800, 1.0998])

$$Q[0] = (0.7593 \times 1.0000) + (-0.1161 \times 0.0000) + (-0.0800 \times 1.0000) + (1.0998 \times 0.0000) = 0.6793$$

$$Q[1] = (0.7593 \times 0.0000) + (-0.1161 \times 2.0000) + (-0.0800 \times 1.0000) + (1.0998 \times 0.0000) = -0.3123$$

[SEP] (X row = [0.1911, -0.9400, 0.1800, 0.9496])

$$Q[0] = (0.1911 \times 1.0000) + (-0.9400 \times 0.0000) + (0.1800 \times 1.0000) + (0.9496 \times 0.0000) = 0.3711$$

$$Q[1] = (0.1911 \times 0.0000) + (-0.9400 \times 2.0000) + (0.1800 \times 1.0000) + (0.9496 \times 0.0000) = -1.7000$$

AI (X row = [-0.5068, -0.9036, 0.0900, 1.1492])

$$Q[0] = (-0.5068 \times 1.0000) + (-0.9036 \times 0.0000) + (0.0900 \times 1.0000) + (1.1492 \times 0.0000) = -0.4168$$

$$Q[1] = (-0.5068 \times 0.0000) + (-0.9036 \times 2.0000) + (0.0900 \times 1.0000) + (1.1492 \times 0.0000) = -1.7173$$

safe (X row = [-0.8589, 0.4337, 0.2500, 0.8988])

$$Q[0] = (-0.8589 \times 1.0000) + (0.4337 \times 0.0000) + (0.2500 \times 1.0000) + (0.8988 \times 0.0000) = -0.6089$$

$$Q[1] = (-0.8589 \times 0.0000) + (0.4337 \times 2.0000) + (0.2500 \times 1.0000) + (0.8988 \times 0.0000) = 1.1173$$

[SEP] (X row = [-0.3294, 0.9102, 0.1100, 0.8482])

$$Q[0] = (-0.3294 \times 1.0000) + (0.9102 \times 0.0000) + (0.1100 \times 1.0000) + (0.8482 \times 0.0000) = -0.2195$$

$$Q[1] = (-0.3294 \times 0.0000) + (0.9102 \times 2.0000) + (0.1100 \times 1.0000) + (0.8482 \times 0.0000) = 1.9303$$

Resulting Q matrix (shape 7×2):

[CLS]	0.2000	2.1500
AI	1.3515	0.9406
risky	0.6793	-0.3123
[SEP]	0.3711	-1.7000
AI	-0.4168	-1.7173
safe	-0.6089	1.1173
[SEP]	-0.2195	1.9303

Wk_1 (shape 4×2):

$$\begin{bmatrix} 2.0000 & 0.0000 \\ 0.0000 & 1.0000 \\ 1.0000 & 0.0000 \\ 0.0000 & 1.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $K = X \cdot Wk_1$, every token:

[CLS] (X row = [0.1500, 1.0500, 0.0500, 1.1500])

$$K[0] = (0.1500 \times 2.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.3500$$

$$K[1] = (0.1500 \times 0.0000) + (1.0500 \times 1.0000) + (0.0500 \times 0.0000) + (1.1500 \times 1.0000) = 2.2000$$

AI (X row = [1.1915, 0.3903, 0.1600, 1.2500])

$$K[0] = (1.1915 \times 2.0000) + (0.3903 \times 0.0000) + (0.1600 \times 1.0000) + (1.2500 \times 0.0000) = 2.5429$$

$$K[1] = (1.1915 \times 0.0000) + (0.3903 \times 1.0000) + (0.1600 \times 0.0000) + (1.2500 \times 1.0000) = 1.6403$$

risky (X row = [0.7593, -0.1161, -0.0800, 1.0998])

$$K[0] = (0.7593 \times 2.0000) + (-0.1161 \times 0.0000) + (-0.0800 \times 1.0000) + (1.0998 \times 0.0000) = 1.4386$$

$$K[1] = (0.7593 \times 0.0000) + (-0.1161 \times 1.0000) + (-0.0800 \times 0.0000) + (1.0998 \times 1.0000) = 0.9837$$

[SEP] (X row = [0.1911, -0.9400, 0.1800, 0.9496])

$$K[0] = (0.1911 \times 2.0000) + (-0.9400 \times 0.0000) + (0.1800 \times 1.0000) + (0.9496 \times 0.0000) = 0.5622$$

$$K[1] = (0.1911 \times 0.0000) + (-0.9400 \times 1.0000) + (0.1800 \times 0.0000) + (0.9496 \times 1.0000) = 0.0096$$

AI (X row = [-0.5068, -0.9036, 0.0900, 1.1492])

$$K[0] = (-0.5068 \times 2.0000) + (-0.9036 \times 0.0000) + (0.0900 \times 1.0000) + (1.1492 \times 0.0000) = -0.9236$$

$$K[1] = (-0.5068 \times 0.0000) + (-0.9036 \times 1.0000) + (0.0900 \times 0.0000) + (1.1492 \times 1.0000) = 0.2456$$

safe (X row = [-0.8589, 0.4337, 0.2500, 0.8988])

$$K[0] = (-0.8589 \times 2.0000) + (0.4337 \times 0.0000) + (0.2500 \times 1.0000) + (0.8988 \times 0.0000) = -1.4679$$

$$K[1] = (-0.8589 \times 0.0000) + (0.4337 \times 1.0000) + (0.2500 \times 0.0000) + (0.8988 \times 1.0000) = 1.3324$$

[SEP] (X row = [-0.3294, 0.9102, 0.1100, 0.8482])

$$K[0] = (-0.3294 \times 2.0000) + (0.9102 \times 0.0000) + (0.1100 \times 1.0000) + (0.8482 \times 0.0000) = -0.5489$$

$$K[1] = (-0.3294 \times 0.0000) + (0.9102 \times 1.0000) + (0.1100 \times 0.0000) + (0.8482 \times 1.0000) = 1.7584$$

Resulting K matrix (shape 7×2):

[CLS]	0.3500	2.2000
AI	2.5429	1.6403
risky	1.4386	0.9837
[SEP]	0.5622	0.0096
AI	-0.9236	0.2456
safe	-1.4679	1.3324
[SEP]	-0.5489	1.7584

Wv_1 (shape 4×2):

0.1000	0.1000
0.1000	0.0000
0.0000	0.2000
0.2000	0.0000

X (**every token as a row**) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $V = X \cdot Wv_1$, every token:

$[CLS]$ (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$V[0] = (0.1500 \times 0.1000) + (1.0500 \times 0.1000) + (0.0500 \times 0.0000) + (1.1500 \times 0.2000) = 0.3500$$

$$V[1] = (0.1500 \times 0.1000) + (1.0500 \times 0.0000) + (0.0500 \times 0.2000) + (1.1500 \times 0.0000) = 0.0250$$

AI (X row = $[1.1915, 0.3903, 0.1600, 1.2500]$)

$$V[0] = (1.1915 \times 0.1000) + (0.3903 \times 0.1000) + (0.1600 \times 0.0000) + (1.2500 \times 0.2000) = 0.4082$$

$$V[1] = (1.1915 \times 0.1000) + (0.3903 \times 0.0000) + (0.1600 \times 0.2000) + (1.2500 \times 0.0000) = 0.1511$$

$risky$ (X row = $[0.7593, -0.1161, -0.0800, 1.0998]$)

$$V[0] = (0.7593 \times 0.1000) + (-0.1161 \times 0.1000) + (-0.0800 \times 0.0000) + (1.0998 \times 0.2000) = 0.2843$$

$$V[1] = (0.7593 \times 0.1000) + (-0.1161 \times 0.0000) + (-0.0800 \times 0.2000) + (1.0998 \times 0.0000) = 0.0599$$

$[SEP]$ (X row = $[0.1911, -0.9400, 0.1800, 0.9496]$)

$$V[0] = (0.1911 \times 0.1000) + (-0.9400 \times 0.1000) + (0.1800 \times 0.0000) + (0.9496 \times 0.2000) = 0.1150$$

$$V[1] = (0.1911 \times 0.1000) + (-0.9400 \times 0.0000) + (0.1800 \times 0.2000) + (0.9496 \times 0.0000) = 0.0551$$

AI (X row = $[-0.5068, -0.9036, 0.0900, 1.1492]$)

$$V[0] = (-0.5068 \times 0.1000) + (-0.9036 \times 0.1000) + (0.0900 \times 0.0000) + (1.1492 \times 0.2000) = 0.0888$$

$$V[1] = (-0.5068 \times 0.1000) + (-0.9036 \times 0.0000) + (0.0900 \times 0.2000) + (1.1492 \times 0.0000) = -0.0327$$

safe (X row = $[-0.8589, 0.4337, 0.2500, 0.8988]$)

$$V[0] = (-0.8589 \times 0.1000) + (0.4337 \times 0.1000) + (0.2500 \times 0.0000) + (0.8988 \times 0.2000) = 0.1372$$

$$V[1] = (-0.8589 \times 0.1000) + (0.4337 \times 0.0000) + (0.2500 \times 0.2000) + (0.8988 \times 0.0000) = -0.0359$$

[SEP] (X row = $[-0.3294, 0.9102, 0.1100, 0.8482]$)

$$V[0] = (-0.3294 \times 0.1000) + (0.9102 \times 0.1000) + (0.1100 \times 0.0000) + (0.8482 \times 0.2000) = 0.2277$$

$$V[1] = (-0.3294 \times 0.1000) + (0.9102 \times 0.0000) + (0.1100 \times 0.2000) + (0.8482 \times 0.0000) = -0.0109$$

Resulting V matrix (shape 7×2):

[CLS]	0.3500	0.0250
AI	0.4082	0.1511
risky	0.2843	0.0599
[SEP]	0.1150	0.0551
AI	0.0888	-0.0327
safe	0.1372	-0.0359
[SEP]	0.2277	-0.0109

Head 2: Q, K, V projections

W_{q_2} (shape 4×2):

0.0000	1.0000
2.0000	0.0000
0.0000	1.0000
1.0000	0.0000

X (every token as a row) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $Q = X \cdot W_{q_2}$, every token:

[CLS] (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$Q[0] = (0.1500 \times 0.0000) + (1.0500 \times 2.0000) + (0.0500 \times 0.0000) + (1.1500 \times 1.0000) = 3.2500$$

$$Q[1] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

AI (X row = $[1.1915, 0.3903, 0.1600, 1.2500]$)

$$Q[0] = (1.1915 \times 0.0000) + (0.3903 \times 2.0000) + (0.1600 \times 0.0000) + (1.2500 \times 1.0000) = 2.0306$$

$$Q[1] = (1.1915 \times 1.0000) + (0.3903 \times 0.0000) + (0.1600 \times 1.0000) + (1.2500 \times 0.0000) = 1.3515$$

risky (X row = [0.7593, -0.1161, -0.0800, 1.0998])

$$Q[0] = (0.7593 \times 0.0000) + (-0.1161 \times 2.0000) + (-0.0800 \times 0.0000) + (1.0998 \times 1.0000) = 0.8675$$

$$Q[1] = (0.7593 \times 1.0000) + (-0.1161 \times 0.0000) + (-0.0800 \times 1.0000) + (1.0998 \times 0.0000) = 0.6793$$

[SEP] (X row = [0.1911, -0.9400, 0.1800, 0.9496])

$$Q[0] = (0.1911 \times 0.0000) + (-0.9400 \times 2.0000) + (0.1800 \times 0.0000) + (0.9496 \times 1.0000) = -0.9304$$

$$Q[1] = (0.1911 \times 1.0000) + (-0.9400 \times 0.0000) + (0.1800 \times 1.0000) + (0.9496 \times 0.0000) = 0.3711$$

AI (X row = [-0.5068, -0.9036, 0.0900, 1.1492])

$$Q[0] = (-0.5068 \times 0.0000) + (-0.9036 \times 2.0000) + (0.0900 \times 0.0000) + (1.1492 \times 1.0000) = -0.6581$$

$$Q[1] = (-0.5068 \times 1.0000) + (-0.9036 \times 0.0000) + (0.0900 \times 1.0000) + (1.1492 \times 0.0000) = -0.4168$$

safe (X row = [-0.8589, 0.4337, 0.2500, 0.8988])

$$Q[0] = (-0.8589 \times 0.0000) + (0.4337 \times 2.0000) + (0.2500 \times 0.0000) + (0.8988 \times 1.0000) = 1.7661$$

$$Q[1] = (-0.8589 \times 1.0000) + (0.4337 \times 0.0000) + (0.2500 \times 1.0000) + (0.8988 \times 0.0000) = -0.6089$$

[SEP] (X row = [-0.3294, 0.9102, 0.1100, 0.8482])

$$Q[0] = (-0.3294 \times 0.0000) + (0.9102 \times 2.0000) + (0.1100 \times 0.0000) + (0.8482 \times 1.0000) = 2.6685$$

$$Q[1] = (-0.3294 \times 1.0000) + (0.9102 \times 0.0000) + (0.1100 \times 1.0000) + (0.8482 \times 0.0000) = -0.2195$$

Resulting Q matrix (shape 7×2):

[CLS]	3.2500	0.2000
AI	2.0306	1.3515
risky	0.8675	0.6793
[SEP]	-0.9304	0.3711
AI	-0.6581	-0.4168
safe	1.7661	-0.6089
[SEP]	2.6685	-0.2195

Wk_2 (shape 4×2):

$$\begin{bmatrix} 1.0000 & 1.0000 \\ 0.0000 & 2.0000 \\ 1.0000 & 0.0000 \\ 0.0000 & 0.0000 \end{bmatrix}$$

X (**every token as a row**) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $K = X \cdot Wk_2$, every token:

[CLS] (X row = [0.1500, 1.0500, 0.0500, 1.1500])

$$K[0] = (0.1500 \times 1.0000) + (1.0500 \times 0.0000) + (0.0500 \times 1.0000) + (1.1500 \times 0.0000) = 0.2000$$

$$K[1] = (0.1500 \times 1.0000) + (1.0500 \times 2.0000) + (0.0500 \times 0.0000) + (1.1500 \times 0.0000) = 2.2500$$

AI (X row = [1.1915, 0.3903, 0.1600, 1.2500])

$$K[0] = (1.1915 \times 1.0000) + (0.3903 \times 0.0000) + (0.1600 \times 1.0000) + (1.2500 \times 0.0000) = 1.3515$$

$$K[1] = (1.1915 \times 1.0000) + (0.3903 \times 2.0000) + (0.1600 \times 0.0000) + (1.2500 \times 0.0000) = 1.9721$$

risky (X row = [0.7593, -0.1161, -0.0800, 1.0998])

$$K[0] = (0.7593 \times 1.0000) + (-0.1161 \times 0.0000) + (-0.0800 \times 1.0000) + (1.0998 \times 0.0000) = 0.6793$$

$$K[1] = (0.7593 \times 1.0000) + (-0.1161 \times 2.0000) + (-0.0800 \times 0.0000) + (1.0998 \times 0.0000) = 0.5270$$

[SEP] (X row = [0.1911, -0.9400, 0.1800, 0.9496])

$$K[0] = (0.1911 \times 1.0000) + (-0.9400 \times 0.0000) + (0.1800 \times 1.0000) + (0.9496 \times 0.0000) = 0.3711$$

$$K[1] = (0.1911 \times 1.0000) + (-0.9400 \times 2.0000) + (0.1800 \times 0.0000) + (0.9496 \times 0.0000) = -1.6889$$

AI (X row = [-0.5068, -0.9036, 0.0900, 1.1492])

$$K[0] = (-0.5068 \times 1.0000) + (-0.9036 \times 0.0000) + (0.0900 \times 1.0000) + (1.1492 \times 0.0000) = -0.4168$$

$$K[1] = (-0.5068 \times 1.0000) + (-0.9036 \times 2.0000) + (0.0900 \times 0.0000) + (1.1492 \times 0.0000) = -2.3141$$

safe (X row = [-0.8589, 0.4337, 0.2500, 0.8988])

$$K[0] = (-0.8589 \times 1.0000) + (0.4337 \times 0.0000) + (0.2500 \times 1.0000) + (0.8988 \times 0.0000) = -0.6089$$

$$K[1] = (-0.8589 \times 1.0000) + (0.4337 \times 2.0000) + (0.2500 \times 0.0000) + (0.8988 \times 0.0000) = 0.0084$$

[SEP] (X row = [-0.3294, 0.9102, 0.1100, 0.8482])

$$K[0] = (-0.3294 \times 1.0000) + (0.9102 \times 0.0000) + (0.1100 \times 1.0000) + (0.8482 \times 0.0000) = -0.2195$$

$$K[1] = (-0.3294 \times 1.0000) + (0.9102 \times 2.0000) + (0.1100 \times 0.0000) + (0.8482 \times 0.0000) = 1.4909$$

Resulting K matrix (shape 7×2):

[CLS]	0.2000	2.2500
AI	1.3515	1.9721
risky	0.6793	0.5270
[SEP]	0.3711	-1.6889
AI	-0.4168	-2.3141
safe	-0.6089	0.0084
[SEP]	-0.2195	1.4909

Wv_2 (shape 4×2):

0.0000	0.2000
0.1000	0.1000
0.0000	0.1000
0.1000	0.0000

X (**every token as a row**) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

Expanded arithmetic, $V = X \cdot Wv_2$, every token:

$[CLS]$ (X row = $[0.1500, 1.0500, 0.0500, 1.1500]$)

$$V[0] = (0.1500 \times 0.0000) + (1.0500 \times 0.1000) + (0.0500 \times 0.0000) + (1.1500 \times 0.1000) = 0.2200$$

$$V[1] = (0.1500 \times 0.2000) + (1.0500 \times 0.1000) + (0.0500 \times 0.1000) + (1.1500 \times 0.0000) = 0.1400$$

AI (X row = $[1.1915, 0.3903, 0.1600, 1.2500]$)

$$V[0] = (1.1915 \times 0.0000) + (0.3903 \times 0.1000) + (0.1600 \times 0.0000) + (1.2500 \times 0.1000) = 0.1640$$

$$V[1] = (1.1915 \times 0.2000) + (0.3903 \times 0.1000) + (0.1600 \times 0.1000) + (1.2500 \times 0.0000) = 0.2933$$

$risky$ (X row = $[0.7593, -0.1161, -0.0800, 1.0998]$)

$$V[0] = (0.7593 \times 0.0000) + (-0.1161 \times 0.1000) + (-0.0800 \times 0.0000) + (1.0998 \times 0.1000) = 0.0984$$

$$V[1] = (0.7593 \times 0.2000) + (-0.1161 \times 0.1000) + (-0.0800 \times 0.1000) + (1.0998 \times 0.0000) = 0.1322$$

$[SEP]$ (X row = $[0.1911, -0.9400, 0.1800, 0.9496]$)

$$V[0] = (0.1911 \times 0.0000) + (-0.9400 \times 0.1000) + (0.1800 \times 0.0000) + (0.9496 \times 0.1000) = 0.0010$$

$$V[1] = (0.1911 \times 0.2000) + (-0.9400 \times 0.1000) + (0.1800 \times 0.1000) + (0.9496 \times 0.0000) = -0.0378$$

AI (X row = $[-0.5068, -0.9036, 0.0900, 1.1492]$)

$$V[0] = (-0.5068 \times 0.0000) + (-0.9036 \times 0.1000) + (0.0900 \times 0.0000) + (1.1492 \times 0.1000) = 0.0246$$

$$V[1] = (-0.5068 \times 0.2000) + (-0.9036 \times 0.1000) + (0.0900 \times 0.1000) + (1.1492 \times 0.0000) = -0.1827$$

$$\text{safe} (X \text{ row} = [-0.8589, 0.4337, 0.2500, 0.8988])$$

$$V[0] = (-0.8589 \times 0.0000) + (0.4337 \times 0.1000) + (0.2500 \times 0.0000) + (0.8988 \times 0.1000) = 0.1332$$

$$V[1] = (-0.8589 \times 0.2000) + (0.4337 \times 0.1000) + (0.2500 \times 0.1000) + (0.8988 \times 0.0000) = -0.1034$$

$$[\text{SEP}] (X \text{ row} = [-0.3294, 0.9102, 0.1100, 0.8482])$$

$$V[0] = (-0.3294 \times 0.0000) + (0.9102 \times 0.1000) + (0.1100 \times 0.0000) + (0.8482 \times 0.1000) = 0.1758$$

$$V[1] = (-0.3294 \times 0.2000) + (0.9102 \times 0.1000) + (0.1100 \times 0.1000) + (0.8482 \times 0.0000) = 0.0361$$

Resulting V matrix (shape 7×2):

[CLS]	0.2200	0.1400
AI	0.1640	0.2933
risky	0.0984	0.1322
[SEP]	0.0010	-0.0378
AI	0.0246	-0.1827
safe	0.1332	-0.1034
[SEP]	0.1758	0.0361

Head 1: full bidirectional attention

Head 1: Q matrix (shape 7×2):

[CLS]	0.2000	2.1500
AI	1.3515	0.9406
risky	0.6793	-0.3123
[SEP]	0.3711	-1.7000
AI	-0.4168	-1.7173
safe	-0.6089	1.1173
[SEP]	-0.2195	1.9303

Head 1: K matrix (shape 7×2):

[CLS]	0.3500	2.2000
AI	2.5429	1.6403
risky	1.4386	0.9837
[SEP]	0.5622	0.0096
AI	-0.9236	0.2456
safe	-1.4679	1.3324
[SEP]	-0.5489	1.7584

Head 1: V matrix (shape 7×2):

[CLS]	0.3500	0.0250
AI	0.4082	0.1511
risky	0.2843	0.0599
[SEP]	0.1150	0.0551
AI	0.0888	-0.0327
safe	0.1372	-0.0359
[SEP]	0.2277	-0.0109

Attention scores (bidirectional) $\text{score}(i, j) = (Q_i \cdot K_j) / \sqrt{2} = /1.4142$:

Query: [CLS]

$$\begin{aligned}\text{score}([\text{CLS}], [\text{CLS}]) &= \frac{(0.2000 \times 0.3500) + (2.1500 \times 2.2000)}{1.4142} = 3.3941 \\ \text{score}([\text{CLS}], \text{AI}) &= \frac{(0.2000 \times 2.5429) + (2.1500 \times 1.6403)}{1.4142} = 2.8533 \\ \text{score}([\text{CLS}], \text{risky}) &= \frac{(0.2000 \times 1.4386) + (2.1500 \times 0.9837)}{1.4142} = 1.6989 \\ \text{score}([\text{CLS}], [\text{SEP}]) &= \frac{(0.2000 \times 0.5622) + (2.1500 \times 0.0096)}{1.4142} = 0.0940 \\ \text{score}([\text{CLS}], \text{AI}) &= \frac{(0.2000 \times -0.9236) + (2.1500 \times 0.2456)}{1.4142} = 0.2427 \\ \text{score}([\text{CLS}], \text{safe}) &= \frac{(0.2000 \times -1.4679) + (2.1500 \times 1.3324)}{1.4142} = 1.8181 \\ \text{score}([\text{CLS}], [\text{SEP}]) &= \frac{(0.2000 \times -0.5489) + (2.1500 \times 1.7584)}{1.4142} = 2.5956\end{aligned}$$

Query: AI

$$\begin{aligned}\text{score}(\text{AI}, [\text{CLS}]) &= \frac{(1.3515 \times 0.3500) + (0.9406 \times 2.2000)}{1.4142} = 1.7977 \\ \text{score}(\text{AI}, \text{AI}) &= \frac{(1.3515 \times 2.5429) + (0.9406 \times 1.6403)}{1.4142} = 3.5211 \\ \text{score}(\text{AI}, \text{risky}) &= \frac{(1.3515 \times 1.4386) + (0.9406 \times 0.9837)}{1.4142} = 2.0290 \\ \text{score}(\text{AI}, [\text{SEP}]) &= \frac{(1.3515 \times 0.5622) + (0.9406 \times 0.0096)}{1.4142} = 0.5436 \\ \text{score}(\text{AI}, \text{AI}) &= \frac{(1.3515 \times -0.9236) + (0.9406 \times 0.2456)}{1.4142} = -0.7193 \\ \text{score}(\text{AI}, \text{safe}) &= \frac{(1.3515 \times -1.4679) + (0.9406 \times 1.3324)}{1.4142} = -0.5165 \\ \text{score}(\text{AI}, [\text{SEP}]) &= \frac{(1.3515 \times -0.5489) + (0.9406 \times 1.7584)}{1.4142} = 0.6450\end{aligned}$$

Query: risky

$$\begin{aligned}\text{score}(\text{risky}, [\text{CLS}]) &= \frac{(0.6793 \times 0.3500) + (-0.3123 \times 2.2000)}{1.4142} = -0.3177 \\ \text{score}(\text{risky}, \text{AI}) &= \frac{(0.6793 \times 2.5429) + (-0.3123 \times 1.6403)}{1.4142} = 0.8593 \\ \text{score}(\text{risky}, \text{risky}) &= \frac{(0.6793 \times 1.4386) + (-0.3123 \times 0.9837)}{1.4142} = 0.4738 \\ \text{score}(\text{risky}, [\text{SEP}]) &= \frac{(0.6793 \times 0.5622) + (-0.3123 \times 0.0096)}{1.4142} = 0.2680 \\ \text{score}(\text{risky}, \text{AI}) &= \frac{(0.6793 \times -0.9236) + (-0.3123 \times 0.2456)}{1.4142} = -0.4979\end{aligned}$$

$$\text{score}(\text{risky}, \text{safe}) = \frac{(0.6793 \times -1.4679) + (-0.3123 \times 1.3324)}{1.4142} = -0.9993$$

$$\text{score}(\text{risky}, [\text{SEP}]) = \frac{(0.6793 \times -0.5489) + (-0.3123 \times 1.7584)}{1.4142} = -0.6519$$

Query: [SEP]

$$\text{score}([\text{SEP}], [\text{CLS}]) = \frac{(0.3711 \times 0.3500) + (-1.7000 \times 2.2000)}{1.4142} = -2.5527$$

$$\text{score}([\text{SEP}], \text{AI}) = \frac{(0.3711 \times 2.5429) + (-1.7000 \times 1.6403)}{1.4142} = -1.3044$$

$$\text{score}([\text{SEP}], \text{risky}) = \frac{(0.3711 \times 1.4386) + (-1.7000 \times 0.9837)}{1.4142} = -0.8049$$

$$\text{score}([\text{SEP}], [\text{SEP}]) = \frac{(0.3711 \times 0.5622) + (-1.7000 \times 0.0096)}{1.4142} = 0.1361$$

$$\text{score}([\text{SEP}], \text{AI}) = \frac{(0.3711 \times -0.9236) + (-1.7000 \times 0.2456)}{1.4142} = -0.5376$$

$$\text{score}([\text{SEP}], \text{safe}) = \frac{(0.3711 \times -1.4679) + (-1.7000 \times 1.3324)}{1.4142} = -1.9869$$

$$\text{score}([\text{SEP}], [\text{SEP}]) = \frac{(0.3711 \times -0.5489) + (-1.7000 \times 1.7584)}{1.4142} = -2.2577$$

Query: AI

$$\text{score}(\text{AI}, [\text{CLS}]) = \frac{(-0.4168 \times 0.3500) + (-1.7173 \times 2.2000)}{1.4142} = -2.7746$$

$$\text{score}(\text{AI}, \text{AI}) = \frac{(-0.4168 \times 2.5429) + (-1.7173 \times 1.6403)}{1.4142} = -2.7413$$

$$\text{score}(\text{AI}, \text{risky}) = \frac{(-0.4168 \times 1.4386) + (-1.7173 \times 0.9837)}{1.4142} = -1.6185$$

$$\text{score}(\text{AI}, [\text{SEP}]) = \frac{(-0.4168 \times 0.5622) + (-1.7173 \times 0.0096)}{1.4142} = -0.1773$$

$$\text{score}(\text{AI}, \text{AI}) = \frac{(-0.4168 \times -0.9236) + (-1.7173 \times 0.2456)}{1.4142} = -0.0260$$

$$\text{score}(\text{AI}, \text{safe}) = \frac{(-0.4168 \times -1.4679) + (-1.7173 \times 1.3324)}{1.4142} = -1.1853$$

$$\text{score}(\text{AI}, [\text{SEP}]) = \frac{(-0.4168 \times -0.5489) + (-1.7173 \times 1.7584)}{1.4142} = -1.9734$$

Query: safe

$$\text{score}(\text{safe}, [\text{CLS}]) = \frac{(-0.6089 \times 0.3500) + (1.1173 \times 2.2000)}{1.4142} = 1.5874$$

$$\text{score}(\text{safe}, \text{AI}) = \frac{(-0.6089 \times 2.5429) + (1.1173 \times 1.6403)}{1.4142} = 0.2009$$

$$\text{score}(\text{safe}, \text{risky}) = \frac{(-0.6089 \times 1.4386) + (1.1173 \times 0.9837)}{1.4142} = 0.1577$$

$$\text{score}(\text{safe}, [\text{SEP}]) = \frac{(-0.6089 \times 0.5622) + (1.1173 \times 0.0096)}{1.4142} = -0.2345$$

$$\begin{aligned}\text{score}(\text{safe}, \text{AI}) &= \frac{(-0.6089 \times -0.9236) + (1.1173 \times 0.2456)}{1.4142} = 0.5917 \\ \text{score}(\text{safe}, \text{safe}) &= \frac{(-0.6089 \times -1.4679) + (1.1173 \times 1.3324)}{1.4142} = 1.6847 \\ \text{score}(\text{safe}, [\text{SEP}]) &= \frac{(-0.6089 \times -0.5489) + (1.1173 \times 1.7584)}{1.4142} = 1.6255\end{aligned}$$

Query: [SEP]

$$\begin{aligned}\text{score}([\text{SEP}], [\text{CLS}]) &= \frac{(-0.2195 \times 0.3500) + (1.9303 \times 2.2000)}{1.4142} = 2.9485 \\ \text{score}([\text{SEP}], \text{AI}) &= \frac{(-0.2195 \times 2.5429) + (1.9303 \times 1.6403)}{1.4142} = 1.8442 \\ \text{score}([\text{SEP}], \text{risky}) &= \frac{(-0.2195 \times 1.4386) + (1.9303 \times 0.9837)}{1.4142} = 1.1194 \\ \text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(-0.2195 \times 0.5622) + (1.9303 \times 0.0096)}{1.4142} = -0.0742 \\ \text{score}([\text{SEP}], \text{AI}) &= \frac{(-0.2195 \times -0.9236) + (1.9303 \times 0.2456)}{1.4142} = 0.4785 \\ \text{score}([\text{SEP}], \text{safe}) &= \frac{(-0.2195 \times -1.4679) + (1.9303 \times 1.3324)}{1.4142} = 2.0464 \\ \text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(-0.2195 \times -0.5489) + (1.9303 \times 1.7584)}{1.4142} = 2.4852\end{aligned}$$

Head 1: raw score matrix (shape 7×7):

	[CLS]	AI	risky	[SEP]	AI	safe	[SEP]
[CLS]	3.3941	2.8533	1.6989	0.0940	0.2427	1.8181	2.5956
AI	1.7977	3.5211	2.0290	0.5436	-0.7193	-0.5165	0.6450
risky	-0.3177	0.8593	0.4738	0.2680	-0.4979	-0.9993	-0.6519
[SEP]	-2.5527	-1.3044	-0.8049	0.1361	-0.5376	-1.9869	-2.2577
AI	-2.7746	-2.7413	-1.6185	-0.1773	-0.0260	-1.1853	-1.9734
safe	1.5874	0.2009	0.1577	-0.2345	0.5917	1.6847	1.6255
[SEP]	2.9485	1.8442	1.1194	-0.0742	0.4785	2.0464	2.4852

Softmax over every row:

[CLS]:

$$\begin{aligned}\text{exponentials} &= [e^{3.3941} = 1.0000, e^{2.8533} = 0.5823, e^{1.6989} = 0.1836, e^{0.0940} = 0.0369, \\ &\quad e^{0.2427} = 0.0428, e^{1.8181} = 0.2068, e^{2.5956} = 0.4500] \\ \text{sum} &= 2.5023 \\ \text{weights} &= [\frac{1.0000}{2.5023} = 0.3996, \frac{0.5823}{2.5023} = 0.2327, \frac{0.1836}{2.5023} = 0.0734, \frac{0.0369}{2.5023} = 0.0147, \\ &\quad \frac{0.0428}{2.5023} = 0.0171, \frac{0.2068}{2.5023} = 0.0826, \frac{0.4500}{2.5023} = 0.1798]\end{aligned}$$

AI:

$$\begin{aligned}\text{exponentials} &= [e^{1.7977} = 0.1785, e^{3.5211} = 1.0000, e^{2.0290} = 0.2249, e^{0.5436} = 0.0509, \\ &\quad e^{-0.7193} = 0.0144, e^{-0.5165} = 0.0176, e^{0.6450} = 0.0564] \\ \text{sum} &= 1.5427\end{aligned}$$

$$\text{weights} = [\frac{0.1785}{1.5427} = 0.1157, \frac{1.0000}{1.5427} = 0.6482, \frac{0.2249}{1.5427} = 0.1458, \frac{0.0509}{1.5427} = 0.0330, \\ \frac{0.0144}{1.5427} = 0.0093, \frac{0.0176}{1.5427} = 0.0114, \frac{0.0564}{1.5427} = 0.0365]$$

risky:

$$\text{exponentials} = [e^{-0.3177} = 0.3082, e^{0.8593} = 1.0000, e^{0.4738} = 0.6801, e^{0.2680} = 0.5536, \\ e^{-0.4979} = 0.2574, e^{-0.9993} = 0.1559, e^{-0.6519} = 0.2206]$$

$$\text{sum} = 3.1759$$

$$\text{weights} = [\frac{0.3082}{3.1759} = 0.0970, \frac{1.0000}{3.1759} = 0.3149, \frac{0.6801}{3.1759} = 0.2142, \frac{0.5536}{3.1759} = 0.1743, \\ \frac{0.2574}{3.1759} = 0.0810, \frac{0.1559}{3.1759} = 0.0491, \frac{0.2206}{3.1759} = 0.0695]$$

[SEP]:

$$\text{exponentials} = [e^{-2.5527} = 0.0680, e^{-1.3044} = 0.2368, e^{-0.8049} = 0.3903, e^{0.1361} = 1.0000, \\ e^{-0.5376} = 0.5099, e^{-1.9869} = 0.1197, e^{-2.2577} = 0.0913]$$

$$\text{sum} = 2.4159$$

$$\text{weights} = [\frac{0.0680}{2.4159} = 0.0281, \frac{0.2368}{2.4159} = 0.0980, \frac{0.3903}{2.4159} = 0.1615, \frac{1.0000}{2.4159} = 0.4139, \\ \frac{0.5099}{2.4159} = 0.2110, \frac{0.1197}{2.4159} = 0.0495, \frac{0.0913}{2.4159} = 0.0378]$$

AI:

$$\text{exponentials} = [e^{-2.7746} = 0.0640, e^{-2.7413} = 0.0662, e^{-1.6185} = 0.2034, e^{-0.1773} = 0.8595, \\ e^{-0.0260} = 1.0000, e^{-1.1853} = 0.3137, e^{-1.9734} = 0.1426]$$

$$\text{sum} = 2.6495$$

$$\text{weights} = [\frac{0.0640}{2.6495} = 0.0242, \frac{0.0662}{2.6495} = 0.0250, \frac{0.2034}{2.6495} = 0.0768, \frac{0.8595}{2.6495} = 0.3244, \\ \frac{1.0000}{2.6495} = 0.3774, \frac{0.3137}{2.6495} = 0.1184, \frac{0.1426}{2.6495} = 0.0538]$$

safe:

$$\text{exponentials} = [e^{1.5874} = 0.9073, e^{0.2009} = 0.2268, e^{0.1577} = 0.2172, e^{-0.2345} = 0.1467, \\ e^{0.5917} = 0.3352, e^{1.6847} = 1.0000, e^{1.6255} = 0.9425]$$

$$\text{sum} = 3.7757$$

$$\text{weights} = [\frac{0.9073}{3.7757} = 0.2403, \frac{0.2268}{3.7757} = 0.0601, \frac{0.2172}{3.7757} = 0.0575, \frac{0.1467}{3.7757} = 0.0389, \\ \frac{0.3352}{3.7757} = 0.0888, \frac{1.0000}{3.7757} = 0.2649, \frac{0.9425}{3.7757} = 0.2496]$$

[SEP]:

$$\text{exponentials} = [e^{2.9485} = 1.0000, e^{1.8442} = 0.3314, e^{1.1194} = 0.1605, e^{-0.0742} = 0.0487, \\ e^{0.4785} = 0.0846, e^{2.0464} = 0.4057, e^{2.4852} = 0.6292]$$

$$\text{sum} = 2.6601$$

$$\text{weights} = [\frac{1.0000}{2.6601} = 0.3759, \frac{0.3314}{2.6601} = 0.1246, \frac{0.1605}{2.6601} = 0.0604, \frac{0.0487}{2.6601} = 0.0183, \\ \frac{0.0846}{2.6601} = 0.0318, \frac{0.4057}{2.6601} = 0.1525, \frac{0.6292}{2.6601} = 0.2365]$$

Head 1: full softmax weight matrix (shape 7×7):

	<i>[CLS]</i>	<i>AI</i>	<i>risky</i>	<i>[SEP]</i>	<i>AI</i>	<i>safe</i>	<i>[SEP]</i>
<i>[CLS]</i>	0.3996	0.2327	0.0734	0.0147	0.0171	0.0826	0.1798
AI	0.1157	0.6482	0.1458	0.0330	0.0093	0.0114	0.0365
risky	0.0970	0.3149	0.2142	0.1743	0.0810	0.0491	0.0695
<i>[SEP]</i>	0.0281	0.0980	0.1615	0.4139	0.2110	0.0495	0.0378
AI	0.0242	0.0250	0.0768	0.3244	0.3774	0.1184	0.0538
safe	0.2403	0.0601	0.0575	0.0389	0.0888	0.2649	0.2496
<i>[SEP]</i>	0.3759	0.1246	0.0604	0.0183	0.0318	0.1525	0.2365

Attention output = weighted sum of V:

$$\begin{aligned} [\text{CLS}]_{[0]} &= (0.3996 \times 0.3500) + (0.2327 \times 0.4082) + (0.0734 \times 0.2843) + (0.0147 \times 0.1150) \\ &\quad + (0.0171 \times 0.0888) + (0.0826 \times 0.1372) + (0.1798 \times 0.2277) = 0.3112 \end{aligned}$$

$$\begin{aligned} [\text{CLS}]_{[1]} &= (0.3996 \times 0.0250) + (0.2327 \times 0.1511) + (0.0734 \times 0.0599) + (0.0147 \times 0.0551) \\ &\quad + (0.0171 \times -0.0327) + (0.0826 \times -0.0359) + (0.1798 \times -0.0109) = 0.0449 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[0]} &= (0.1157 \times 0.3500) + (0.6482 \times 0.4082) + (0.1458 \times 0.2843) + (0.0330 \times 0.1150) \\ &\quad + (0.0093 \times 0.0888) + (0.0114 \times 0.1372) + (0.0365 \times 0.2277) = 0.3610 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[1]} &= (0.1157 \times 0.0250) + (0.6482 \times 0.1511) + (0.1458 \times 0.0599) + (0.0330 \times 0.0551) \\ &\quad + (0.0093 \times -0.0327) + (0.0114 \times -0.0359) + (0.0365 \times -0.0109) = 0.1103 \end{aligned}$$

$$\begin{aligned} \text{risky}_{[0]} &= (0.0970 \times 0.3500) + (0.3149 \times 0.4082) + (0.2142 \times 0.2843) + (0.1743 \times 0.1150) \\ &\quad + (0.0810 \times 0.0888) + (0.0491 \times 0.1372) + (0.0695 \times 0.2277) = 0.2732 \end{aligned}$$

$$\begin{aligned} \text{risky}_{[1]} &= (0.0970 \times 0.0250) + (0.3149 \times 0.1511) + (0.2142 \times 0.0599) + (0.1743 \times 0.0551) \\ &\quad + (0.0810 \times -0.0327) + (0.0491 \times -0.0359) + (0.0695 \times -0.0109) = 0.0673 \end{aligned}$$

$$\begin{aligned} [\text{SEP}]_{[0]} &= (0.0281 \times 0.3500) + (0.0980 \times 0.4082) + (0.1615 \times 0.2843) + (0.4139 \times 0.1150) \\ &\quad + (0.2110 \times 0.0888) + (0.0495 \times 0.1372) + (0.0378 \times 0.2277) = 0.1775 \end{aligned}$$

$$\begin{aligned} [\text{SEP}]_{[1]} &= (0.0281 \times 0.0250) + (0.0980 \times 0.1511) + (0.1615 \times 0.0599) + (0.4139 \times 0.0551) \\ &\quad + (0.2110 \times -0.0327) + (0.0495 \times -0.0359) + (0.0378 \times -0.0109) = 0.0389 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[0]} &= (0.0242 \times 0.3500) + (0.0250 \times 0.4082) + (0.0768 \times 0.2843) + (0.3244 \times 0.1150) \\ &\quad + (0.3774 \times 0.0888) + (0.1184 \times 0.1372) + (0.0538 \times 0.2277) = 0.1398 \end{aligned}$$

$$\begin{aligned} \text{AI}_{[1]} &= (0.0242 \times 0.0250) + (0.0250 \times 0.1511) + (0.0768 \times 0.0599) + (0.3244 \times 0.0551) \\ &\quad + (0.3774 \times -0.0327) + (0.1184 \times -0.0359) + (0.0538 \times -0.0109) = 0.0097 \end{aligned}$$

$$\begin{aligned} \text{safe}_{[0]} &= (0.2403 \times 0.3500) + (0.0601 \times 0.4082) + (0.0575 \times 0.2843) + (0.0389 \times 0.1150) \\ &\quad + (0.0888 \times 0.0888) + (0.2649 \times 0.1372) + (0.2496 \times 0.2277) = 0.2305 \end{aligned}$$

$$\begin{aligned} \text{safe}_{[1]} &= (0.2403 \times 0.0250) + (0.0601 \times 0.1511) + (0.0575 \times 0.0599) + (0.0389 \times 0.0551) \\ &\quad + (0.0888 \times -0.0327) + (0.2649 \times -0.0359) + (0.2496 \times -0.0109) = 0.0055 \end{aligned}$$

$$\begin{aligned} [\text{SEP}]_{[0]} &= (0.3759 \times 0.3500) + (0.1246 \times 0.4082) + (0.0604 \times 0.2843) + (0.0183 \times 0.1150) \\ &\quad + (0.0318 \times 0.0888) + (0.1525 \times 0.1372) + (0.2365 \times 0.2277) = 0.2793 \end{aligned}$$

$$[\text{SEP}]_{[1]} = (0.3759 \times 0.0250) + (0.1246 \times 0.1511) + (0.0604 \times 0.0599) + (0.0183 \times 0.0551) \\ + (0.0318 \times -0.0327) + (0.1525 \times -0.0359) + (0.2365 \times -0.0109) = 0.0238$$

Head 1: resulting attention-output matrix (shape 7×2):

[CLS]	0.3112	0.0449
AI	0.3610	0.1103
risky	0.2732	0.0673
[SEP]	0.1775	0.0389
AI	0.1398	0.0097
safe	0.2305	0.0055
[SEP]	0.2793	0.0238

Head 2: full bidirectional attention

Head 2: Q matrix (shape 7×2):

[CLS]	3.2500	0.2000
AI	2.0306	1.3515
risky	0.8675	0.6793
[SEP]	-0.9304	0.3711
AI	-0.6581	-0.4168
safe	1.7661	-0.6089
[SEP]	2.6685	-0.2195

Head 2: K matrix (shape 7×2):

[CLS]	0.2000	2.2500
AI	1.3515	1.9721
risky	0.6793	0.5270
[SEP]	0.3711	-1.6889
AI	-0.4168	-2.3141
safe	-0.6089	0.0084
[SEP]	-0.2195	1.4909

Head 2: V matrix (shape 7×2):

[CLS]	0.2200	0.1400
AI	0.1640	0.2933
risky	0.0984	0.1322
[SEP]	0.0010	-0.0378
AI	0.0246	-0.1827
safe	0.1332	-0.1034
[SEP]	0.1758	0.0361

Attention scores (bidirectional) $\text{score}(i, j) = (Q_i \cdot K_j) / \sqrt{2} = /1.4142$:

Query: [CLS]

$$\text{score}([\text{CLS}], [\text{CLS}]) = \frac{(3.2500 \times 0.2000) + (0.2000 \times 2.2500)}{1.4142} = 0.7778$$

$$\text{score}([\text{CLS}], \text{AI}) = \frac{(3.2500 \times 1.3515) + (0.2000 \times 1.9721)}{1.4142} = 3.3847$$

$$\begin{aligned}
\text{score}([\text{CLS}], \text{risky}) &= \frac{(3.2500 \times 0.6793) + (0.2000 \times 0.5270)}{1.4142} = 1.6356 \\
\text{score}([\text{CLS}], [\text{SEP}]) &= \frac{(3.2500 \times 0.3711) + (0.2000 \times -1.6889)}{1.4142} = 0.6140 \\
\text{score}([\text{CLS}], \text{AI}) &= \frac{(3.2500 \times -0.4168) + (0.2000 \times -2.3141)}{1.4142} = -1.2851 \\
\text{score}([\text{CLS}], \text{safe}) &= \frac{(3.2500 \times -0.6089) + (0.2000 \times 0.0084)}{1.4142} = -1.3982 \\
\text{score}([\text{CLS}], [\text{SEP}]) &= \frac{(3.2500 \times -0.2195) + (0.2000 \times 1.4909)}{1.4142} = -0.2935
\end{aligned}$$

Query: AI

$$\begin{aligned}
\text{score}(\text{AI}, [\text{CLS}]) &= \frac{(2.0306 \times 0.2000) + (1.3515 \times 2.2500)}{1.4142} = 2.4373 \\
\text{score}(\text{AI}, \text{AI}) &= \frac{(2.0306 \times 1.3515) + (1.3515 \times 1.9721)}{1.4142} = 3.8251 \\
\text{score}(\text{AI}, \text{risky}) &= \frac{(2.0306 \times 0.6793) + (1.3515 \times 0.5270)}{1.4142} = 1.4790 \\
\text{score}(\text{AI}, [\text{SEP}]) &= \frac{(2.0306 \times 0.3711) + (1.3515 \times -1.6889)}{1.4142} = -1.0811 \\
\text{score}(\text{AI}, \text{AI}) &= \frac{(2.0306 \times -0.4168) + (1.3515 \times -2.3141)}{1.4142} = -2.8099 \\
\text{score}(\text{AI}, \text{safe}) &= \frac{(2.0306 \times -0.6089) + (1.3515 \times 0.0084)}{1.4142} = -0.8663 \\
\text{score}(\text{AI}, [\text{SEP}]) &= \frac{(2.0306 \times -0.2195) + (1.3515 \times 1.4909)}{1.4142} = 1.1097
\end{aligned}$$

Query: risky

$$\begin{aligned}
\text{score}(\text{risky}, [\text{CLS}]) &= \frac{(0.8675 \times 0.2000) + (0.6793 \times 2.2500)}{1.4142} = 1.2034 \\
\text{score}(\text{risky}, \text{AI}) &= \frac{(0.8675 \times 1.3515) + (0.6793 \times 1.9721)}{1.4142} = 1.7763 \\
\text{score}(\text{risky}, \text{risky}) &= \frac{(0.8675 \times 0.6793) + (0.6793 \times 0.5270)}{1.4142} = 0.6698 \\
\text{score}(\text{risky}, [\text{SEP}]) &= \frac{(0.8675 \times 0.3711) + (0.6793 \times -1.6889)}{1.4142} = -0.5836 \\
\text{score}(\text{risky}, \text{AI}) &= \frac{(0.8675 \times -0.4168) + (0.6793 \times -2.3141)}{1.4142} = -1.3672 \\
\text{score}(\text{risky}, \text{safe}) &= \frac{(0.8675 \times -0.6089) + (0.6793 \times 0.0084)}{1.4142} = -0.3695 \\
\text{score}(\text{risky}, [\text{SEP}]) &= \frac{(0.8675 \times -0.2195) + (0.6793 \times 1.4909)}{1.4142} = 0.5815
\end{aligned}$$

Query: [SEP]

$$\text{score}([\text{SEP}], [\text{CLS}]) = \frac{(-0.9304 \times 0.2000) + (0.3711 \times 2.2500)}{1.4142} = 0.4589$$

$$\begin{aligned}
\text{score}([\text{SEP}], \text{AI}) &= \frac{(-0.9304 \times 1.3515) + (0.3711 \times 1.9721)}{1.4142} = -0.3716 \\
\text{score}([\text{SEP}], \text{risky}) &= \frac{(-0.9304 \times 0.6793) + (0.3711 \times 0.5270)}{1.4142} = -0.3086 \\
\text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(-0.9304 \times 0.3711) + (0.3711 \times -1.6889)}{1.4142} = -0.6874 \\
\text{score}([\text{SEP}], \text{AI}) &= \frac{(-0.9304 \times -0.4168) + (0.3711 \times -2.3141)}{1.4142} = -0.3330 \\
\text{score}([\text{SEP}], \text{safe}) &= \frac{(-0.9304 \times -0.6089) + (0.3711 \times 0.0084)}{1.4142} = 0.4028 \\
\text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(-0.9304 \times -0.2195) + (0.3711 \times 1.4909)}{1.4142} = 0.5356
\end{aligned}$$

Query: AI

$$\begin{aligned}
\text{score}(\text{AI}, [\text{CLS}]) &= \frac{(-0.6581 \times 0.2000) + (-0.4168 \times 2.2500)}{1.4142} = -0.7562 \\
\text{score}(\text{AI}, \text{AI}) &= \frac{(-0.6581 \times 1.3515) + (-0.4168 \times 1.9721)}{1.4142} = -1.2101 \\
\text{score}(\text{AI}, \text{risky}) &= \frac{(-0.6581 \times 0.6793) + (-0.4168 \times 0.5270)}{1.4142} = -0.4714 \\
\text{score}(\text{AI}, [\text{SEP}]) &= \frac{(-0.6581 \times 0.3711) + (-0.4168 \times -1.6889)}{1.4142} = 0.3251 \\
\text{score}(\text{AI}, \text{AI}) &= \frac{(-0.6581 \times -0.4168) + (-0.4168 \times -2.3141)}{1.4142} = 0.8760 \\
\text{score}(\text{AI}, \text{safe}) &= \frac{(-0.6581 \times -0.6089) + (-0.4168 \times 0.0084)}{1.4142} = 0.2809 \\
\text{score}(\text{AI}, [\text{SEP}]) &= \frac{(-0.6581 \times -0.2195) + (-0.4168 \times 1.4909)}{1.4142} = -0.3373
\end{aligned}$$

Query: safe

$$\begin{aligned}
\text{score}(\text{safe}, [\text{CLS}]) &= \frac{(1.7661 \times 0.2000) + (-0.6089 \times 2.2500)}{1.4142} = -0.7191 \\
\text{score}(\text{safe}, \text{AI}) &= \frac{(1.7661 \times 1.3515) + (-0.6089 \times 1.9721)}{1.4142} = 0.8386 \\
\text{score}(\text{safe}, \text{risky}) &= \frac{(1.7661 \times 0.6793) + (-0.6089 \times 0.5270)}{1.4142} = 0.6214 \\
\text{score}(\text{safe}, [\text{SEP}]) &= \frac{(1.7661 \times 0.3711) + (-0.6089 \times -1.6889)}{1.4142} = 1.1907 \\
\text{score}(\text{safe}, \text{AI}) &= \frac{(1.7661 \times -0.4168) + (-0.6089 \times -2.3141)}{1.4142} = 0.4759 \\
\text{score}(\text{safe}, \text{safe}) &= \frac{(1.7661 \times -0.6089) + (-0.6089 \times 0.0084)}{1.4142} = -0.7641 \\
\text{score}(\text{safe}, [\text{SEP}]) &= \frac{(1.7661 \times -0.2195) + (-0.6089 \times 1.4909)}{1.4142} = -0.9160
\end{aligned}$$

Query: [SEP]

$$\begin{aligned}
\text{score}([\text{SEP}], [\text{CLS}]) &= \frac{(2.6685 \times 0.2000) + (-0.2195 \times 2.2500)}{1.4142} = 0.0282 \\
\text{score}([\text{SEP}], \text{AI}) &= \frac{(2.6685 \times 1.3515) + (-0.2195 \times 1.9721)}{1.4142} = 2.2441 \\
\text{score}([\text{SEP}], \text{risky}) &= \frac{(2.6685 \times 0.6793) + (-0.2195 \times 0.5270)}{1.4142} = 1.2000 \\
\text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(2.6685 \times 0.3711) + (-0.2195 \times -1.6889)}{1.4142} = 0.9623 \\
\text{score}([\text{SEP}], \text{AI}) &= \frac{(2.6685 \times -0.4168) + (-0.2195 \times -2.3141)}{1.4142} = -0.4274 \\
\text{score}([\text{SEP}], \text{safe}) &= \frac{(2.6685 \times -0.6089) + (-0.2195 \times 0.0084)}{1.4142} = -1.1503 \\
\text{score}([\text{SEP}], [\text{SEP}]) &= \frac{(2.6685 \times -0.2195) + (-0.2195 \times 1.4909)}{1.4142} = -0.6454
\end{aligned}$$

Head 2: raw score matrix (shape 7×7):

	<i>[CLS]</i>	<i>AI</i>	<i>risky</i>	<i>[SEP]</i>	<i>AI</i>	<i>safe</i>	<i>[SEP]</i>
<i>[CLS]</i>	0.7778	3.3847	1.6356	0.6140	-1.2851	-1.3982	-0.2935
<i>AI</i>	2.4373	3.8251	1.4790	-1.0811	-2.8099	-0.8663	1.1097
<i>risky</i>	1.2034	1.7763	0.6698	-0.5836	-1.3672	-0.3695	0.5815
<i>[SEP]</i>	0.4589	-0.3716	-0.3086	-0.6874	-0.3330	0.4028	0.5356
<i>AI</i>	-0.7562	-1.2101	-0.4714	0.3251	0.8760	0.2809	-0.3373
<i>safe</i>	-0.7191	0.8386	0.6214	1.1907	0.4759	-0.7641	-0.9160
<i>[SEP]</i>	0.0282	2.2441	1.2000	0.9623	-0.4274	-1.1503	-0.6454

Softmax over every row:

[CLS]:

$$\begin{aligned}
\text{exponentials} &= [e^{0.7778} = 0.0738, e^{3.3847} = 1.0000, e^{1.6356} = 0.1739, e^{0.6140} = 0.0626, \\
&\quad e^{-1.2851} = 0.0094, e^{-1.3982} = 0.0084, e^{-0.2935} = 0.0253] \\
\text{sum} &= 1.3533 \\
\text{weights} &= [\frac{0.0738}{1.3533} = 0.0545, \frac{1.0000}{1.3533} = 0.7389, \frac{0.1739}{1.3533} = 0.1285, \frac{0.0626}{1.3533} = 0.0463, \\
&\quad \frac{0.0094}{1.3533} = 0.0069, \frac{0.0084}{1.3533} = 0.0062, \frac{0.0253}{1.3533} = 0.0187]
\end{aligned}$$

AI:

$$\begin{aligned}
\text{exponentials} &= [e^{2.4373} = 0.2496, e^{3.8251} = 1.0000, e^{1.4790} = 0.0957, e^{-1.0811} = 0.0074, \\
&\quad e^{-2.8099} = 0.0013, e^{-0.8663} = 0.0092, e^{1.1097} = 0.0662] \\
\text{sum} &= 1.4295 \\
\text{weights} &= [\frac{0.2496}{1.4295} = 0.1746, \frac{1.0000}{1.4295} = 0.6996, \frac{0.0957}{1.4295} = 0.0670, \frac{0.0074}{1.4295} = 0.0052, \\
&\quad \frac{0.0013}{1.4295} = 0.0009, \frac{0.0092}{1.4295} = 0.0064, \frac{0.0662}{1.4295} = 0.0463]
\end{aligned}$$

risky:

$$\begin{aligned}
\text{exponentials} &= [e^{1.2034} = 0.5639, e^{1.7763} = 1.0000, e^{0.6698} = 0.3307, e^{-0.5836} = 0.0944, \\
&\quad e^{-1.3672} = 0.0431, e^{-0.3695} = 0.1170, e^{0.5815} = 0.3028] \\
\text{sum} &= 2.4520
\end{aligned}$$

$$\text{weights} = [\frac{0.5639}{2.4520} = 0.2300, \frac{1.0000}{2.4520} = 0.4078, \frac{0.3307}{2.4520} = 0.1349, \frac{0.0944}{2.4520} = 0.0385, \\ \frac{0.0431}{2.4520} = 0.0176, \frac{0.1170}{2.4520} = 0.0477, \frac{0.3028}{2.4520} = 0.1235]$$

[SEP]:

$$\text{exponentials} = [e^{0.4589} = 0.9261, e^{-0.3716} = 0.4036, e^{-0.3086} = 0.4299, e^{-0.6874} = 0.2944, \\ e^{-0.3330} = 0.4195, e^{0.4028} = 0.8757, e^{0.5356} = 1.0000]$$

$$\text{sum} = 4.3491$$

$$\text{weights} = [\frac{0.9261}{4.3491} = 0.2129, \frac{0.4036}{4.3491} = 0.0928, \frac{0.4299}{4.3491} = 0.0988, \frac{0.2944}{4.3491} = 0.0677, \\ \frac{0.4195}{4.3491} = 0.0965, \frac{0.8757}{4.3491} = 0.2013, \frac{1.0000}{4.3491} = 0.2299]$$

AI:

$$\text{exponentials} = [e^{-0.7562} = 0.1955, e^{-1.2101} = 0.1242, e^{-0.4714} = 0.2599, e^{0.3251} = 0.5764, \\ e^{0.8760} = 1.0000, e^{0.2809} = 0.5515, e^{-0.3373} = 0.2972]$$

$$\text{sum} = 3.0047$$

$$\text{weights} = [\frac{0.1955}{3.0047} = 0.0651, \frac{0.1242}{3.0047} = 0.0413, \frac{0.2599}{3.0047} = 0.0865, \frac{0.5764}{3.0047} = 0.1918, \\ \frac{1.0000}{3.0047} = 0.3328, \frac{0.5515}{3.0047} = 0.1835, \frac{0.2972}{3.0047} = 0.0989]$$

safe:

$$\text{exponentials} = [e^{-0.7191} = 0.1481, e^{0.8386} = 0.7032, e^{0.6214} = 0.5659, e^{1.1907} = 1.0000, \\ e^{0.4759} = 0.4893, e^{-0.7641} = 0.1416, e^{-0.9160} = 0.1216]$$

$$\text{sum} = 3.1698$$

$$\text{weights} = [\frac{0.1481}{3.1698} = 0.0467, \frac{0.7032}{3.1698} = 0.2218, \frac{0.5659}{3.1698} = 0.1785, \frac{1.0000}{3.1698} = 0.3155, \\ \frac{0.4893}{3.1698} = 0.1544, \frac{0.1416}{3.1698} = 0.0447, \frac{0.1216}{3.1698} = 0.0384]$$

[SEP]:

$$\text{exponentials} = [e^{0.0282} = 0.1091, e^{2.2441} = 1.0000, e^{1.2000} = 0.3520, e^{0.9623} = 0.2775, \\ e^{-0.4274} = 0.0691, e^{-1.1503} = 0.0336, e^{-0.6454} = 0.0556]$$

$$\text{sum} = 1.8969$$

$$\text{weights} = [\frac{0.1091}{1.8969} = 0.0575, \frac{1.0000}{1.8969} = 0.5272, \frac{0.3520}{1.8969} = 0.1856, \frac{0.2775}{1.8969} = 0.1463, \\ \frac{0.0691}{1.8969} = 0.0365, \frac{0.0336}{1.8969} = 0.0177, \frac{0.0556}{1.8969} = 0.0293]$$

Head 2: full softmax weight matrix (shape 7×7):

	[CLS]	AI	risky	[SEP]	AI	safe	[SEP]
[CLS]	0.0545	0.7389	0.1285	0.0463	0.0069	0.0062	0.0187
AI	0.1746	0.6996	0.0670	0.0052	0.0009	0.0064	0.0463
risky	0.2300	0.4078	0.1349	0.0385	0.0176	0.0477	0.1235
[SEP]	0.2129	0.0928	0.0988	0.0677	0.0965	0.2013	0.2299
AI	0.0651	0.0413	0.0865	0.1918	0.3328	0.1835	0.0989
safe	0.0467	0.2218	0.1785	0.3155	0.1544	0.0447	0.0384
[SEP]	0.0575	0.5272	0.1856	0.1463	0.0365	0.0177	0.0293

Attention output = weighted sum of V:

$$\begin{aligned}
[\text{CLS}]_{[0]} &= (0.0545 \times 0.2200) + (0.7389 \times 0.1640) + (0.1285 \times 0.0984) + (0.0463 \times 0.0010) \\
&\quad + (0.0069 \times 0.0246) + (0.0062 \times 0.1332) + (0.0187 \times 0.1758) = 0.1502 \\
[\text{CLS}]_{[1]} &= (0.0545 \times 0.1400) + (0.7389 \times 0.2933) + (0.1285 \times 0.1322) + (0.0463 \times -0.0378) \\
&\quad + (0.0069 \times -0.1827) + (0.0062 \times -0.1034) + (0.0187 \times 0.0361) = 0.2384 \\
\text{AI}_{[0]} &= (0.1746 \times 0.2200) + (0.6996 \times 0.1640) + (0.0670 \times 0.0984) + (0.0052 \times 0.0010) \\
&\quad + (0.0009 \times 0.0246) + (0.0064 \times 0.1332) + (0.0463 \times 0.1758) = 0.1688 \\
\text{AI}_{[1]} &= (0.1746 \times 0.1400) + (0.6996 \times 0.2933) + (0.0670 \times 0.1322) + (0.0052 \times -0.0378) \\
&\quad + (0.0009 \times -0.1827) + (0.0064 \times -0.1034) + (0.0463 \times 0.0361) = 0.2392 \\
\text{risky}_{[0]} &= (0.2300 \times 0.2200) + (0.4078 \times 0.1640) + (0.1349 \times 0.0984) + (0.0385 \times 0.0010) \\
&\quad + (0.0176 \times 0.0246) + (0.0477 \times 0.1332) + (0.1235 \times 0.1758) = 0.1593 \\
\text{risky}_{[1]} &= (0.2300 \times 0.1400) + (0.4078 \times 0.2933) + (0.1349 \times 0.1322) + (0.0385 \times -0.0378) \\
&\quad + (0.0176 \times -0.1827) + (0.0477 \times -0.1034) + (0.1235 \times 0.0361) = 0.1645 \\
[\text{SEP}]_{[0]} &= (0.2129 \times 0.2200) + (0.0928 \times 0.1640) + (0.0988 \times 0.0984) + (0.0677 \times 0.0010) \\
&\quad + (0.0965 \times 0.0246) + (0.2013 \times 0.1332) + (0.2299 \times 0.1758) = 0.1415 \\
[\text{SEP}]_{[1]} &= (0.2129 \times 0.1400) + (0.0928 \times 0.2933) + (0.0988 \times 0.1322) + (0.0677 \times -0.0378) \\
&\quad + (0.0965 \times -0.1827) + (0.2013 \times -0.1034) + (0.2299 \times 0.0361) = 0.0374 \\
\text{AI}_{[0]} &= (0.0651 \times 0.2200) + (0.0413 \times 0.1640) + (0.0865 \times 0.0984) + (0.1918 \times 0.0010) \\
&\quad + (0.3328 \times 0.0246) + (0.1835 \times 0.1332) + (0.0989 \times 0.1758) = 0.0798 \\
\text{AI}_{[1]} &= (0.0651 \times 0.1400) + (0.0413 \times 0.2933) + (0.0865 \times 0.1322) + (0.1918 \times -0.0378) \\
&\quad + (0.3328 \times -0.1827) + (0.1835 \times -0.1034) + (0.0989 \times 0.0361) = -0.0508 \\
\text{safe}_{[0]} &= (0.0467 \times 0.2200) + (0.2218 \times 0.1640) + (0.1785 \times 0.0984) + (0.3155 \times 0.0010) \\
&\quad + (0.1544 \times 0.0246) + (0.0447 \times 0.1332) + (0.0384 \times 0.1758) = 0.0810 \\
\text{safe}_{[1]} &= (0.0467 \times 0.1400) + (0.2218 \times 0.2933) + (0.1785 \times 0.1322) + (0.3155 \times -0.0378) \\
&\quad + (0.1544 \times -0.1827) + (0.0447 \times -0.1034) + (0.0384 \times 0.0361) = 0.0519 \\
[\text{SEP}]_{[0]} &= (0.0575 \times 0.2200) + (0.5272 \times 0.1640) + (0.1856 \times 0.0984) + (0.1463 \times 0.0010) \\
&\quad + (0.0365 \times 0.0246) + (0.0177 \times 0.1332) + (0.0293 \times 0.1758) = 0.1259 \\
[\text{SEP}]_{[1]} &= (0.0575 \times 0.1400) + (0.5272 \times 0.2933) + (0.1856 \times 0.1322) + (0.1463 \times -0.0378) \\
&\quad + (0.0365 \times -0.1827) + (0.0177 \times -0.1034) + (0.0293 \times 0.0361) = 0.1743
\end{aligned}$$

Head 2: resulting attention-output matrix (shape 7×2):

[CLS]	0.1502	0.2384
AI	0.1688	0.2392
risky	0.1593	0.1645
[SEP]	0.1415	0.0374
AI	0.0798	-0.0508
safe	0.0810	0.0519
[SEP]	0.1259	0.1743

Concatenate heads, project through W_o

Concat(head1, head2) (shape 7×4):

[CLS]	0.3112	0.0449	0.1502	0.2384
AI	0.3610	0.1103	0.1688	0.2392
risky	0.2732	0.0673	0.1593	0.1645
[SEP]	0.1775	0.0389	0.1415	0.0374
AI	0.1398	0.0097	0.0798	-0.0508
safe	0.2305	0.0055	0.0810	0.0519
[SEP]	0.2793	0.0238	0.1259	0.1743

W_o (shape 4×4):

0.3000	0.1000	0.0000	0.2000
0.1000	0.2000	0.1000	0.0000
0.2000	0.0000	0.3000	0.1000
0.0000	0.3000	0.1000	0.2000

concat (**every token as a row**) (shape 7×4):

[CLS]	0.3112	0.0449	0.1502	0.2384
AI	0.3610	0.1103	0.1688	0.2392
risky	0.2732	0.0673	0.1593	0.1645
[SEP]	0.1775	0.0389	0.1415	0.0374
AI	0.1398	0.0097	0.0798	-0.0508
safe	0.2305	0.0055	0.0810	0.0519
[SEP]	0.2793	0.0238	0.1259	0.1743

Expanded arithmetic, $MultiHead = concat \cdot W_o$, every token:

[CLS] (*concat row* = [0.3112, 0.0449, 0.1502, 0.2384])

$$MultiHead[0] = (0.3112 \times 0.3000) + (0.0449 \times 0.1000) + (0.1502 \times 0.2000) + (0.2384 \times 0.0000) = 0.1279$$

$$MultiHead[1] = (0.3112 \times 0.1000) + (0.0449 \times 0.2000) + (0.1502 \times 0.0000) + (0.2384 \times 0.3000) = 0.1116$$

$$MultiHead[2] = (0.3112 \times 0.0000) + (0.0449 \times 0.1000) + (0.1502 \times 0.3000) + (0.2384 \times 0.1000) = 0.0734$$

$$MultiHead[3] = (0.3112 \times 0.2000) + (0.0449 \times 0.0000) + (0.1502 \times 0.1000) + (0.2384 \times 0.2000) = 0.1249$$

AI (*concat row* = [0.3610, 0.1103, 0.1688, 0.2392])

$$MultiHead[0] = (0.3610 \times 0.3000) + (0.1103 \times 0.1000) + (0.1688 \times 0.2000) + (0.2392 \times 0.0000) = 0.1531$$

$$MultiHead[1] = (0.3610 \times 0.1000) + (0.1103 \times 0.2000) + (0.1688 \times 0.0000) + (0.2392 \times 0.3000) = 0.1299$$

$$MultiHead[2] = (0.3610 \times 0.0000) + (0.1103 \times 0.1000) + (0.1688 \times 0.3000) + (0.2392 \times 0.1000) = 0.0856$$

$$MultiHead[3] = (0.3610 \times 0.2000) + (0.1103 \times 0.0000) + (0.1688 \times 0.1000) + (0.2392 \times 0.2000) = 0.1369$$

risky (*concat row* = [0.2732, 0.0673, 0.1593, 0.1645])

$$MultiHead[0] = (0.2732 \times 0.3000) + (0.0673 \times 0.1000) + (0.1593 \times 0.2000) + (0.1645 \times 0.0000) = 0.1205$$

$$\begin{aligned}
MultiHead[1] &= (0.2732 \times 0.1000) + (0.0673 \times 0.2000) + (0.1593 \times 0.0000) + (0.1645 \times 0.3000) = 0.0901 \\
MultiHead[2] &= (0.2732 \times 0.0000) + (0.0673 \times 0.1000) + (0.1593 \times 0.3000) + (0.1645 \times 0.1000) = 0.0710 \\
MultiHead[3] &= (0.2732 \times 0.2000) + (0.0673 \times 0.0000) + (0.1593 \times 0.1000) + (0.1645 \times 0.2000) = 0.1035
\end{aligned}$$

$$[SEP] \text{ (concat row} = [0.1775, 0.0389, 0.1415, 0.0374])$$

$$\begin{aligned}
MultiHead[0] &= (0.1775 \times 0.3000) + (0.0389 \times 0.1000) + (0.1415 \times 0.2000) + (0.0374 \times 0.0000) = 0.0854 \\
MultiHead[1] &= (0.1775 \times 0.1000) + (0.0389 \times 0.2000) + (0.1415 \times 0.0000) + (0.0374 \times 0.3000) = 0.0368 \\
MultiHead[2] &= (0.1775 \times 0.0000) + (0.0389 \times 0.1000) + (0.1415 \times 0.3000) + (0.0374 \times 0.1000) = 0.0501 \\
MultiHead[3] &= (0.1775 \times 0.2000) + (0.0389 \times 0.0000) + (0.1415 \times 0.1000) + (0.0374 \times 0.2000) = 0.0571
\end{aligned}$$

$$AI \text{ (concat row} = [0.1398, 0.0097, 0.0798, -0.0508])$$

$$\begin{aligned}
MultiHead[0] &= (0.1398 \times 0.3000) + (0.0097 \times 0.1000) + (0.0798 \times 0.2000) + (-0.0508 \times 0.0000) = 0.0589 \\
MultiHead[1] &= (0.1398 \times 0.1000) + (0.0097 \times 0.2000) + (0.0798 \times 0.0000) + (-0.0508 \times 0.3000) = 0.0007 \\
MultiHead[2] &= (0.1398 \times 0.0000) + (0.0097 \times 0.1000) + (0.0798 \times 0.3000) + (-0.0508 \times 0.1000) = 0.0198 \\
MultiHead[3] &= (0.1398 \times 0.2000) + (0.0097 \times 0.0000) + (0.0798 \times 0.1000) + (-0.0508 \times 0.2000) = 0.0258
\end{aligned}$$

$$safe \text{ (concat row} = [0.2305, 0.0055, 0.0810, 0.0519])$$

$$\begin{aligned}
MultiHead[0] &= (0.2305 \times 0.3000) + (0.0055 \times 0.1000) + (0.0810 \times 0.2000) + (0.0519 \times 0.0000) = 0.0859 \\
MultiHead[1] &= (0.2305 \times 0.1000) + (0.0055 \times 0.2000) + (0.0810 \times 0.0000) + (0.0519 \times 0.3000) = 0.0397 \\
MultiHead[2] &= (0.2305 \times 0.0000) + (0.0055 \times 0.1000) + (0.0810 \times 0.3000) + (0.0519 \times 0.1000) = 0.0300 \\
MultiHead[3] &= (0.2305 \times 0.2000) + (0.0055 \times 0.0000) + (0.0810 \times 0.1000) + (0.0519 \times 0.2000) = 0.0646
\end{aligned}$$

$$[SEP] \text{ (concat row} = [0.2793, 0.0238, 0.1259, 0.1743])$$

$$\begin{aligned}
MultiHead[0] &= (0.2793 \times 0.3000) + (0.0238 \times 0.1000) + (0.1259 \times 0.2000) + (0.1743 \times 0.0000) = 0.1113 \\
MultiHead[1] &= (0.2793 \times 0.1000) + (0.0238 \times 0.2000) + (0.1259 \times 0.0000) + (0.1743 \times 0.3000) = 0.0850 \\
MultiHead[2] &= (0.2793 \times 0.0000) + (0.0238 \times 0.1000) + (0.1259 \times 0.3000) + (0.1743 \times 0.1000) = 0.0576 \\
MultiHead[3] &= (0.2793 \times 0.2000) + (0.0238 \times 0.0000) + (0.1259 \times 0.1000) + (0.1743 \times 0.2000) = 0.1033
\end{aligned}$$

Resulting *MultiHead* matrix (shape 7×4):

[CLS]	0.1279	0.1116	0.0734	0.1249
AI	0.1531	0.1299	0.0856	0.1369
risky	0.1205	0.0901	0.0710	0.1035
[SEP]	0.0854	0.0368	0.0501	0.0571
AI	0.0589	0.0007	0.0198	0.0258
safe	0.0859	0.0397	0.0300	0.0646
[SEP]	0.1113	0.0850	0.0576	0.1033

Residual add: $\mathbf{X} + \text{MultiHead}(\mathbf{X})$

$\mathbf{X}$ (original embedding) (shape 7×4):

[CLS]	0.1500	1.0500	0.0500	1.1500
AI	1.1915	0.3903	0.1600	1.2500
risky	0.7593	-0.1161	-0.0800	1.0998
[SEP]	0.1911	-0.9400	0.1800	0.9496
AI	-0.5068	-0.9036	0.0900	1.1492
safe	-0.8589	0.4337	0.2500	0.8988
[SEP]	-0.3294	0.9102	0.1100	0.8482

MultiHead($\mathbf{X}$) (shape 7×4):

[CLS]	0.1279	0.1116	0.0734	0.1249
AI	0.1531	0.1299	0.0856	0.1369
risky	0.1205	0.0901	0.0710	0.1035
[SEP]	0.0854	0.0368	0.0501	0.0571
AI	0.0589	0.0007	0.0198	0.0258
safe	0.0859	0.0397	0.0300	0.0646
[SEP]	0.1113	0.0850	0.0576	0.1033

Resulting residual_1 matrix (shape 7×4):

[CLS]	0.2779	1.1616	0.1234	1.2749
AI	1.3446	0.5202	0.2456	1.3869
risky	0.8798	-0.0260	-0.0090	1.2033
[SEP]	0.2766	-0.9032	0.2301	1.0067
AI	-0.4479	-0.9030	0.1098	1.1750
safe	-0.7730	0.4734	0.2800	0.9633
[SEP]	-0.2181	0.9951	0.1675	0.9515

LayerNorm(residual_1) -> $\mathbf{Z}$

Input matrix (every token as a row) (shape 7×4):

[CLS]	0.2779	1.1616	0.1234	1.2749
AI	1.3446	0.5202	0.2456	1.3869
risky	0.8798	-0.0260	-0.0090	1.2033
[SEP]	0.2766	-0.9032	0.2301	1.0067
AI	-0.4479	-0.9030	0.1098	1.1750
safe	-0.7730	0.4734	0.2800	0.9633
[SEP]	-0.2181	0.9951	0.1675	0.9515

$\gamma = 1$, $\beta = 0$ (identity affine, to isolate the normalization math)

$[CLS]: v = [0.2779, 1.1616, 0.1234, 1.2749]$

$$\text{mean} = \frac{0.2779 + 1.1616 + 0.1234 + 1.2749}{4} = 0.7095$$

$$\text{deviations} = [-0.4316, 0.4522, -0.5861, 0.5655] \Rightarrow \text{variance} = \frac{1.0540}{4} = 0.2635$$

$$\text{std} = \sqrt{0.2635 + 10^{-5}} = 0.5133$$

$$\text{normalized} = [\frac{-0.4316}{0.5133} = -0.8407, \frac{0.4522}{0.5133} = 0.8809, \frac{-0.5861}{0.5133} = -1.1417, \frac{0.5655}{0.5133} = 1.1016]$$

$$AI: v = [1.3446, 0.5202, 0.2456, 1.3869]$$

$$\text{mean} = \frac{1.3446 + 0.5202 + 0.2456 + 1.3869}{4} = 0.8743$$

$$\text{deviations} = [0.4703, -0.3541, -0.6287, 0.5126] \Rightarrow \text{variance} = \frac{1.0045}{4} = 0.2511$$

$$\text{std} = \sqrt{0.2511 + 10^{-5}} = 0.5011$$

$$\text{normalized} = [\frac{0.4703}{0.5011} = 0.9384, \frac{-0.3541}{0.5011} = -0.7066, \frac{-0.6287}{0.5011} = -1.2546, \frac{0.5126}{0.5011} = 1.0228]$$

$$risky: v = [0.8798, -0.0260, -0.0090, 1.2033]$$

$$\text{mean} = \frac{0.8798 + -0.0260 + -0.0090 + 1.2033}{4} = 0.5120$$

$$\text{deviations} = [0.3678, -0.5380, -0.5210, 0.6913] \Rightarrow \text{variance} = \frac{1.1741}{4} = 0.2935$$

$$\text{std} = \sqrt{0.2935 + 10^{-5}} = 0.5418$$

$$\text{normalized} = [\frac{0.3678}{0.5418} = 0.6789, \frac{-0.5380}{0.5418} = -0.9931, \frac{-0.5210}{0.5418} = -0.9617, \frac{0.6913}{0.5418} = 1.2759]$$

$$[SEP]: v = [0.2766, -0.9032, 0.2301, 1.0067]$$

$$\text{mean} = \frac{0.2766 + -0.9032 + 0.2301 + 1.0067}{4} = 0.1525$$

$$\text{deviations} = [0.1240, -1.0558, 0.0775, 0.8542] \Rightarrow \text{variance} = \frac{1.8656}{4} = 0.4664$$

$$\text{std} = \sqrt{0.4664 + 10^{-5}} = 0.6829$$

$$\text{normalized} = [\frac{0.1240}{0.6829} = 0.1816, \frac{-1.0558}{0.6829} = -1.5459, \frac{0.0775}{0.6829} = 0.1136, \frac{0.8542}{0.6829} = 1.2507]$$

$$AI: v = [-0.4479, -0.9030, 0.1098, 1.1750]$$

$$\text{mean} = \frac{-0.4479 + -0.9030 + 0.1098 + 1.1750}{4} = -0.0165$$

$$\text{deviations} = [-0.4314, -0.8864, 0.1263, 1.1915] \Rightarrow \text{variance} = \frac{2.4075}{4} = 0.6019$$

$$\text{std} = \sqrt{0.6019 + 10^{-5}} = 0.7758$$

$$\text{normalized} = [\frac{-0.4314}{0.7758} = -0.5561, \frac{-0.8864}{0.7758} = -1.1426, \frac{0.1263}{0.7758} = 0.1629, \frac{1.1915}{0.7758} = 1.5358]$$

$$safe: v = [-0.7730, 0.4734, 0.2800, 0.9633]$$

$$\text{mean} = \frac{-0.7730 + 0.4734 + 0.2800 + 0.9633}{4} = 0.2359$$

$$\text{deviations} = [-1.0089, 0.2374, 0.0441, 0.7274] \Rightarrow \text{variance} = \frac{1.6054}{4} = 0.4014$$

$$\text{std} = \sqrt{0.4014 + 10^{-5}} = 0.6335$$

$$\text{normalized} = [\frac{-1.0089}{0.6335} = -1.5926, \frac{0.2374}{0.6335} = 0.3748, \frac{0.0441}{0.6335} = 0.0696, \frac{0.7274}{0.6335} = 1.1482]$$

[SEP]: $v = [-0.2181, 0.9951, 0.1675, 0.9515]$

$$\text{mean} = \frac{-0.2181 + 0.9951 + 0.1675 + 0.9515}{4} = 0.4740$$

$$\text{deviations} = [-0.6921, 0.5211, -0.3065, 0.4775] \Rightarrow \text{variance} = \frac{1.0725}{4} = 0.2681$$

$$\text{std} = \sqrt{0.2681 + 10^{-5}} = 0.5178$$

$$\text{normalized} = [\frac{-0.6921}{0.5178} = -1.3366, \frac{0.5211}{0.5178} = 1.0064, \frac{-0.3065}{0.5178} = -0.5919, \frac{0.4775}{0.5178} = 0.9221]$$

Resulting normalized matrix (shape 7×4):

[CLS]	-0.8407	0.8809	-1.1417	1.1016
AI	0.9384	-0.7066	-1.2546	1.0228
risky	0.6789	-0.9931	-0.9617	1.2759
[SEP]	0.1816	-1.5459	0.1136	1.2507
AI	-0.5561	-1.1426	0.1629	1.5358
safe	-1.5926	0.3748	0.0696	1.1482
[SEP]	-1.3366	1.0064	-0.5919	0.9221

FFN first linear layer: hidden = Z . W1

W_1 (shape 4×8):

$$\begin{bmatrix} -0.0750 & 0.2700 & 0.1390 & 0.0590 & -0.2060 & -0.2060 & -0.2650 & 0.2200 \\ 0.0610 & 0.1250 & -0.2880 & 0.2820 & 0.1990 & -0.1730 & -0.1910 & -0.1900 \\ -0.1170 & 0.0150 & -0.0410 & -0.1250 & 0.0670 & -0.2160 & -0.1250 & -0.0800 \\ -0.0260 & 0.1710 & -0.1800 & 0.0090 & 0.0550 & -0.2720 & 0.0650 & -0.1980 \end{bmatrix}$$

Z (every token as a row) (shape 7×4):

[CLS]	-0.8407	0.8809	-1.1417	1.1016
AI	0.9384	-0.7066	-1.2546	1.0228
risky	0.6789	-0.9931	-0.9617	1.2759
[SEP]	0.1816	-1.5459	0.1136	1.2507
AI	-0.5561	-1.1426	0.1629	1.5358
safe	-1.5926	0.3748	0.0696	1.1482
[SEP]	-1.3366	1.0064	-0.5919	0.9221

Expanded arithmetic, hidden = Z · W1, every token:

[CLS] (Z row = $[-0.8407, 0.8809, -1.1417, 1.1016]$)

$$\text{hidden}[0] = (-0.8407 \times -0.0750) + (0.8809 \times 0.0610) + (-1.1417 \times -0.1170) + (1.1016 \times -0.0260) = 0.2217$$

$$\text{hidden}[1] = (-0.8407 \times 0.2700) + (0.8809 \times 0.1250) + (-1.1417 \times 0.0150) + (1.1016 \times 0.1710) = 0.0544$$

$$\text{hidden}[2] = (-0.8407 \times 0.1390) + (0.8809 \times -0.2880) + (-1.1417 \times -0.0410) + (1.1016 \times -0.1800) = -0.5220$$

$$\begin{aligned}
hidden[3] &= (-0.8407 \times 0.0590) + (0.8809 \times 0.2820) + (-1.1417 \times -0.1250) + (1.1016 \times 0.0090) = 0.3514 \\
hidden[4] &= (-0.8407 \times -0.2060) + (0.8809 \times 0.1990) + (-1.1417 \times 0.0670) + (1.1016 \times 0.0550) = 0.3326 \\
hidden[5] &= (-0.8407 \times -0.2060) + (0.8809 \times -0.1730) + (-1.1417 \times -0.2160) + (1.1016 \times -0.2720) = -0.0322 \\
hidden[6] &= (-0.8407 \times -0.2650) + (0.8809 \times -0.1910) + (-1.1417 \times -0.1250) + (1.1016 \times 0.0650) = 0.2689 \\
hidden[7] &= (-0.8407 \times 0.2200) + (0.8809 \times -0.1900) + (-1.1417 \times -0.0800) + (1.1016 \times -0.1980) = -0.4791 \\
AI \text{ (} Z \text{ row} &= [0.9384, -0.7066, -1.2546, 1.0228])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.9384 \times -0.0750) + (-0.7066 \times 0.0610) + (-1.2546 \times -0.1170) + (1.0228 \times -0.0260) = 0.0067 \\
hidden[1] &= (0.9384 \times 0.2700) + (-0.7066 \times 0.1250) + (-1.2546 \times 0.0150) + (1.0228 \times 0.1710) = 0.3211 \\
hidden[2] &= (0.9384 \times 0.1390) + (-0.7066 \times -0.2880) + (-1.2546 \times -0.0410) + (1.0228 \times -0.1800) = 0.2013 \\
hidden[3] &= (0.9384 \times 0.0590) + (-0.7066 \times 0.2820) + (-1.2546 \times -0.1250) + (1.0228 \times 0.0090) = 0.0221 \\
hidden[4] &= (0.9384 \times -0.2060) + (-0.7066 \times 0.1990) + (-1.2546 \times 0.0670) + (1.0228 \times 0.0550) = -0.3617 \\
hidden[5] &= (0.9384 \times -0.2060) + (-0.7066 \times -0.1730) + (-1.2546 \times -0.2160) + (1.0228 \times -0.2720) = -0.0783 \\
hidden[6] &= (0.9384 \times -0.2650) + (-0.7066 \times -0.1910) + (-1.2546 \times -0.1250) + (1.0228 \times 0.0650) = 0.1096 \\
hidden[7] &= (0.9384 \times 0.2200) + (-0.7066 \times -0.1900) + (-1.2546 \times -0.0800) + (1.0228 \times -0.1980) = 0.2385 \\
risky \text{ (} Z \text{ row} &= [0.6789, -0.9931, -0.9617, 1.2759])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.6789 \times -0.0750) + (-0.9931 \times 0.0610) + (-0.9617 \times -0.1170) + (1.2759 \times -0.0260) = -0.0321 \\
hidden[1] &= (0.6789 \times 0.2700) + (-0.9931 \times 0.1250) + (-0.9617 \times 0.0150) + (1.2759 \times 0.1710) = 0.2629 \\
hidden[2] &= (0.6789 \times 0.1390) + (-0.9931 \times -0.2880) + (-0.9617 \times -0.0410) + (1.2759 \times -0.1800) = 0.1901 \\
hidden[3] &= (0.6789 \times 0.0590) + (-0.9931 \times 0.2820) + (-0.9617 \times -0.1250) + (1.2759 \times 0.0090) = -0.1083 \\
hidden[4] &= (0.6789 \times -0.2060) + (-0.9931 \times 0.1990) + (-0.9617 \times 0.0670) + (1.2759 \times 0.0550) = -0.3317 \\
hidden[5] &= (0.6789 \times -0.2060) + (-0.9931 \times -0.1730) + (-0.9617 \times -0.2160) + (1.2759 \times -0.2720) = -0.1074 \\
hidden[6] &= (0.6789 \times -0.2650) + (-0.9931 \times -0.1910) + (-0.9617 \times -0.1250) + (1.2759 \times 0.0650) = 0.2129 \\
hidden[7] &= (0.6789 \times 0.2200) + (-0.9931 \times -0.1900) + (-0.9617 \times -0.0800) + (1.2759 \times -0.1980) = 0.1624 \\
[SEP] \text{ (} Z \text{ row} &= [0.1816, -1.5459, 0.1136, 1.2507])
\end{aligned}$$

$$\begin{aligned}
hidden[0] &= (0.1816 \times -0.0750) + (-1.5459 \times 0.0610) + (0.1136 \times -0.1170) + (1.2507 \times -0.0260) = -0.1537 \\
hidden[1] &= (0.1816 \times 0.2700) + (-1.5459 \times 0.1250) + (0.1136 \times 0.0150) + (1.2507 \times 0.1710) = 0.0714 \\
hidden[2] &= (0.1816 \times 0.1390) + (-1.5459 \times -0.2880) + (0.1136 \times -0.0410) + (1.2507 \times -0.1800) = 0.2407 \\
hidden[3] &= (0.1816 \times 0.0590) + (-1.5459 \times 0.2820) + (0.1136 \times -0.1250) + (1.2507 \times 0.0090) = -0.4282 \\
hidden[4] &= (0.1816 \times -0.2060) + (-1.5459 \times 0.1990) + (0.1136 \times 0.0670) + (1.2507 \times 0.0550) = -0.2687 \\
hidden[5] &= (0.1816 \times -0.2060) + (-1.5459 \times -0.1730) + (0.1136 \times -0.2160) + (1.2507 \times -0.2720) = -0.1347 \\
hidden[6] &= (0.1816 \times -0.2650) + (-1.5459 \times -0.1910) + (0.1136 \times -0.1250) + (1.2507 \times 0.0650) = 0.3142 \\
hidden[7] &= (0.1816 \times 0.2200) + (-1.5459 \times -0.1900) + (0.1136 \times -0.0800) + (1.2507 \times -0.1980) = 0.0770
\end{aligned}$$

AI (Z row = $[-0.5561, -1.1426, 0.1629, 1.5358]$)

$$hidden[0] = (-0.5561 \times -0.0750) + (-1.1426 \times 0.0610) + (0.1629 \times -0.1170) + (1.5358 \times -0.0260) = -0.0870$$

$$hidden[1] = (-0.5561 \times 0.2700) + (-1.1426 \times 0.1250) + (0.1629 \times 0.0150) + (1.5358 \times 0.1710) = -0.0279$$

$$hidden[2] = (-0.5561 \times 0.1390) + (-1.1426 \times -0.2880) + (0.1629 \times -0.0410) + (1.5358 \times -0.1800) = -0.0313$$

$$hidden[3] = (-0.5561 \times 0.0590) + (-1.1426 \times 0.2820) + (0.1629 \times -0.1250) + (1.5358 \times 0.0090) = -0.3616$$

$$hidden[4] = (-0.5561 \times -0.2060) + (-1.1426 \times 0.1990) + (0.1629 \times 0.0670) + (1.5358 \times 0.0550) = -0.0174$$

$$hidden[5] = (-0.5561 \times -0.2060) + (-1.1426 \times -0.1730) + (0.1629 \times -0.2160) + (1.5358 \times -0.2720) = -0.1407$$

$$hidden[6] = (-0.5561 \times -0.2650) + (-1.1426 \times -0.1910) + (0.1629 \times -0.1250) + (1.5358 \times 0.0650) = 0.4451$$

$$hidden[7] = (-0.5561 \times 0.2200) + (-1.1426 \times -0.1900) + (0.1629 \times -0.0800) + (1.5358 \times -0.1980) = -0.2224$$

safe (Z row = $[-1.5926, 0.3748, 0.0696, 1.1482]$)

$$hidden[0] = (-1.5926 \times -0.0750) + (0.3748 \times 0.0610) + (0.0696 \times -0.1170) + (1.1482 \times -0.0260) = 0.1043$$

$$hidden[1] = (-1.5926 \times 0.2700) + (0.3748 \times 0.1250) + (0.0696 \times 0.0150) + (1.1482 \times 0.1710) = -0.1858$$

$$hidden[2] = (-1.5926 \times 0.1390) + (0.3748 \times -0.2880) + (0.0696 \times -0.0410) + (1.1482 \times -0.1800) = -0.5388$$

$$hidden[3] = (-1.5926 \times 0.0590) + (0.3748 \times 0.2820) + (0.0696 \times -0.1250) + (1.1482 \times 0.0090) = 0.0134$$

$$hidden[4] = (-1.5926 \times -0.2060) + (0.3748 \times 0.1990) + (0.0696 \times 0.0670) + (1.1482 \times 0.0550) = 0.4705$$

$$hidden[5] = (-1.5926 \times -0.2060) + (0.3748 \times -0.1730) + (0.0696 \times -0.2160) + (1.1482 \times -0.2720) = -0.0641$$

$$hidden[6] = (-1.5926 \times -0.2650) + (0.3748 \times -0.1910) + (0.0696 \times -0.1250) + (1.1482 \times 0.0650) = 0.4164$$

$$hidden[7] = (-1.5926 \times 0.2200) + (0.3748 \times -0.1900) + (0.0696 \times -0.0800) + (1.1482 \times -0.1980) = -0.6545$$

[SEP] (Z row = $[-1.3366, 1.0064, -0.5919, 0.9221]$)

$$hidden[0] = (-1.3366 \times -0.0750) + (1.0064 \times 0.0610) + (-0.5919 \times -0.1170) + (0.9221 \times -0.0260) = 0.2069$$

$$hidden[1] = (-1.3366 \times 0.2700) + (1.0064 \times 0.1250) + (-0.5919 \times 0.0150) + (0.9221 \times 0.1710) = -0.0863$$

$$hidden[2] = (-1.3366 \times 0.1390) + (1.0064 \times -0.2880) + (-0.5919 \times -0.0410) + (0.9221 \times -0.1800) = -0.6173$$

$$hidden[3] = (-1.3366 \times 0.0590) + (1.0064 \times 0.2820) + (-0.5919 \times -0.1250) + (0.9221 \times 0.0090) = 0.2872$$

$$hidden[4] = (-1.3366 \times -0.2060) + (1.0064 \times 0.1990) + (-0.5919 \times 0.0670) + (0.9221 \times 0.0550) = 0.4867$$

$$hidden[5] = (-1.3366 \times -0.2060) + (1.0064 \times -0.1730) + (-0.5919 \times -0.2160) + (0.9221 \times -0.2720) = -0.0217$$

$$hidden[6] = (-1.3366 \times -0.2650) + (1.0064 \times -0.1910) + (-0.5919 \times -0.1250) + (0.9221 \times 0.0650) = 0.2959$$

$$hidden[7] = (-1.3366 \times 0.2200) + (1.0064 \times -0.1900) + (-0.5919 \times -0.0800) + (0.9221 \times -0.1980) = -0.6205$$

Resulting hidden matrix (shape 7×8):

[CLS]	0.2217	0.0544	-0.5220	0.3514	0.3326	-0.0322	0.2689	-0.4791
AI	0.0067	0.3211	0.2013	0.0221	-0.3617	-0.0783	0.1096	0.2385
risky	-0.0321	0.2629	0.1901	-0.1083	-0.3317	-0.1074	0.2129	0.1624
[SEP]	-0.1537	0.0714	0.2407	-0.4282	-0.2687	-0.1347	0.3142	0.0770
AI	-0.0870	-0.0279	-0.0313	-0.3616	-0.0174	-0.1407	0.4451	-0.2224
safe	0.1043	-0.1858	-0.5388	0.0134	0.4705	-0.0641	0.4164	-0.6545
[SEP]	0.2069	-0.0863	-0.6173	0.2872	0.4867	-0.0217	0.2959	-0.6205

GELU activation

Hidden matrix BEFORE GELU (shape 7×8):

[CLS]	0.2217	0.0544	-0.5220	0.3514	0.3326	-0.0322	0.2689	-0.4791
AI	0.0067	0.3211	0.2013	0.0221	-0.3617	-0.0783	0.1096	0.2385
risky	-0.0321	0.2629	0.1901	-0.1083	-0.3317	-0.1074	0.2129	0.1624
[SEP]	-0.1537	0.0714	0.2407	-0.4282	-0.2687	-0.1347	0.3142	0.0770
AI	-0.0870	-0.0279	-0.0313	-0.3616	-0.0174	-0.1407	0.4451	-0.2224
safe	0.1043	-0.1858	-0.5388	0.0134	0.4705	-0.0641	0.4164	-0.6545
[SEP]	0.2069	-0.0863	-0.6173	0.2872	0.4867	-0.0217	0.2959	-0.6205

$$\text{GELU}(x) = 0.5x \left(1 + \tanh \left(\sqrt{2/\pi} (x + 0.044715x^3) \right) \right)$$

[CLS]:

$$[0.2217 \rightarrow 0.1303, 0.0544 \rightarrow 0.0284, -0.5220 \rightarrow -0.1571, 0.3514 \rightarrow 0.2240, \\ 0.3326 \rightarrow 0.2096, -0.0322 \rightarrow -0.0157, 0.2689 \rightarrow 0.1629, -0.4791 \rightarrow -0.1514]$$

AI:

$$[0.0067 \rightarrow 0.0034, 0.3211 \rightarrow 0.2010, 0.2013 \rightarrow 0.1167, 0.0221 \rightarrow 0.0113, \\ -0.3617 \rightarrow -0.1298, -0.0783 \rightarrow -0.0367, 0.1096 \rightarrow 0.0596, 0.2385 \rightarrow 0.1418]$$

risky:

$$[-0.0321 \rightarrow -0.0157, 0.2629 \rightarrow 0.1587, 0.1901 \rightarrow 0.1094, -0.1083 \rightarrow -0.0495, \\ -0.3317 \rightarrow -0.1228, -0.1074 \rightarrow -0.0491, 0.2129 \rightarrow 0.1624, 0.1624 \rightarrow 0.0916]$$

[SEP]:

$$[-0.1537 \rightarrow -0.0675, 0.0714 \rightarrow 0.0377, 0.2407 \rightarrow 0.1432, -0.4282 \rightarrow -0.1431, \\ -0.2687 \rightarrow -0.1059, -0.1347 \rightarrow -0.0601, 0.3142 \rightarrow 0.1959, 0.0770 \rightarrow 0.0408]$$

AI:

$$[-0.0870 \rightarrow -0.0405, -0.0279 \rightarrow -0.0136, -0.0313 \rightarrow -0.0153, -0.3616 \rightarrow -0.1297, \\ -0.0174 \rightarrow -0.0086, -0.1407 \rightarrow -0.0625, 0.4451 \rightarrow 0.2990, -0.2224 \rightarrow -0.0916]$$

safe:

$$[0.1043 \rightarrow 0.0565, -0.1858 \rightarrow -0.0792, -0.5388 \rightarrow -0.1590, 0.0134 \rightarrow 0.0068, \\ 0.4705 \rightarrow 0.3204, -0.0641 \rightarrow -0.0304, 0.4164 \rightarrow 0.2754, -0.6545 \rightarrow -0.1679]$$

[SEP]:

$$[0.2069 \rightarrow 0.1204, -0.0863 \rightarrow -0.0402, -0.6173 \rightarrow -0.1658, 0.2872 \rightarrow 0.1761, \\ 0.4867 \rightarrow 0.3342, -0.0217 \rightarrow -0.0107, 0.2959 \rightarrow 0.1824, -0.6205 \rightarrow -0.1660]$$

Resulting activated matrix (shape 7×8):

[CLS]	0.1303	0.0284	-0.1571	0.2240	0.2096	-0.0157	0.1629	-0.1514
AI	0.0034	0.2010	0.1167	0.0113	-0.1298	-0.0367	0.0596	0.1418
risky	-0.0157	0.1587	0.1094	-0.0495	-0.1228	-0.0491	0.1244	0.0916
[SEP]	-0.0675	0.0377	0.1432	-0.1431	-0.1059	-0.0601	0.1959	0.0408
AI	-0.0405	-0.0136	-0.0153	-0.1297	-0.0086	-0.0625	0.2990	-0.0916
safe	0.0565	-0.0792	-0.1590	0.0068	0.3204	-0.0304	0.2754	-0.1679
[SEP]	0.1204	-0.0402	-0.1658	0.1761	0.3342	-0.0107	0.1824	-0.1660

FFN second linear layer: $\text{FFN}(\mathbf{Z}) = \text{activated} \cdot \mathbf{W}_2$

$\mathbf{W}_2$ (shape 8×4):

$$\begin{bmatrix} -0.2610 & 0.2690 & 0.2790 & 0.1850 \\ -0.1170 & -0.2410 & 0.1110 & -0.0360 \\ -0.2270 & -0.0030 & -0.2790 & 0.2460 \\ -0.1450 & 0.0980 & -0.1130 & 0.0120 \\ 0.0280 & -0.1890 & 0.2820 & 0.1650 \\ 0.2640 & 0.2370 & 0.0590 & 0.2530 \\ -0.2470 & -0.1820 & -0.2730 & -0.1050 \\ -0.0670 & -0.1370 & 0.1970 & -0.0860 \end{bmatrix}$$

activated (every token as a row) (shape 7×8):

[CLS]	0.1303	0.0284	-0.1571	0.2240	0.2096	-0.0157	0.1629	-0.1514
AI	0.0034	0.2010	0.1167	0.0113	-0.1298	-0.0367	0.0596	0.1418
risky	-0.0157	0.1587	0.1094	-0.0495	-0.1228	-0.0491	0.1244	0.0916
[SEP]	-0.0675	0.0377	0.1432	-0.1431	-0.1059	-0.0601	0.1959	0.0408
AI	-0.0405	-0.0136	-0.0153	-0.1297	-0.0086	-0.0625	0.2990	-0.0916
safe	0.0565	-0.0792	-0.1590	0.0068	0.3204	-0.0304	0.2754	-0.1679
[SEP]	0.1204	-0.0402	-0.1658	0.1761	0.3342	-0.0107	0.1824	-0.1660

Expanded arithmetic, $\text{FFN} = \text{activated} \cdot \mathbf{W}_2$, every token:

[CLS] (*activated* row = [0.1303, 0.0284, -0.1571, 0.2240, 0.2096, -0.0157, 0.1629, -0.1514])

$$\begin{aligned} \text{FFN}[0] &= (0.1303 \times -0.2610) + (0.0284 \times -0.1170) + (-0.1571 \times -0.2270) + (0.2240 \times -0.1450) \\ &\quad + (0.2096 \times 0.0280) + (-0.0157 \times 0.2640) + (0.1629 \times -0.2470) + (-0.1514 \times -0.0670) = -0.0625 \end{aligned}$$

$$\begin{aligned} \text{FFN}[1] &= (0.1303 \times 0.2690) + (0.0284 \times -0.2410) + (-0.1571 \times -0.0030) + (0.2240 \times 0.0980) \\ &\quad + (0.2096 \times -0.1890) + (-0.0157 \times 0.2370) + (0.1629 \times -0.1820) + (-0.1514 \times -0.1370) = -0.0016 \end{aligned}$$

$$\begin{aligned} \text{FFN}[2] &= (0.1303 \times 0.2790) + (0.0284 \times 0.1110) + (-0.1571 \times -0.2790) + (0.2240 \times -0.1130) \\ &\quad + (0.2096 \times 0.2820) + (-0.0157 \times 0.0590) + (0.1629 \times -0.2730) + (-0.1514 \times 0.1970) = 0.0419 \end{aligned}$$

$$\begin{aligned} \text{FFN}[3] &= (0.1303 \times 0.1850) + (0.0284 \times -0.0360) + (-0.1571 \times 0.2460) + (0.2240 \times 0.0120) \\ &\quad + (0.2096 \times 0.1650) + (-0.0157 \times 0.2530) + (0.1629 \times -0.1050) + (-0.1514 \times -0.0860) = 0.0137 \end{aligned}$$

AI (*activated* row = [0.0034, 0.2010, 0.1167, 0.0113, -0.1298, -0.0367, 0.0596, 0.1418])

$$\begin{aligned} \text{FFN}[0] &= (0.0034 \times -0.2610) + (0.2010 \times -0.1170) + (0.1167 \times -0.2270) + (0.0113 \times -0.1450) \\ &\quad + (-0.1298 \times 0.0280) + (-0.0367 \times 0.2640) + (0.0596 \times -0.2470) + (0.1418 \times -0.0670) = -0.0901 \end{aligned}$$

$$\begin{aligned} \text{FFN}[1] &= (0.0034 \times 0.2690) + (0.2010 \times -0.2410) + (0.1167 \times -0.0030) + (0.0113 \times 0.0980) \\ &\quad + (-0.1298 \times -0.1890) + (-0.0367 \times 0.2370) + (0.0596 \times -0.1820) + (0.1418 \times -0.1370) = -0.0612 \end{aligned}$$

$$\begin{aligned} \text{FFN}[2] &= (0.0034 \times 0.2790) + (0.2010 \times 0.1110) + (0.1167 \times -0.2790) + (0.0113 \times -0.1130) \\ &\quad + (-0.1298 \times 0.2820) + (-0.0367 \times 0.0590) + (0.0596 \times -0.2730) + (0.1418 \times 0.1970) = -0.0377 \end{aligned}$$

$$\begin{aligned} \text{FFN}[3] &= (0.0034 \times 0.1850) + (0.2010 \times -0.0360) + (0.1167 \times 0.2460) + (0.0113 \times 0.0120) \\ &\quad + (-0.1298 \times 0.1650) + (-0.0367 \times 0.2530) + (0.0596 \times -0.1050) + (0.1418 \times -0.0860) = -0.0269 \end{aligned}$$

risky (activated row = $[-0.0157, 0.1587, 0.1094, -0.0495, -0.1228, -0.0491, 0.1244, 0.0916]$)

$$FFN[0] = (-0.0157 \times -0.2610) + (0.1587 \times -0.1170) + (0.1094 \times -0.2270) + (-0.0495 \times -0.1450) \\ + (-0.1228 \times 0.0280) + (-0.0491 \times 0.2640) + (0.1244 \times -0.2470) + (0.0916 \times -0.0670) = -0.0854$$

$$FFN[1] = (-0.0157 \times 0.2690) + (0.1587 \times -0.2410) + (0.1094 \times -0.0030) + (-0.0495 \times 0.0980) \\ + (-0.1228 \times -0.1890) + (-0.0491 \times 0.2370) + (0.1244 \times -0.1820) + (0.0916 \times -0.1370) = -0.0713$$

$$FFN[2] = (-0.0157 \times 0.2790) + (0.1587 \times 0.1110) + (0.1094 \times -0.2790) + (-0.0495 \times -0.1130) \\ + (-0.1228 \times 0.2820) + (-0.0491 \times 0.0590) + (0.1244 \times -0.2730) + (0.0916 \times 0.1970) = -0.0651$$

$$FFN[3] = (-0.0157 \times 0.1850) + (0.1587 \times -0.0360) + (0.1094 \times 0.2460) + (-0.0495 \times 0.0120) \\ + (-0.1228 \times 0.1650) + (-0.0491 \times 0.2530) + (0.1244 \times -0.1050) + (0.0916 \times -0.0860) = -0.0359$$

[SEP] (activated row = $[-0.0675, 0.0377, 0.1432, -0.1431, -0.1059, -0.0601, 0.1959, 0.0408]$)

$$FFN[0] = (-0.0675 \times -0.2610) + (0.0377 \times -0.1170) + (0.1432 \times -0.2270) + (-0.1431 \times -0.1450) \\ + (-0.1059 \times 0.0280) + (-0.0601 \times 0.2640) + (0.1959 \times -0.2470) + (0.0408 \times -0.0670) = -0.0685$$

$$FFN[1] = (-0.0675 \times 0.2690) + (0.0377 \times -0.2410) + (0.1432 \times -0.0030) + (-0.1431 \times 0.0980) \\ + (-0.1059 \times -0.1890) + (-0.0601 \times 0.2370) + (0.1959 \times -0.1820) + (0.0408 \times -0.1370) = -0.0772$$

$$FFN[2] = (-0.0675 \times 0.2790) + (0.0377 \times 0.1110) + (0.1432 \times -0.2790) + (-0.1431 \times -0.1130) \\ + (-0.1059 \times 0.2820) + (-0.0601 \times 0.0590) + (0.1959 \times -0.2730) + (0.0408 \times 0.1970) = -0.1173$$

$$FFN[3] = (-0.0675 \times 0.1850) + (0.0377 \times -0.0360) + (0.1432 \times 0.2460) + (-0.1431 \times 0.0120) \\ + (-0.1059 \times 0.1650) + (-0.0601 \times 0.2530) + (0.1959 \times -0.1050) + (0.0408 \times -0.0860) = -0.0371$$

AI (activated row = $[-0.0405, -0.0136, -0.0153, -0.1297, -0.0086, -0.0625, 0.2990, -0.0916]$)

$$FFN[0] = (-0.0405 \times -0.2610) + (-0.0136 \times -0.1170) + (-0.0153 \times -0.2270) + (-0.1297 \times -0.1450) \\ + (-0.0086 \times 0.0280) + (-0.0625 \times 0.2640) + (0.2990 \times -0.2470) + (-0.0916 \times -0.0670) = -0.0500$$

$$FFN[1] = (-0.0405 \times 0.2690) + (-0.0136 \times -0.2410) + (-0.0153 \times -0.0030) + (-0.1297 \times 0.0980) \\ + (-0.0086 \times -0.1890) + (-0.0625 \times 0.2370) + (0.2990 \times -0.1820) + (-0.0916 \times -0.1370) = -0.0753$$

$$FFN[2] = (-0.0405 \times 0.2790) + (-0.0136 \times 0.1110) + (-0.0153 \times -0.2790) + (-0.1297 \times -0.1130) \\ + (-0.0086 \times 0.2820) + (-0.0625 \times 0.0590) + (0.2990 \times -0.2730) + (-0.0916 \times 0.1970) = -0.0997$$

$$FFN[3] = (-0.0405 \times 0.1850) + (-0.0136 \times -0.0360) + (-0.0153 \times 0.2460) + (-0.1297 \times 0.0120) \\ + (-0.0086 \times 0.1650) + (-0.0625 \times 0.2530) + (0.2990 \times -0.1050) + (-0.0916 \times -0.0860) = -0.0531$$

safe (activated row = $[0.0565, -0.0792, -0.1590, 0.0068, 0.3204, -0.0304, 0.2754, -0.1679]$)

$$FFN[0] = (0.0565 \times -0.2610) + (-0.0792 \times -0.1170) + (-0.1590 \times -0.2270) + (0.0068 \times -0.1450) \\ + (0.3204 \times 0.0280) + (-0.0304 \times 0.2640) + (0.2754 \times -0.2470) + (-0.1679 \times -0.0670) = -0.0262$$

$$FFN[1] = (0.0565 \times 0.2690) + (-0.0792 \times -0.2410) + (-0.1590 \times -0.0030) + (0.0068 \times 0.0980) \\ + (0.3204 \times -0.1890) + (-0.0304 \times 0.2370) + (0.2754 \times -0.1820) + (-0.1679 \times -0.1370) = -0.0595$$

$$FFN[2] = (0.0565 \times 0.2790) + (-0.0792 \times 0.1110) + (-0.1590 \times -0.2790) + (0.0068 \times -0.1130) \\ + (0.3204 \times 0.2820) + (-0.0304 \times 0.0590) + (0.2754 \times -0.2730) + (-0.1679 \times 0.1970) = 0.0309$$

$$FFN[3] = (0.0565 \times 0.1850) + (-0.0792 \times -0.0360) + (-0.1590 \times 0.2460) + (0.0068 \times 0.0120) \\ + (0.3204 \times 0.1650) + (-0.0304 \times 0.2530) + (0.2754 \times -0.1050) + (-0.1679 \times -0.0860) = 0.0050$$

$$[SEP] \text{ (activated row} = [0.1204, -0.0402, -0.1658, 0.1761, 0.3342, -0.0107, 0.1824, -0.1660])$$

$$FFN[0] = (0.1204 \times -0.2610) + (-0.0402 \times -0.1170) + (-0.1658 \times -0.2270) + (0.1761 \times -0.1450) \\ + (0.3342 \times 0.0280) + (-0.0107 \times 0.2640) + (0.1824 \times -0.2470) + (-0.1660 \times -0.0670) = -0.0420$$

$$FFN[1] = (0.1204 \times 0.2690) + (-0.0402 \times -0.2410) + (-0.1658 \times -0.0030) + (0.1761 \times 0.0980) \\ + (0.3342 \times -0.1890) + (-0.0107 \times 0.2370) + (0.1824 \times -0.1820) + (-0.1660 \times -0.1370) = -0.0163$$

$$FFN[2] = (0.1204 \times 0.2790) + (-0.0402 \times 0.1110) + (-0.1658 \times -0.2790) + (0.1761 \times -0.1130) \\ + (0.3342 \times 0.2820) + (-0.0107 \times 0.0590) + (0.1824 \times -0.2730) + (-0.1660 \times 0.1970) = 0.0666$$

$$FFN[3] = (0.1204 \times 0.1850) + (-0.0402 \times -0.0360) + (-0.1658 \times 0.2460) + (0.1761 \times 0.0120) \\ + (0.3342 \times 0.1650) + (-0.0107 \times 0.2530) + (0.1824 \times -0.1050) + (-0.1660 \times -0.0860) = 0.0326$$

Resulting FFN matrix (shape 7×4):

[CLS]	-0.0625	-0.0016	0.0419	0.0137
AI	-0.0901	-0.0612	-0.0377	-0.0269
risky	-0.0854	-0.0713	-0.0651	-0.0359
[SEP]	-0.0685	-0.0772	-0.1173	-0.0371
AI	-0.0500	-0.0753	-0.0997	-0.0531
safe	-0.0262	-0.0595	0.0309	0.0050
[SEP]	-0.0420	-0.0163	0.0666	0.0326

Residual add: $Z + FFN(Z)$

Z (shape 7×4):

[CLS]	-0.8407	0.8809	-1.1417	1.1016
AI	0.9384	-0.7066	-1.2546	1.0228
risky	0.6789	-0.9931	-0.9617	1.2759
[SEP]	0.1816	-1.5459	0.1136	1.2507
AI	-0.5561	-1.1426	0.1629	1.5358
safe	-1.5926	0.3748	0.0696	1.1482
[SEP]	-1.3366	1.0064	-0.5919	0.9221

$FFN(Z)$ (shape 7×4):

[CLS]	-0.0625	-0.0016	0.0419	0.0137
AI	-0.0901	-0.0612	-0.0377	-0.0269
risky	-0.0854	-0.0713	-0.0651	-0.0359
[SEP]	-0.0685	-0.0772	-0.1173	-0.0371
AI	-0.0500	-0.0753	-0.0997	-0.0531
safe	-0.0262	-0.0595	0.0309	0.0050
[SEP]	-0.0420	-0.0163	0.0666	0.0326

Resulting residual_2 matrix (shape 7×4):

[CLS]	−0.9033	0.8792	−1.0998	1.1153
AI	0.8483	−0.7678	−1.2923	0.9959
risky	0.5935	−1.0643	−1.0268	1.2400
[SEP]	0.1131	−1.6231	−0.0037	1.2136
AI	−0.6061	−1.2179	0.0632	1.4827
safe	−1.6188	0.3153	0.1005	1.1531
[SEP]	−1.3786	0.9900	−0.5253	0.9547

Final LayerNorm(residual_2) -> FINAL LAYER OUTPUT

Input matrix (every token as a row) (shape 7×4):

[CLS]	−0.9033	0.8792	−1.0998	1.1153
AI	0.8483	−0.7678	−1.2923	0.9959
risky	0.5935	−1.0643	−1.0268	1.2400
[SEP]	0.1131	−1.6231	−0.0037	1.2136
AI	−0.6061	−1.2179	0.0632	1.4827
safe	−1.6188	0.3153	0.1005	1.1531
[SEP]	−1.3786	0.9900	−0.5253	0.9547

$\gamma = 1$, $\beta = 0$ (identity affine, to isolate the normalization math)

[CLS]: $v = [-0.9033, 0.8792, -1.0998, 1.1153]$

$$\text{mean} = \frac{-0.9033 + 0.8792 + -1.0998 + 1.1153}{4} = -0.0021$$

$$\text{deviations} = [-0.9011, 0.8814, -1.0977, 1.1174] \Rightarrow \text{variance} = \frac{4.0424}{4} = 1.0106$$

$$\text{std} = \sqrt{1.0106 + 10^{-5}} = 1.0053$$

$$\text{normalized} = \left[\frac{-0.9011}{1.0053} = -0.8964, \frac{0.8814}{1.0053} = 0.8767, \frac{-1.0977}{1.0053} = -1.0919, \frac{1.1174}{1.0053} = 1.1115 \right]$$

AI: $v = [0.8483, -0.7678, -1.2923, 0.9959]$

$$\text{mean} = \frac{0.8483 + -0.7678 + -1.2923 + 0.9959}{4} = -0.0540$$

$$\text{deviations} = [0.9023, -0.7138, -1.2383, 1.0498] \Rightarrow \text{variance} = \frac{3.9592}{4} = 0.9898$$

$$\text{std} = \sqrt{0.9898 + 10^{-5}} = 0.9949$$

$$\text{normalized} = \left[\frac{0.9023}{0.9949} = 0.9069, \frac{-0.7138}{0.9949} = -0.7175, \frac{-1.2383}{0.9949} = -1.2447, \frac{1.0498}{0.9949} = 1.0552 \right]$$

risky: $v = [0.5935, -1.0643, -1.0268, 1.2400]$

$$\text{mean} = \frac{0.5935 + -1.0643 + -1.0268 + 1.2400}{4} = -0.0644$$

$$\text{deviations} = [0.6579, -0.9999, -0.9624, 1.3044] \Rightarrow \text{variance} = \frac{4.0603}{4} = 1.0151$$

$$\text{std} = \sqrt{1.0151 + 10^{-5}} = 1.0075$$

$$\text{normalized} = \left[\frac{0.6579}{1.0075} = 0.6530, \frac{-0.9999}{1.0075} = -0.9925, \frac{-0.9624}{1.0075} = -0.9552, \frac{1.3044}{1.0075} = 1.2947 \right]$$

$$[SEP]: v = [0.1131, -1.6231, -0.0037, 1.2136]$$

$$\text{mean} = \frac{0.1131 + -1.6231 + -0.0037 + 1.2136}{4} = -0.0750$$

$$\text{deviations} = [0.1881, -1.5481, 0.0713, 1.2886] \Rightarrow \text{variance} = \frac{4.0975}{4} = 1.0244$$

$$\text{std} = \sqrt{1.0244 + 10^{-5}} = 1.0121$$

$$\text{normalized} = [\frac{0.1881}{1.0121} = 0.1859, \frac{-1.5481}{1.0121} = -1.5295, \frac{0.0713}{1.0121} = 0.0705, \frac{1.2886}{1.0121} = 1.2732]$$

$$AI: v = [-0.6061, -1.2179, 0.0632, 1.4827]$$

$$\text{mean} = \frac{-0.6061 + -1.2179 + 0.0632 + 1.4827}{4} = -0.0695$$

$$\text{deviations} = [-0.5366, -1.1484, 0.1327, 1.5523] \Rightarrow \text{variance} = \frac{4.0338}{4} = 1.0085$$

$$\text{std} = \sqrt{1.0085 + 10^{-5}} = 1.0042$$

$$\text{normalized} = [\frac{-0.5366}{1.0042} = -0.5343, \frac{-1.1484}{1.0042} = -1.1436, \frac{0.1327}{1.0042} = 0.1321, \frac{1.5523}{1.0042} = 1.5457]$$

$$safe: v = [-1.6188, 0.3153, 0.1005, 1.1531]$$

$$\text{mean} = \frac{-1.6188 + 0.3153 + 0.1005 + 1.1531}{4} = -0.0125$$

$$\text{deviations} = [-1.6063, 0.3278, 0.1129, 1.1656] \Rightarrow \text{variance} = \frac{4.0591}{4} = 1.0148$$

$$\text{std} = \sqrt{1.0148 + 10^{-5}} = 1.0074$$

$$\text{normalized} = [\frac{-1.6063}{1.0074} = -1.5946, \frac{0.3278}{1.0074} = 0.3254, \frac{0.1129}{1.0074} = 0.1121, \frac{1.1656}{1.0074} = 1.1571]$$

$$[SEP]: v = [-1.3786, 0.9900, -0.5253, 0.9547]$$

$$\text{mean} = \frac{-1.3786 + 0.9900 + -0.5253 + 0.9547}{4} = 0.0102$$

$$\text{deviations} = [-1.3888, 0.9798, -0.5355, 0.9445] \Rightarrow \text{variance} = \frac{4.0677}{4} = 1.0169$$

$$\text{std} = \sqrt{1.0169 + 10^{-5}} = 1.0084$$

$$\text{normalized} = [\frac{-1.3888}{1.0084} = -1.3772, \frac{0.9798}{1.0084} = 0.9716, \frac{-0.5355}{1.0084} = -0.5310, \frac{0.9445}{1.0084} = 0.9366]$$

Resulting normalized matrix (shape 7×4):

[CLS]	-0.8964	0.8767	-1.0919	1.1115
AI	0.9069	-0.7175	-1.2447	1.0552
risky	0.6530	-0.9925	-0.9552	1.2947
[SEP]	0.1859	-1.5295	0.0705	1.2732
AI	-0.5343	-1.1436	0.1321	1.5457
safe	-1.5946	0.3254	0.1121	1.1571
[SEP]	-1.3772	0.9716	-0.5310	0.9366

Classifier head on [CLS] -> relevance score

[CLS] final vector (only this row is used) (shape 1×4):

$$[\text{CLS}] \mid \begin{bmatrix} -0.8964 & 0.8767 & -1.0919 & 1.1115 \end{bmatrix}$$

W_{clf} (shape 4×1):

$$\begin{bmatrix} 0.8000 \\ -0.5000 \\ 0.6000 \\ 0.3000 \end{bmatrix}$$

CLS (every token as a row) (shape 1×4):

$$[\text{CLS}] \mid \begin{bmatrix} -0.8964 & 0.8767 & -1.0919 & 1.1115 \end{bmatrix}$$

Expanded arithmetic, $raw_logit = CLS \cdot W_{clf}$, every token:

$[CLS]$ (CLS row = $[-0.8964, 0.8767, -1.0919, 1.1115]$)

$$raw_logit[0] = (-0.8964 \times 0.8000) + (0.8767 \times -0.5000) + (-1.0919 \times 0.6000) + (1.1115 \times 0.3000) = -1.4772$$

Resulting raw_logit matrix (shape 1×1):

$$[\text{CLS}] \mid -1.4772$$

$$logit = raw_logit + b_{clf} = -1.4772 + 0.1 = -1.3772$$

$$score = \text{sigmoid}(-1.3772) = \frac{1}{1 + e^{1.3772}} = 0.2015$$

AI" encoder-layer example in code — tokenization through the final LayerNorm output — with exercises at the end for extending it (new sentences, extra heads, a second stacked layer, and confirming that an untrained network's attention starts out near-uniform).

BERT Encoder Layer — Full Worked Example: "I love AI"

This notebook replicates, in executable code, every calculation from the walkthrough: tokenization -> embeddings -> multi-head self-attention (2 heads) -> residual + LayerNorm -> feed-forward network -> residual + LayerNorm. All weight matrices are the exact same hand-picked toy values used in the manual walkthrough, so every number below matches.

In [1]:

```
import numpy as np
np.set_printoptions(precision=4, suppress=True)

tokens = ["[CLS]", "I", "love", "AI", "[SEP]"]
n = len(tokens)
d_model = 4
num_heads = 2
d_k = d_model // num_heads
d_ff = 8
print(f"Tokens: {tokens}")
print(f"d_model={d_model}, num_heads={num_heads}, d_k per head={d_k}, d_ff={d_ff}")
```

```
Tokens: ['[CLS]', 'I', 'love', 'AI', '[SEP]']
d_model=4, num_heads=2, d_k per head=2, d_ff=8
```

Step 1-2: Token Embeddings

Looked up from a (toy) vocabulary table. In real BERT this is a learned 30522x768 matrix.

In [2]:

```
token_emb = {
    "[CLS]": np.array([0.10, 0.00, 0.00, 0.10]),
    "I":      np.array([0.20, 0.10, -0.10, 0.00]),
    "love":   np.array([-0.10, 0.30, 0.20, 0.10]),
    "AI":     np.array([0.30, -0.20, 0.10, 0.20]),
    "[SEP]":  np.array([0.00, 0.00, 0.10, -0.10]),
}
E_tok = np.array([token_emb[t] for t in tokens])
for t, v in zip(tokens, E_tok):
    print(f"{t:8s} -> {v}")
```

```
[CLS]    -> [0.1 0.  0.  0.1]
I        -> [ 0.2  0.1 -0.1  0. ]
love     -> [-0.1  0.3  0.2  0.1]
AI       -> [ 0.3 -0.2  0.1  0.2]
[SEP]    -> [ 0.  0.  0.1 -0.1]
```

Step 3: Segment Embeddings

Single sentence -> every token gets Segment A.

In [3]:

```
seg_A = np.array([0.05, 0.05, 0.05, 0.05])
E_seg = np.tile(seg_A, (n, 1))
print(E_seg)
```

```
[[0.05 0.05 0.05 0.05]
 [0.05 0.05 0.05 0.05]
 [0.05 0.05 0.05 0.05]
 [0.05 0.05 0.05 0.05]
 [0.05 0.05 0.05 0.05]]
```

Step 4: Positional Embeddings

Using the closed-form sinusoidal formula (real BERT uses a *learned* position table instead, added the same way):

$$PE(pos, 2i) = \sin(pos / 10000^{(2i/d_model)})$$

$$PE(pos, 2i+1) = \cos(pos / 10000^{(2i/d_model)})$$

In [4]:

```
def positional_encoding(pos, d_model):
    pe = np.zeros(d_model)
    for i in range(0, d_model, 2):
        div_term = 10000 ** (i / d_model)
        pe[i] = np.sin(pos / div_term)
        if i + 1 < d_model:
            pe[i+1] = np.cos(pos / div_term)
    return pe

E_pos = np.array([positional_encoding(pos, d_model) for pos in range(n)])
for pos, t in enumerate(tokens):
    print(f"pos={pos} ({t:8s}) -> {E_pos[pos]}")
```

```
pos=0 ([CLS] ) -> [0. 1. 0. 1.]
pos=1 (I ) -> [0.8415 0.5403 0.01 1. ]
pos=2 (love ) -> [ 0.9093 -0.4161 0.02 0.9998]
pos=3 (AI ) -> [ 0.1411 -0.99 0.03 0.9996]
pos=4 ([SEP] ) -> [-0.7568 -0.6536 0.04 0.9992]
```

Step 5: Sum -> Input Embedding X

In [5]:

```
X = E_tok + E_seg + E_pos
for t, v in zip(tokens, X):
    print(f"{t:8s} -> {v}")
```

```
[CLS] -> [0.15 1.05 0.05 1.15]
I -> [ 1.0915 0.6903 -0.04 1.05 ]
love -> [ 0.8593 -0.0661 0.27 1.1498]
AI -> [ 0.4911 -1.14 0.18 1.2496]
[SEP] -> [-0.7068 -0.6036 0.19 0.9492]
```

Step 6: Project X into Q, K, V for both heads

Each head has its own learned W_q , W_k , W_v (4x2 here). These are the exact toy weights used in the manual walkthrough.

In [6]:

```
Wq = [
    np.array([[1.0, 0.0], [0.0, 2.0], [1.0, 1.0], [0.0, 0.0]]),
    np.array([[0.0, 1.0], [2.0, 0.0], [0.0, 1.0], [1.0, 0.0]]),
]
Wk = [
    np.array([[2.0, 0.0], [0.0, 1.0], [1.0, 0.0], [0.0, 1.0]]),
    np.array([[1.0, 1.0], [0.0, 2.0], [1.0, 0.0], [0.0, 0.0]]),
]
Wv = [
    np.array([[0.1, 0.1], [0.1, 0.0], [0.0, 0.2], [0.2, 0.0]]),
    np.array([[0.0, 0.2], [0.1, 0.1], [0.0, 0.1], [0.1, 0.0]]),
]

heads_Q, heads_K, heads_V = [], [], []
for h in range(num_heads):
    Q = X @ Wq[h]; K = X @ Wk[h]; V = X @ Wv[h]
    heads_Q.append(Q); heads_K.append(K); heads_V.append(V)
    print(f"--- HEAD {h+1} ---")
    print("Q =\n", Q)
    print("K =\n", K)
    print("V =\n", V)
    print()
```

```
--- HEAD 1 ---
Q =
[[ 0.2      2.15   ]
 [ 1.0515   1.3406]
 [ 1.1293   0.1377]
 [ 0.6711  -2.1    ]
 [-0.5168  -1.0173]]
K =
[[ 0.35     2.2    ]
 [ 2.1429   1.7403]
 [ 1.9886   1.0837]
 [ 1.1622   0.1096]
 [-1.2236   0.3456]]
V =
[[ 0.35     0.025 ]
 [ 0.3882   0.1011]
 [ 0.3093   0.1399]
 [ 0.185    0.0851]
 [ 0.0588  -0.0327]]
```

```
--- HEAD 2 ---
Q =
[[ 3.25     0.2    ]
 [ 2.4306   1.0515]
 [ 1.0175   1.1293]
 [-1.0304   0.6711]
 [-0.2581  -0.5168]]
K =
[[ 0.2      2.25   ]
 [ 1.0515   2.4721]
 [ 1.1293   0.727  ]
 [ 0.6711  -1.7889]
 [-0.5168  -1.9141]]
V =
[[ 0.22     0.14   ]
 [ 0.174    0.2833]
 [ 0.1084   0.1922]
 [ 0.011    0.0022]
 [ 0.0346  -0.1827]]
```

Step 7: Scaled Dot-Product Attention (per head)

$\text{score}(i,j) = (Q_i \cdot K_j) / \sqrt{d_k}$, then softmax over j , then weighted sum of V .

In [7]:

```
def softmax(x, axis=-1):
    e = np.exp(x - np.max(x, axis=axis, keepdims=True))
    return e / e.sum(axis=axis, keepdims=True)

head_outputs, attn_weights_all = [], []
for h in range(num_heads):
    Q, K, V = heads_Q[h], heads_K[h], heads_V[h]
    scores = (Q @ K.T) / np.sqrt(d_k)
    weights = softmax(scores, axis=-1)
    out = weights @ V
    head_outputs.append(out); attn_weights_all.append(weights)
    print(f"--- HEAD {h+1} ---")
    print("Raw scores =\n", scores)
    print("Softmax weights (rows sum to 1) =\n", weights)
    print("Row sums check:", weights.sum(axis=1))
    print("Head output =\n", out)
    print()
```

--- HEAD 1 ---

Raw scores =

```
[[ 3.3941  2.9487  1.9287  0.3309  0.3523]
 [ 2.3457  3.243   2.5058  0.968   -0.5822]
 [ 0.4937  1.8807  1.6935  0.9388  -0.9434]
 [-3.1007 -1.5672 -0.6654  0.3889 -1.0938]
 [-1.7104 -2.035   -1.5062 -0.5035  0.1986]]
```

Softmax weights (rows sum to 1) =

```
[[0.5086 0.3258 0.1175 0.0238 0.0243]
 [0.2028 0.4973 0.2379 0.0511 0.0108]
 [0.0988 0.3955 0.328  0.1542 0.0235]
 [0.0175 0.0809 0.1994 0.5723 0.1299]
 [0.0767 0.0554 0.0941 0.2564 0.5174]]
```

Row sums check: [1. 1. 1. 1. 1.]

Head output =

```
[[0.3467 0.0633]
 [0.3477 0.0927]
 [0.3195 0.1007]
 [0.2127 0.081 ]
 [0.1553 0.0256]]
```

--- HEAD 2 ---

Raw scores =

```
[[ 0.7778  2.766   2.698   1.2893 -1.4584]
 [ 2.0166  3.6451  2.4814 -0.1766 -2.3114]
 [ 1.9406  2.7306  1.393   -0.9456 -1.9003]
 [ 0.922   0.407   -0.4778 -1.3379 -0.5318]
 [-0.8587 -1.0953 -0.4718  0.5313  0.7938]]
```

Softmax weights (rows sum to 1) =

```
[[0.0592 0.4321 0.4037 0.0987 0.0063]
 [0.128  0.6523 0.2037 0.0143 0.0017]
 [0.2591 0.571  0.1499 0.0145 0.0056]
 [0.4583 0.2738 0.113  0.0478 0.1071]
 [0.08   0.0632 0.1178 0.3213 0.4177]]
```

Row sums check: [1. 1. 1. 1. 1.]

Head output =

```
[[ 0.1333  0.2074]
 [ 0.164   0.2416]
 [ 0.173   0.2259]
 [ 0.1649  0.144 ]
 [ 0.0593 -0.0239]]
```

Step 8: Concatenate heads, then project through Wo

In [8]:

```
concat = np.concatenate([head_outputs[0], head_outputs[1]], axis=1)
Wo = np.array([
    [0.3, 0.1, 0.0, 0.2],
    [0.1, 0.2, 0.1, 0.0],
    [0.2, 0.0, 0.3, 0.1],
    [0.0, 0.3, 0.1, 0.2],
])
multihead_out = concat @ Wo
print("Concat(head1, head2) =\n", concat)
print("\nMultiHead(X) = Concat @ Wo =\n", multihead_out)
```

```
Concat(head1, head2) =
[[ 0.3467  0.0633  0.1333  0.2074]
 [ 0.3477  0.0927  0.164   0.2416]
 [ 0.3195  0.1007  0.173   0.2259]
 [ 0.2127  0.081   0.1649  0.144 ]
 [ 0.1553  0.0256  0.0593 -0.0239]]
```

```
MultiHead(X) = Concat @ Wo =
[[0.137  0.1095 0.0671 0.1241]
 [0.1464 0.1258 0.0826 0.1343]
 [0.1405 0.1199 0.0845 0.1264]
 [0.1049 0.0807 0.072  0.0878]
 [0.061  0.0135 0.018  0.0322]]
```

Step 9: Residual connection + LayerNorm

$Z = \text{LayerNorm}(X + \text{MultiHead}(X))$. $\gamma=1$, $\beta=0$ to isolate the normalization math.

In [9]:

```
def layer_norm(v, eps=1e-5, gamma=1.0, beta=0.0):
    mean = v.mean(axis=-1, keepdims=True)
    var = v.var(axis=-1, keepdims=True)
    return gamma * (v - mean) / np.sqrt(var + eps) + beta

residual_1 = X + multihead_out
Z = layer_norm(residual_1)
print("X + MultiHead(X) =\n", residual_1)
print("\nZ = LayerNorm(residual_1) =\n", Z)
print("\nSanity check -- row means (~0):", Z.mean(axis=-1))
print("Sanity check -- row stds (~1):", Z.std(axis=-1))
```

```
X + MultiHead(X) =
[[ 0.287  1.1595  0.1171  1.2741]
 [ 1.2378  0.8161  0.0426  1.1842]
 [ 0.9998  0.0537  0.3545  1.2762]
 [ 0.596  -1.0593  0.252   1.3374]
 [-0.6458 -0.5902  0.208   0.9814]]
```

```
Z = LayerNorm(residual_1) =
[[-0.8242  0.8782 -1.1557  1.1017]
 [ 0.8749 -0.0086 -1.6288  0.7625]
 [ 0.6726 -1.2631 -0.6476  1.238 ]
 [ 0.3624 -1.545  -0.034   1.2167]
 [-0.953  -0.8694  0.33    1.4923]]
```

```
Sanity check -- row means (~0): [ 0. -0. -0.  0.  0.]
Sanity check -- row stds (~1): [1. 1. 1. 1. 1.]
```

Step 10: Feed-Forward Network (4 -> 8 -> 4, applied per token)

$\text{FFN}(z) = \text{GELU}(z W_1 + b_1) W_2 + b_2$

In [10]:

```
def gelu(x):
    return 0.5 * x * (1 + np.tanh(np.sqrt(2/np.pi) * (x + 0.044715 * x**3)))

np.random.seed(42)
W1 = np.round(np.random.uniform(-0.3, 0.3, size=(d_model, d_ff)), 3)
b1 = np.zeros(d_ff)
W2 = np.round(np.random.uniform(-0.3, 0.3, size=(d_ff, d_model)), 3)
b2 = np.zeros(d_model)

hidden = Z @ W1 + b1
activated = gelu(hidden)
ffn_out = activated @ W2 + b2

print("W1 (4x8) =\n", W1)
print("\nhidden = Z @ W1 =\n", hidden)
print("\nactivated = GELU(hidden) =\n", activated)
print("\nW2 (8x4) =\n", W2)
print("\nFFN(Z) = activated @ W2 =\n", ffn_out)
```

```
W1 (4x8) =
[[-0.075  0.27   0.139  0.059 -0.206 -0.206 -0.265  0.22 ]
 [ 0.061  0.125 -0.288  0.282  0.199 -0.173 -0.191 -0.19 ]
 [-0.117  0.015 -0.041 -0.125  0.067 -0.216 -0.125 -0.08 ]
 [-0.026  0.171 -0.18   0.009  0.055 -0.272  0.065 -0.198]]

hidden = Z @ W1 =
[[ 0.222  0.0583 -0.5184  0.3534  0.3277 -0.0322  0.2668 -0.4739]
 [ 0.1046  0.3411  0.0536  0.2597 -0.2491 -0.0343  0.023  0.1734]
 [-0.0839  0.2257  0.261  -0.2244 -0.3652 -0.1169  0.2244  0.1946]
 [-0.1491  0.1123  0.2777 -0.3991 -0.3175 -0.1309  0.2824  0.1351]
 [-0.059  -0.1058 -0.1642 -0.3292  0.1275 -0.1305  0.4743 -0.3664]]

activated = GELU(hidden) =
[[ 0.1305  0.0305 -0.1566  0.2255  0.2059 -0.0157  0.1614 -0.1506]
 [ 0.0567  0.2161  0.0279  0.1564 -0.1001 -0.0167  0.0117  0.0987]
 [-0.0392  0.133  0.1573 -0.0923 -0.1306 -0.053  0.1321  0.1123]
 [-0.0657  0.0611  0.1693 -0.1377 -0.1192 -0.0587  0.1726  0.0748]
 [-0.0281 -0.0485 -0.0714 -0.1221  0.0702 -0.0585  0.3237 -0.1308]]

W2 (8x4) =
[[-0.261  0.269  0.279  0.185]
 [-0.117 -0.241  0.111 -0.036]
 [-0.227 -0.003 -0.279  0.246]
 [-0.145  0.098 -0.113  0.012]
 [ 0.028 -0.189  0.282  0.165]
 [ 0.264  0.237  0.059  0.253]
 [-0.247 -0.182 -0.273 -0.105]
 [-0.067 -0.137  0.197 -0.086]]

FFN(Z) = activated @ W2 =
[[-0.0629 -0.0011  0.0414  0.0132]
 [-0.0858 -0.0223  0.0014 -0.019 ]
 [-0.0855 -0.0794 -0.0835 -0.0329]
 [-0.0749 -0.0794 -0.1127 -0.0334]
 [-0.0377 -0.0757 -0.0773 -0.0484]]
```

Step 11: Second residual + LayerNorm -> Final layer output

In [11]:

```
residual_2 = Z + ffn_out
final_output = layer_norm(residual_2)

print("Z + FFN(Z) =\n", residual_2)
print("\nFinal_Output = LayerNorm(Z + FFN(Z)) =\n", final_output)
print()
print("="*60)
print("FINAL CONTEXTUALIZED REPRESENTATIONS")
print("="*60)
for t, v in zip(tokens, final_output):
    print(f"{t:8s} -> {v}")
print()
print("[CLS] vector (fed to a classifier / reranker head):")
print(final_output[0])
```

```
Z + FFN(Z) =
[[-0.8871  0.8771 -1.1143  1.1149]
 [ 0.7891 -0.0309 -1.6274  0.7435]
 [ 0.5871 -1.3425 -0.7311  1.2051]
 [ 0.2875 -1.6245 -0.1467  1.1832]
 [-0.9907 -0.9451  0.2527  1.4439]]
```

```
Final_Output = LayerNorm(Z + FFN(Z)) =
[[-0.8803  0.875  -1.1063  1.1116]
 [ 0.8395  0.0006 -1.633  0.7929]
 [ 0.6483 -1.2544 -0.6515  1.2577]
 [ 0.3572 -1.5265 -0.0705  1.2398]
 [-0.9298 -0.8843  0.3122  1.502  ]]
```

```
=====
FINAL CONTEXTUALIZED REPRESENTATIONS
=====
```

```
[CLS]    -> [-0.8803  0.875  -1.1063  1.1116]
I         -> [ 0.8395  0.0006 -1.633  0.7929]
love      -> [ 0.6483 -1.2544 -0.6515  1.2577]
AI        -> [ 0.3572 -1.5265 -0.0705  1.2398]
[SEP]     -> [-0.9298 -0.8843  0.3122  1.502  ]
```

```
[CLS] vector (fed to a classifier / reranker head):
[-0.8803  0.875  -1.1063  1.1116]
```

Exercise: Try it yourself

Now that every step is defined as a function above, try modifying the example to build your own intuition:

1. **Change the sentence structure** — swap in different toy token embeddings (e.g. for "AI loves me") and re-run cells 3 onward. Notice how the attention weight matrices change shape in meaning but not in mechanics.
2. **Add a third head** — extend W_q , W_k , W_v lists with a third 4×2 (or smaller, e.g. 4×1 if you reduce d_k) matrix, update `num_heads`, and adjust W_o to be $4 \times (2 * \text{num_heads})$... -> back to 4. Confirm the shapes still work out.
3. **Verify the untrained-attention claim** — replace W_q/W_k with `np.random.uniform(-0.05, 0.05, size=(4,2))` (small random values) and re-run the attention step. Confirm the softmax output rows are all close to `[0.2, 0.2, 0.2, 0.2, 0.2]` — this is the near-uniform attention an untrained network actually starts with.
4. **Stack a second layer** — feed `final_output` back in as the new X and re-run Steps 6-11 (skip Steps 2-5, embeddings are only computed once). Compare how much the `[CLS]` vector shifts after 2 layers vs. 1.

In [12]:

```
# Your experiments here
# e.g. try changing d_k, adding a head, or re-feeding final_output as new X
```